

OSCILLATIONS

17.1 Simple harmonic oscillations

Candidates should be able to:

- 1 understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency
- 2 understand that simple harmonic motion occurs when acceleration is proportional to displacement from a fixed point and in the opposite direction
- 3 use $a = -\omega^2 x$ and recall and use, as a solution to this equation, $x = x_0 \sin \omega t$
- 4 use the equations $v = v_0 \cos \omega t$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
- 5 analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion

17.2 Energy in simple harmonic motion

Candidates should be able to:

- 1 describe the interchange between kinetic and potential energy during simple harmonic motion
- 2 recall and use $E = \frac{1}{2} m \omega^2 x_0^2$ for the total energy of a system undergoing simple harmonic motion

17.3 Damped and forced oscillations, resonance

Candidates should be able to:

- 1 understand that a resistive force acting on an oscillating system causes damping
- 2 understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping
- 3 understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency

1. O/N/07/Q3

A spring is hung from a fixed point. A mass of 130 g is hung from the free end of the spring, as shown in Fig. 3.1.

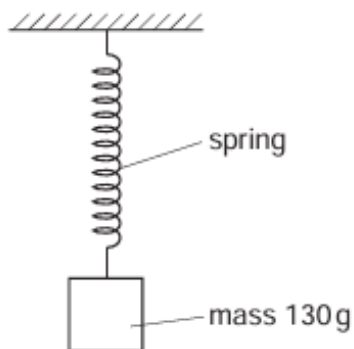


Fig. 3.1

The mass is pulled downwards from its equilibrium position through a small distance d and is released. The mass undergoes simple harmonic motion.

Fig. 3.2 shows the variation with displacement x from the equilibrium position of the kinetic energy of the mass.

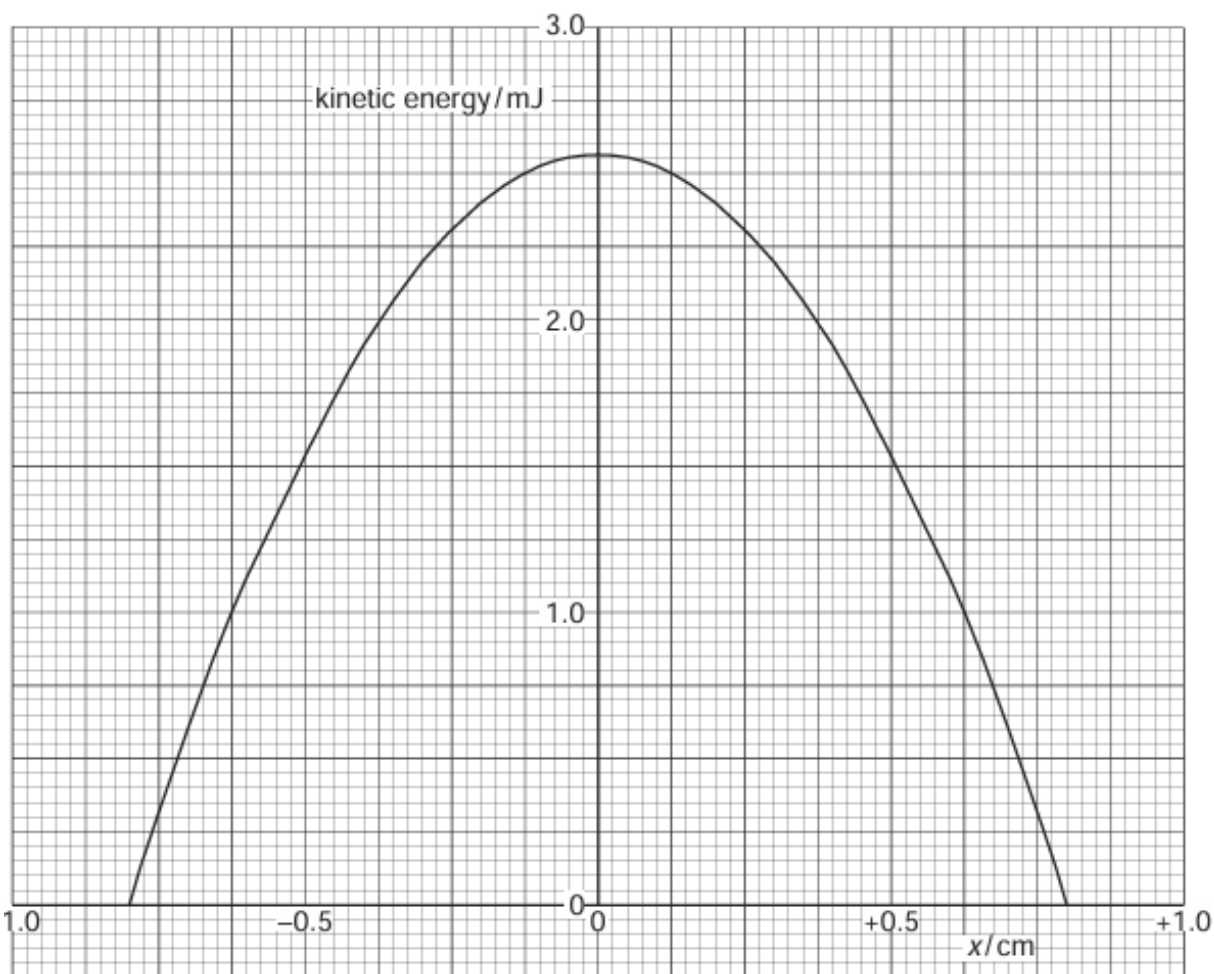


Fig. 3.2

(a) Use Fig. 3.2 to

(i) determine the distance d through which the mass was displaced initially,

$d = \dots\dots\dots$ cm [1]

(ii) show that the frequency of oscillation of the mass is approximately 4.0 Hz.

[6]

(b) (i) On Fig. 3.2, draw a line to represent the total energy of the oscillating mass. [1]

(ii) After many oscillations, damping reduces the total energy of the mass to 1.0 mJ.
For the oscillations with reduced energy,

1. state the frequency,

frequency = $\dots\dots\dots$ Hz

2. using the graph, or otherwise, state the amplitude.

amplitude = $\dots\dots\dots$ cm [2]

2. M/J/08/Q3

A tube, closed at one end, has a uniform area of cross-section. The tube contains some sand so that the tube floats upright in a liquid, as shown in Fig. 3.1.

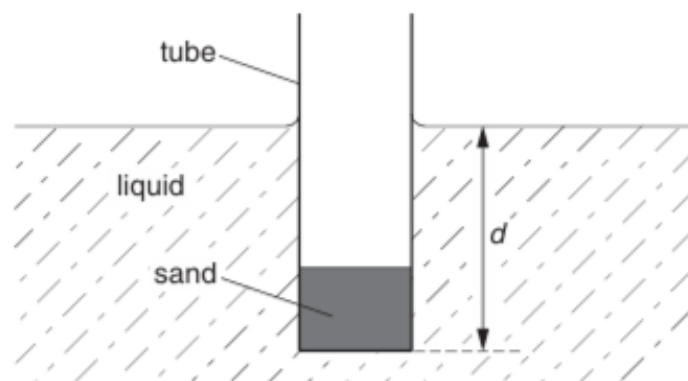
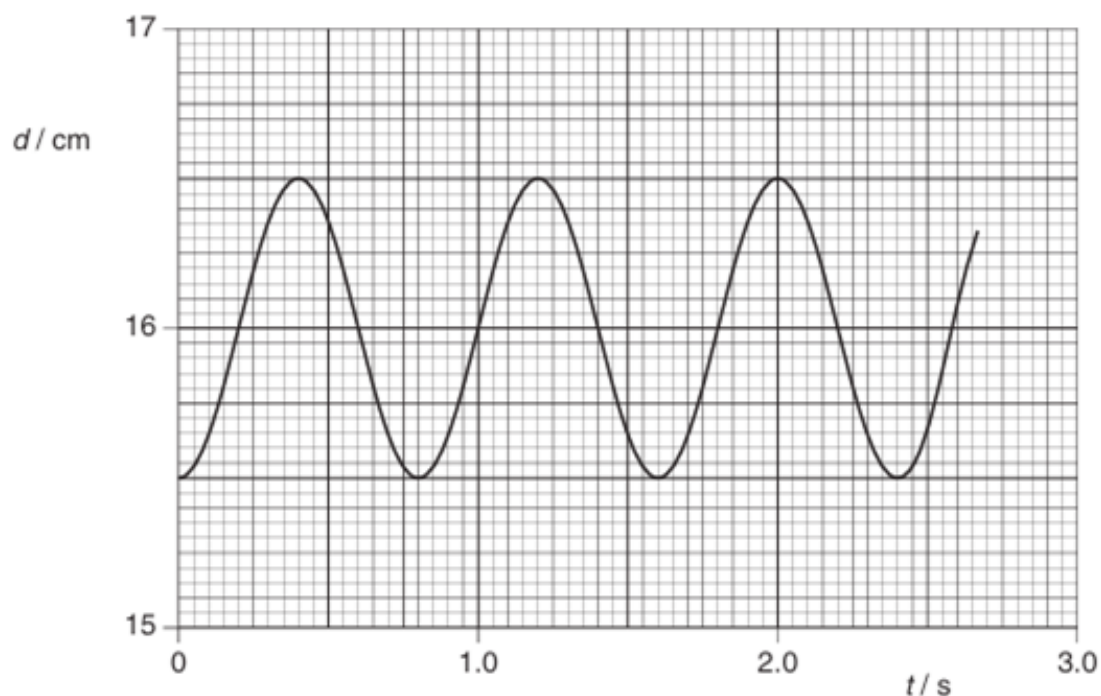


Fig. 3.1

When the tube is at rest, the depth d of immersion of the base of the tube is 16 cm. The tube is displaced vertically and then released.

The variation with time t of the depth d of the base of the tube is shown in Fig. 3.2.



(a) Use Fig. 3.2 to determine, for the oscillations of the tube,

(i) the amplitude,

amplitude = cm [1]

(ii) the period.

period = s [1]

- (b) (i) Calculate the vertical speed of the tube at a point where the depth d is 16.2 cm.

speed = cm s⁻¹ [3]

- (ii) State **one** other depth d where the speed will be equal to that calculated in (i).

d = cm [1]

- (c) (i) Explain what is meant by *damping*.

.....
.....
.....
..... [2]

- (ii) The liquid in (b) is now cooled so that, although the density is unchanged, there is friction between the liquid and the tube as it oscillates. Having been displaced, the tube completes approximately 10 oscillations before coming to rest.
On Fig. 3.2, draw a line to show the variation with time t of depth d for the first 2.5 s of the motion. [3]

3. O/N/08/Q3

The needle of a sewing machine is made to oscillate vertically through a total distance of 22 mm, as shown in Fig. 3.1.

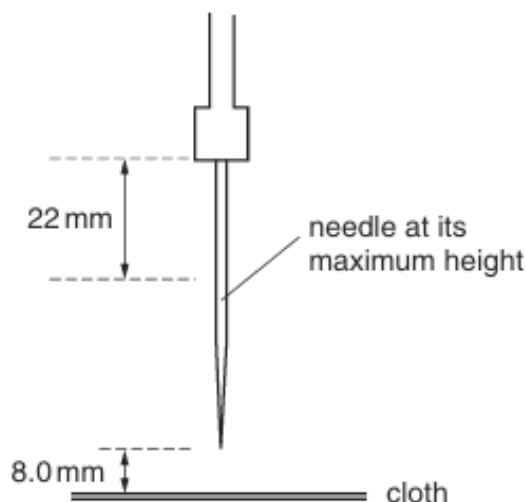


Fig. 3.1

The oscillations are simple harmonic with a frequency of 4.5 Hz.

The cloth that is being sewn is positioned 8.0 mm below the point of the needle when the needle is at its maximum height.

(a) State what is meant by *simple harmonic motion*.

.....

 [2]

(b) The displacement y of the point of the needle may be represented by the equation

$$y = a \cos \omega t.$$

(i) Suggest the position of the point of the needle at time $t = 0$.

..... [1]

(ii) Determine the values of

1. a ,

$a = \dots\dots\dots$ mm [1]

2. ω .

$\omega = \dots\dots\dots$ rads^{-1} [2]

(c) Calculate, for the point of the needle,

(i) its maximum speed,

speed = ms^{-1} [2]

(ii) its speed as it moves downwards through the cloth.

speed = ms^{-1} [3]

4. M/J/09/Q4

- I A vertical peg is attached to the edge of a horizontal disc of radius r , as shown in Fig. 4.1.

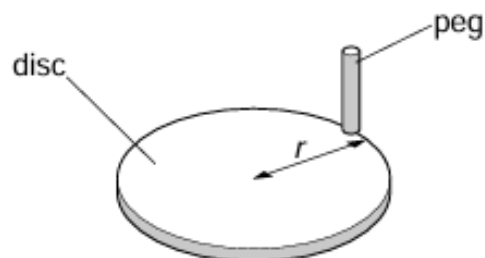


Fig. 4.1

The disc rotates at constant angular speed ω . A horizontal beam of parallel light produces a shadow of the peg on a screen, as shown in Fig. 4.2.

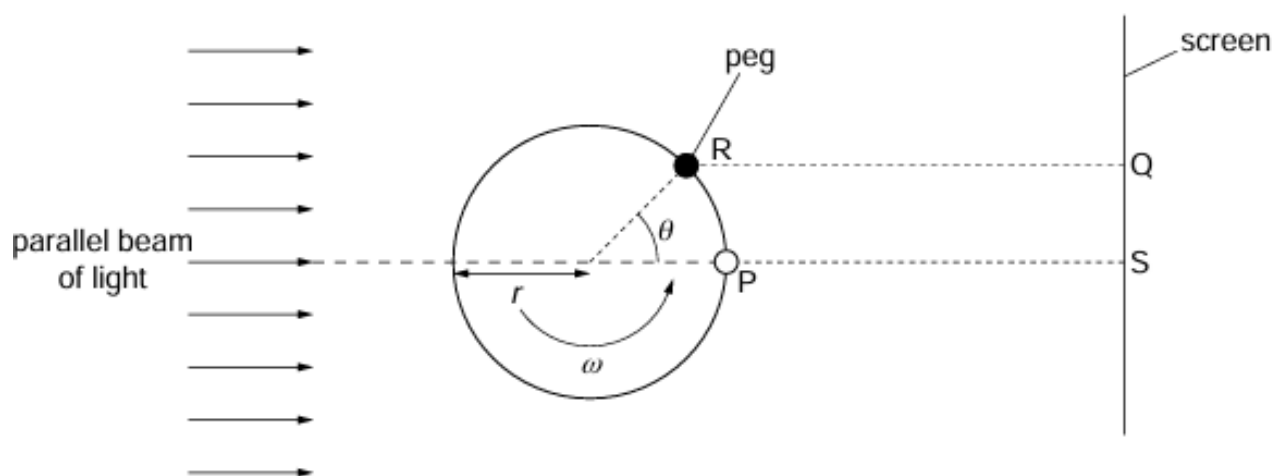


Fig. 4.2 (plan view)

At time zero, the peg is at P, producing a shadow on the screen at S.

At time t , the disc has rotated through angle θ . The peg is now at R, producing a shadow at Q.

(a) Determine,

- (i) in terms of ω and t , the angle θ ,

..... [1]

- (ii) in terms of ω , t and r , the distance SQ.

..... [1]

- (b) Use your answer to (a)(ii) to show that the shadow on the screen performs simple harmonic motion.

.....
.....
..... [2]

- (c) The disc has radius r of 12 cm and is rotating with angular speed ω of 4.7 rad s^{-1} .

Determine, for the shadow on the screen,

- (i) the frequency of oscillation,

frequency = Hz [2]

- (ii) its maximum speed.

speed = cm s^{-1} [2]

5. O/N/09/41/Q4

The variation with time t of the displacement x of the cone of a loudspeaker is shown in Fig. 4.1.

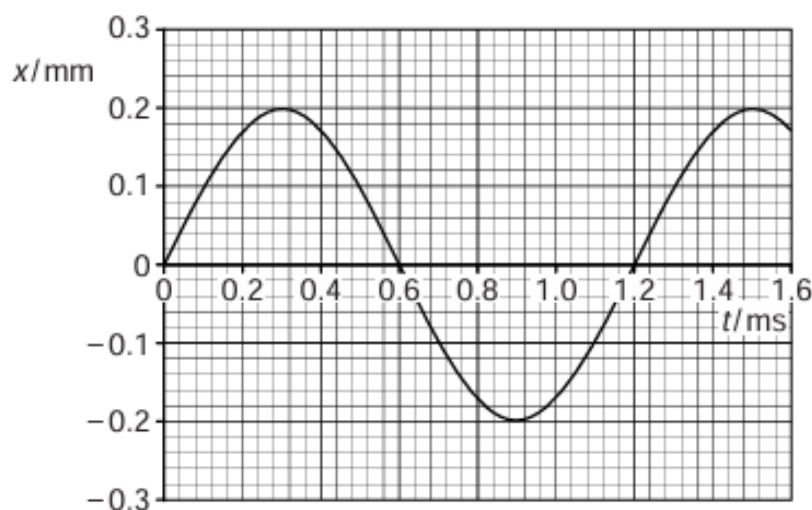


Fig. 4.1

(a) Use Fig. 4.1 to determine, for these oscillations,

(i) the amplitude,

amplitude = mm [1]

(ii) the frequency.

frequency = Hz [2]

(b) State two times at which

(i) the speed of the cone is maximum,

time ms and time ms [1]

(ii) the acceleration of the cone is maximum.

time ms and time ms [1]

- (c) The effective mass of the cone is 2.5 g.

Use your answers in (a) to determine the maximum kinetic energy of the cone.

kinetic energy = J [3]

- (d) The loudspeaker must be designed so that resonance of the cone is avoided.

- (i) State what is meant by *resonance*.

.....
.....
..... [2]

- (ii) State and briefly explain one other situation in which resonance should be avoided.

.....
.....
.....
..... [2]

6. O/N/09/42/Q3

The variation with displacement x of the acceleration a of the centre of the cone of a loudspeaker is shown in Fig. 3.1.

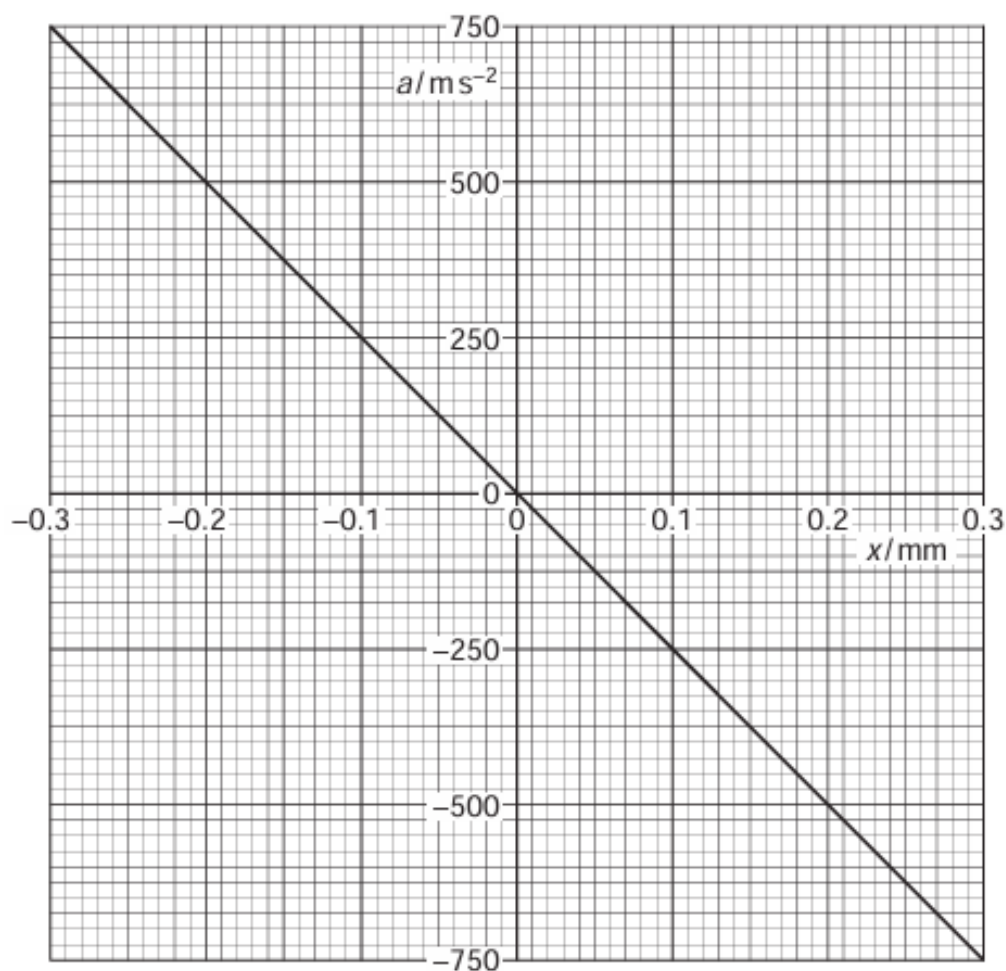


Fig. 3.1

- (a) State the two features of Fig. 3.1 that show that the motion of the cone is simple harmonic.

1.

2.

[2]

- (b) Use data from Fig. 3.1 to determine the frequency, in hertz, of vibration of the cone.

frequency = Hz [3]

- (c) The frequency of vibration of the cone is now reduced to one half of that calculated in (b).

The amplitude of vibration remains unchanged.

On the axes of Fig. 3.1, draw a line to represent the variation with displacement x of the acceleration a of the centre of the loudspeaker cone.

7. M/J/10/41/Q3

(a) State what is meant by

(i) *oscillations*,

.....
.....[1]

(ii) *free oscillations*,

.....
.....[1]

(iii) *simple harmonic motion*.

.....
.....
.....[2]

(b) Two inclined planes RA and LA each have the same constant gradient. They meet at their lower edges, as shown in Fig. 3.1.

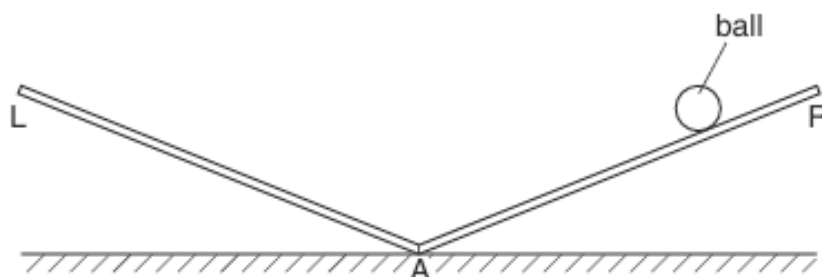


Fig. 3.1

A small ball moves from rest down plane RA and then rises up plane LA. It then moves down plane LA and rises up plane RA to its original height. The motion repeats itself.

State and explain whether the motion of the ball is simple harmonic.

.....
.....
.....[2]

8. M/J/10/42/Q2

A long strip of springy steel is clamped at one end so that the strip is vertical. A mass of 65 g is attached to the free end of the strip, as shown in Fig. 2.1.

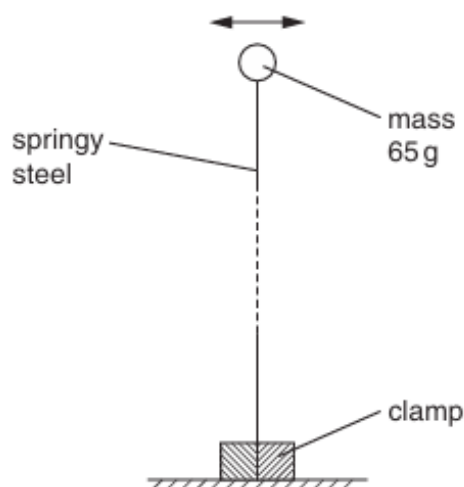


Fig. 2.1

The mass is pulled to one side and then released. The variation with time t of the horizontal displacement of the mass is shown in Fig. 2.2.

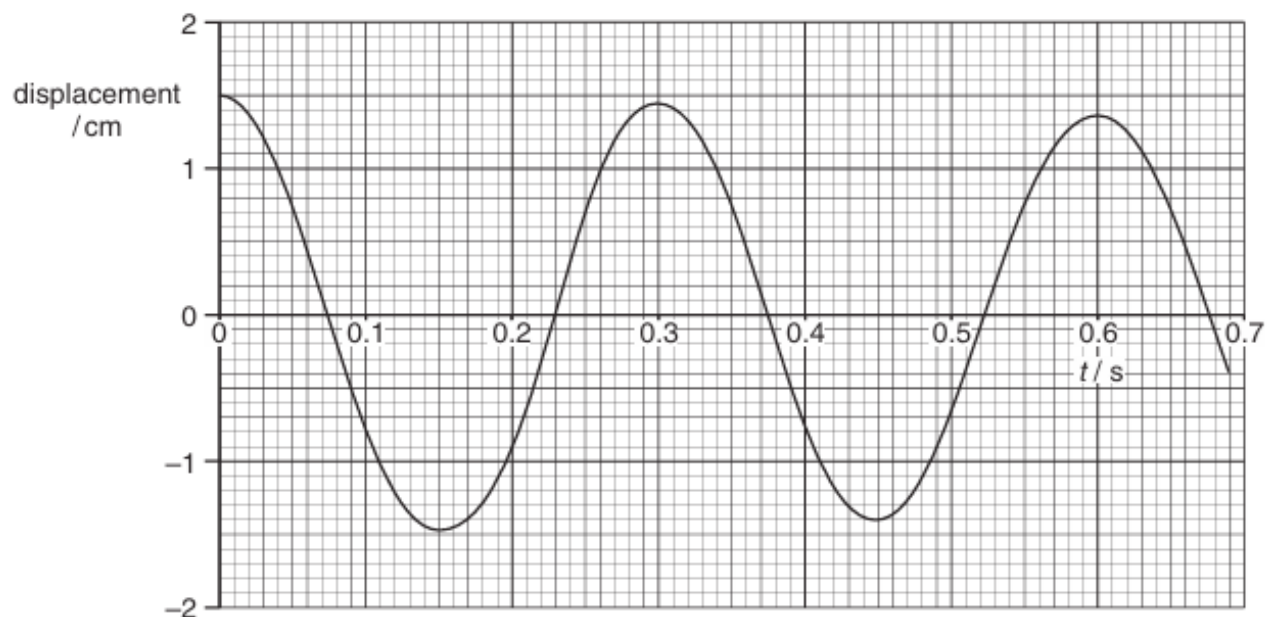


Fig. 2.2

The mass undergoes damped simple harmonic motion.

(a) (i) Explain what is meant by *damping*.

.....

.....

..... [2]

- (ii) Suggest, with a reason, whether the damping is light, critical or heavy.

.....
.....
..... [2]

- (b) (i) Use Fig. 2.2 to determine the frequency of vibration of the mass.

frequency = Hz [1]

- (ii) Hence show that the initial energy stored in the steel strip before the mass is released is approximately 3.2 mJ.

[2]

- (c) After eight complete oscillations of the mass, the amplitude of vibration is reduced from 1.5 cm to 1.1 cm. State and explain whether, after a further eight complete oscillations, the amplitude will be 0.7 cm.

.....
.....
..... [2]

9. O/N/10/41/Q3

A student sets up the apparatus illustrated in Fig. 3.1 in order to investigate the oscillations of a metal cube suspended on a spring.

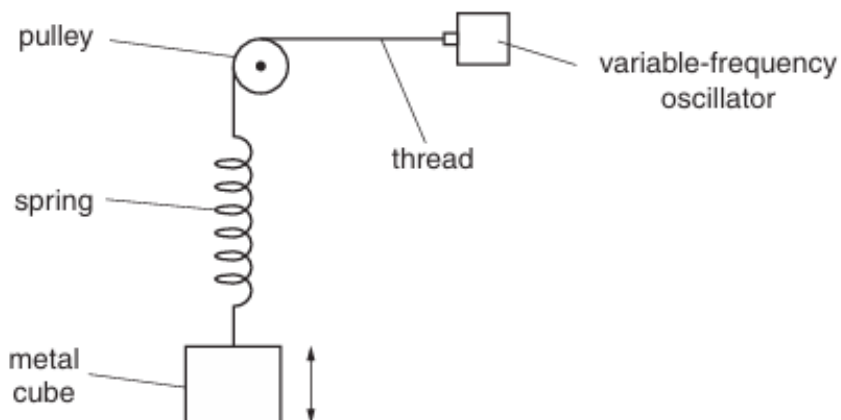


Fig. 3.1

The amplitude of the vibrations produced by the oscillator is constant. The variation with frequency of the amplitude of the oscillations of the metal cube is shown in Fig. 3.2.

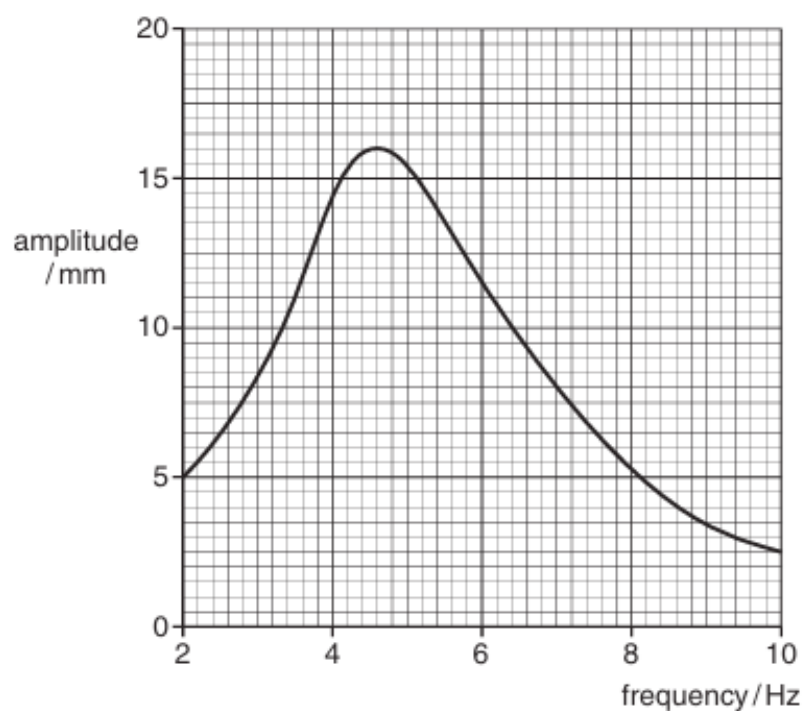


Fig. 3.2

- (a) (i)** State the phenomenon illustrated in Fig. 3.2.

..... [1]

- (ii)** For the maximum amplitude of vibration, state the magnitudes of the amplitude and the frequency.

amplitude = mm

frequency = Hz
[1]

- (b)** The oscillations of the metal cube of mass 150g may be assumed to be simple harmonic.

Use your answers in **(a)(ii)** to determine, for the metal cube,

- (i)** its maximum acceleration,

acceleration = m s^{-2} [3]

- (ii)** the maximum resultant force on the cube.

force = N [2]

- (c)** Some very light feathers are attached to the top surface of the cube so that the feathers extend outwards, beyond the vertical sides of the cube.

The investigation is now repeated.

On Fig. 3.2, draw a line to show the new variation with frequency of the amplitude of vibration for frequencies between 2 Hz and 10 Hz. [2]

10. O/N/10/43/Q3

A cylinder and piston, used in a car engine, are illustrated in Fig. 3.1.

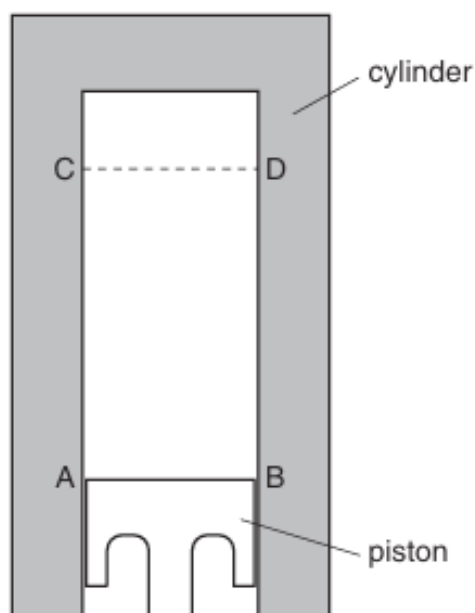


Fig. 3.1

The vertical motion of the piston in the cylinder is assumed to be simple harmonic. The top surface of the piston is at AB when it is at its lowest position; it is at CD when at its highest position, as marked in Fig. 3.1.

- (a) The displacement d of the piston may be represented by the equation

$$d = -4.0 \cos(220t)$$

where d is measured in centimetres.

- (i) State the distance between the lowest position AB and the highest position CD of the top surface of the piston.

distance = cm [1]

- (ii) Determine the number of oscillations made per second by the piston.

number = [2]

- (iii) On Fig. 3.1, draw a line to represent the top surface of the piston in the position where the speed of the piston is maximum. [1]

- (iv) Calculate the maximum speed of the piston.

speed = cm s^{-1} [2]

- (b) The engine of a car has several cylinders. Three of these cylinders are shown in Fig. 3.2.

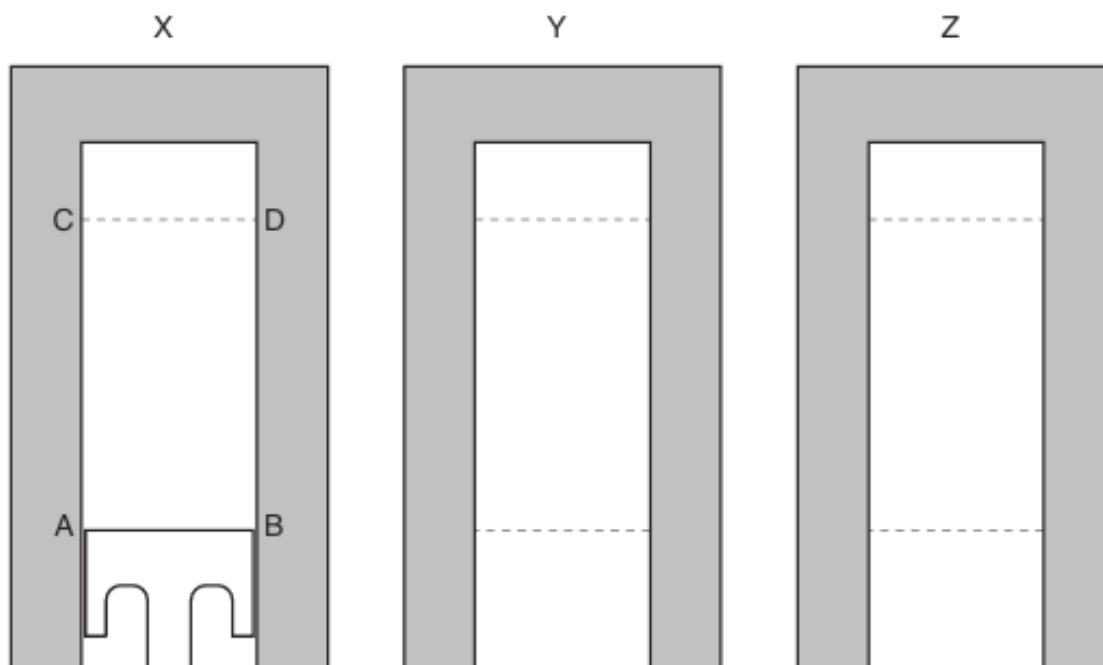


Fig. 3.2

X is the same cylinder and piston as in Fig. 3.1.

Y and Z are two further cylinders, with the lowest and the highest positions of the top surface of each piston indicated.

The pistons in the cylinders each have the same frequency of oscillation, but they are not in phase.

At a particular instant in time, the position of the top of the piston in cylinder X is as shown.

- (i) In cylinder Y, the oscillations of the piston lead those of the piston in cylinder X by a phase angle of 120° ($\frac{2}{3}\pi$ rad).

Complete the diagram of cylinder Y, for this instant, by drawing

1. a line to show the top surface of the piston, [1]
2. an arrow to show the direction of movement of the piston. [1]

- (ii) In cylinder Z, the oscillations of the piston lead those of the piston in cylinder X by a phase angle of 240° ($\frac{4}{3}\pi$ rad).

Complete the diagram of cylinder Z, for this instant, by drawing

1. a line to show the top surface of the piston, [1]
2. an arrow to show the direction of movement of the piston. [1]

- (iii) For the piston in cylinder Y, calculate its speed for this instant.

speed = cm s^{-1} [2]

11. M/J/11/41/Q3

(a) Define *simple harmonic motion*.

.....

 [2]

(b) A tube, sealed at one end, has a total mass m and a uniform area of cross-section A . The tube floats upright in a liquid of density ρ with length L submerged, as shown in Fig. 3.1a.

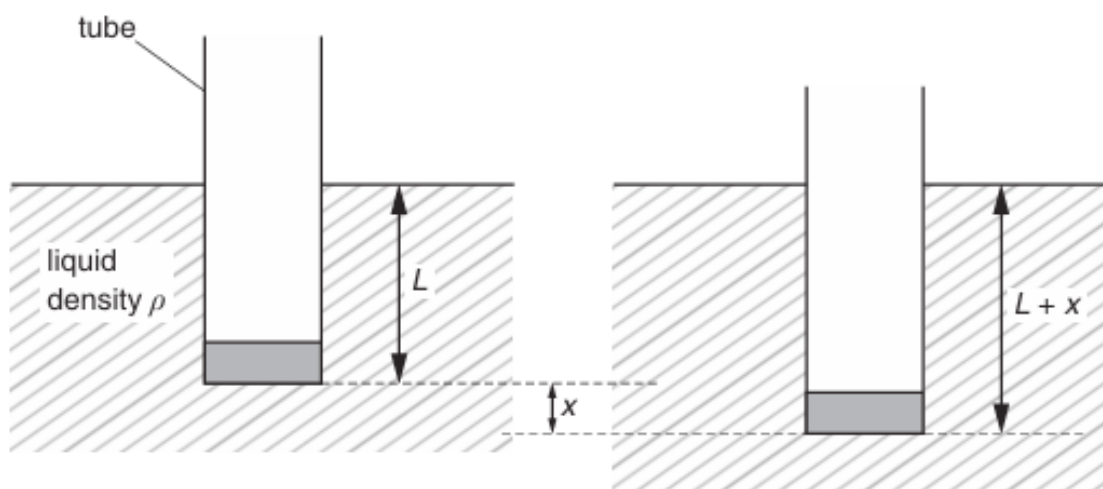


Fig. 3.1a

Fig. 3.1b

The tube is displaced vertically and then released. The tube oscillates vertically in the liquid.

At one time, the displacement is x , as shown in Fig. 3.1b.

Theory shows that the acceleration a of the tube is given by the expression

$$a = -\frac{A\rho g}{m}x.$$

- (i) Explain how it can be deduced from the expression that the tube is moving with simple harmonic motion.

.....
.....
.....[2]

- (ii) The tube, of area of cross-section 4.5cm^2 , is floating in water of density $1.0 \times 10^3\text{kgm}^{-3}$.

Calculate the mass of the tube that would give rise to oscillations of frequency 1.5 Hz.

mass = g [4]

12. O/N/11/41/Q3

A bar magnet is suspended from the free end of a helical spring, as illustrated in Fig. 3.1.

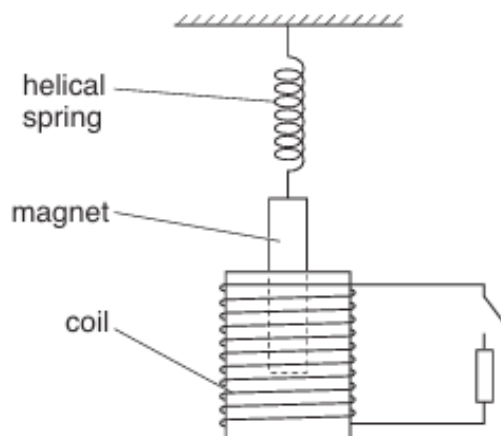


Fig. 3.1

One pole of the magnet is situated in a coil of wire. The coil is connected in series with a switch and a resistor. The switch is open.

The magnet is displaced vertically and then released. As the magnet passes through its rest position, a timer is started. The variation with time t of the vertical displacement y of the magnet from its rest position is shown in Fig. 3.2.

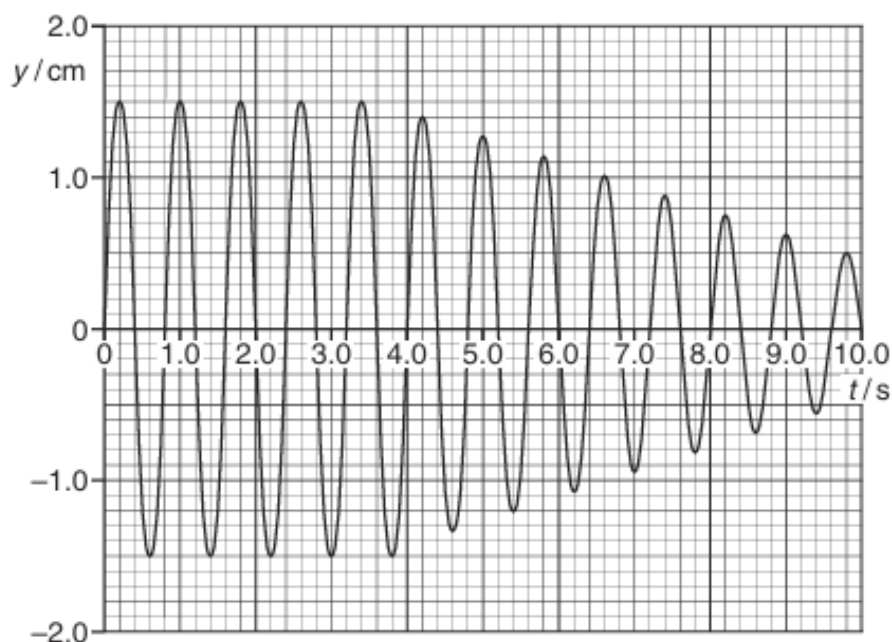


Fig. 3.2

At time $t = 4.0$ s, the switch is closed.

(a) Use Fig. 3.2 to

- (i)** state the evidence for the magnet to be undergoing free oscillations during the period $t = 0$ to $t = 4.0$ s,

.....
.....[1]

- (ii)** state, with a reason, whether the damping after time $t = 4.0$ s is light, critical or heavy,

.....
.....
.....[2]

- (iii)** determine the natural frequency of vibration of the magnet on the spring.

frequency = Hz [2]

(b) (i) State Faraday's law of electromagnetic induction.

.....
.....
.....[2]

- (ii)** Explain why, after time $t = 4.0$ s, the amplitude of vibration of the magnet is seen to decrease.

.....
.....
.....
.....
.....
.....[4]

13. O/N/11/43/Q3

- (a) Define *simple harmonic motion*.

.....

 [2]

- (b) A horizontal plate is vibrating vertically, as shown in Fig. 3.1.

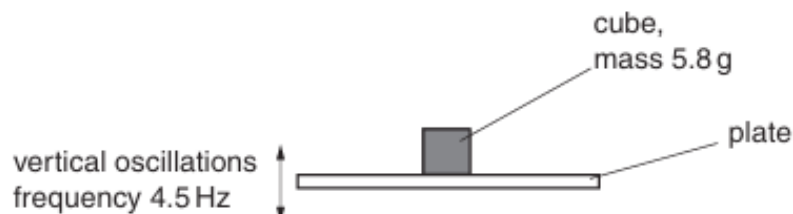


Fig. 3.1

The plate undergoes simple harmonic motion with a frequency of 4.5 Hz and amplitude 3.0 mm.

A metal cube of mass 5.8 g rests on the plate.

Calculate, for the cube, the energy of oscillation.

energy = J [3]

- (c) The amplitude of oscillation of the plate in (b) is gradually increased. The frequency remains constant.

At one particular amplitude, the cube just loses contact momentarily with the plate.

- (i) State the position of the plate in its oscillation at the point when the cube loses contact.

.....

 [2]

(ii) Calculate this amplitude of oscillation.

amplitude = m [2]

14. M/J/12/41/Q4

A small metal ball is suspended from a fixed point by means of a string, as shown in Fig. 4.1.

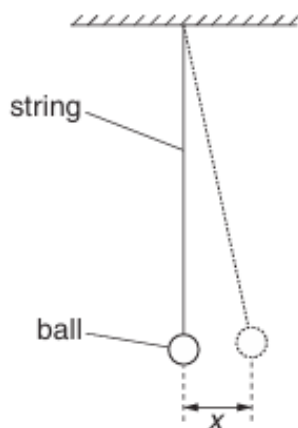


Fig. 4.1

The ball is pulled a small distance to one side and then released. The variation with time t of the horizontal displacement x of the ball is shown in Fig. 4.2.

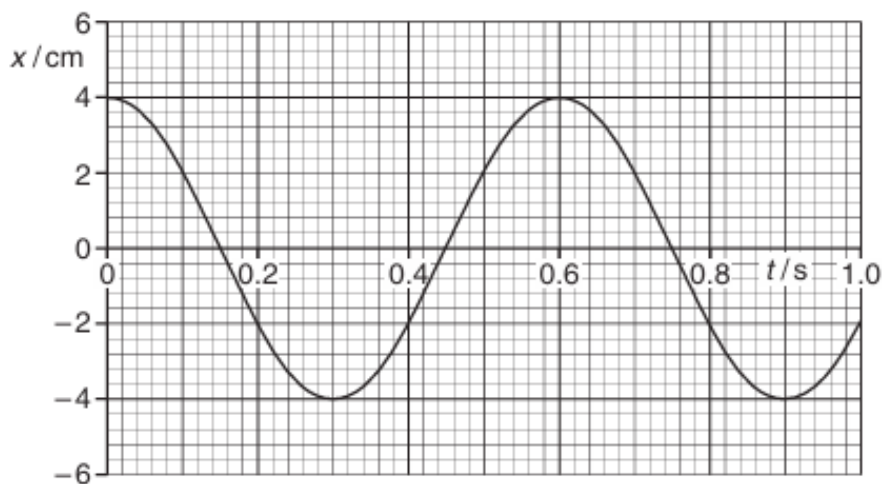


Fig. 4.2

The motion of the ball is simple harmonic.

- (a) Use data from Fig. 4.2 to determine the horizontal acceleration of the ball for a displacement x of 2.0 cm.

acceleration = ms^{-2} [3]

- (b) The maximum kinetic energy of the ball is E_K .
On the axes of Fig. 4.3, sketch a graph to show the variation with time t of the kinetic energy of the ball for the first 1.0 s of its motion.

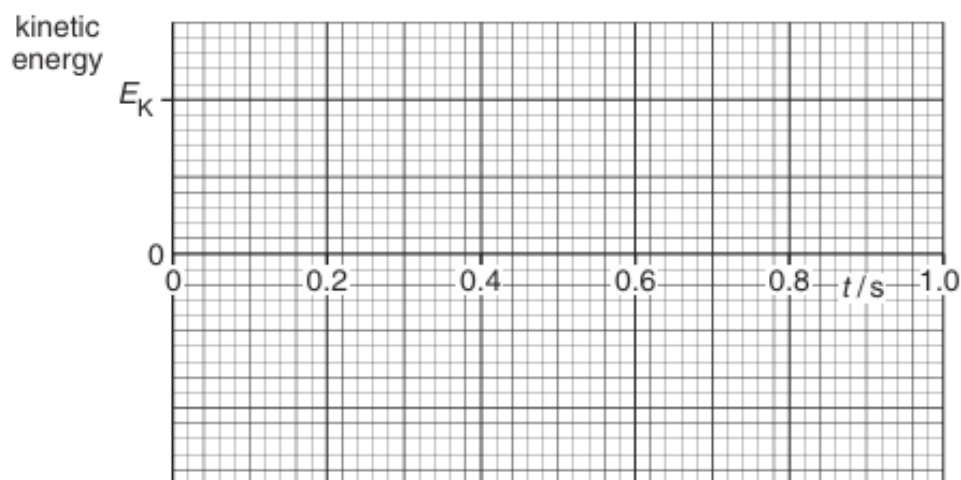


Fig. 4.3

[3]

15. M/J/12/42/Q2

A ball of mass 37 g is held between two fixed points A and B by two stretched helical springs, as shown in Fig. 2.1.

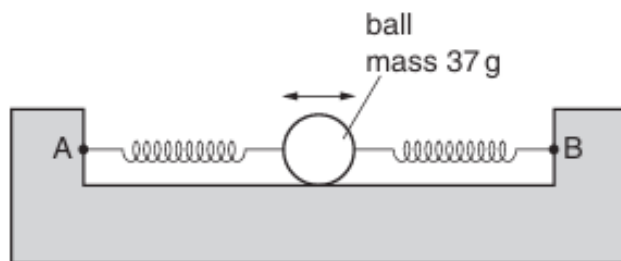


Fig. 2.1

The ball oscillates along the line AB with simple harmonic motion of frequency 3.5 Hz and amplitude 2.8 cm.

- (a)** Show that the total energy of the oscillations is 7.0 mJ.

[2]

- (b)** At two points in the oscillation of the ball, its kinetic energy is equal to the potential energy stored in the springs.
Calculate the magnitude of the displacement at which this occurs.

displacement = cm [3]

- (c) On the axes of Fig. 2.2 and using your answers in (a) and (b), sketch a graph to show the variation with displacement x of
- (i) the total energy of the system (label this line T), [1]
 - (ii) the kinetic energy of the ball (label this line K), [2]
 - (iii) the potential energy stored in the springs (label this line P). [2]

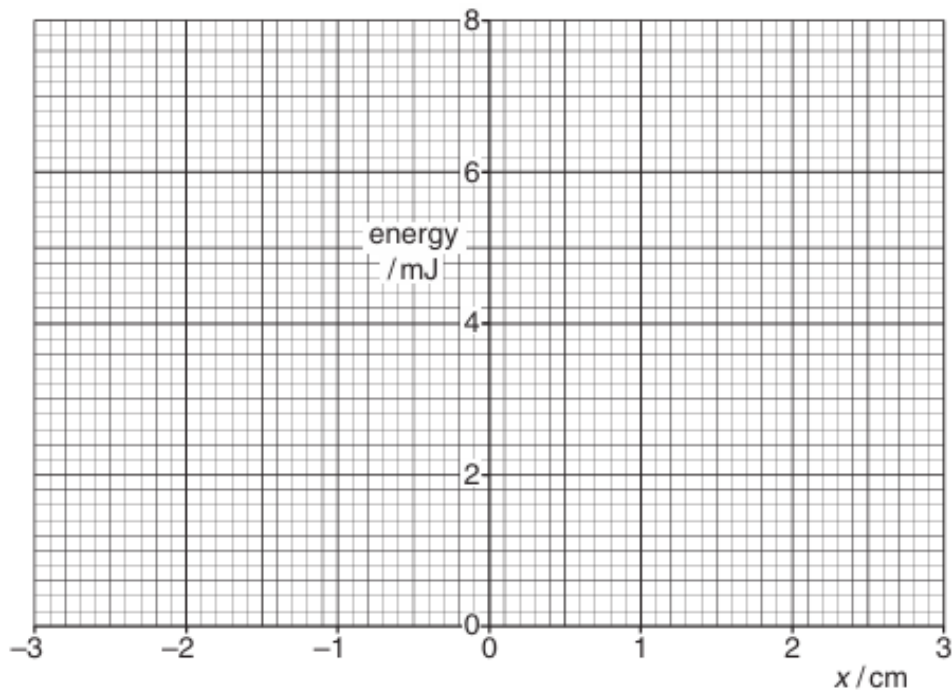


Fig. 2.2

- (d) The arrangement in Fig. 2.1 is now rotated through 90° so that the line AB is vertical and the ball oscillates in a vertical plane.

Suggest one form of energy, other than those in (c), that must be taken into consideration when plotting new graphs to show energy changes with displacement.

..... [1]

16. O/N/12/41/Q4

A ball is held between two fixed points A and B by means of two stretched springs, as shown in Fig. 4.1.

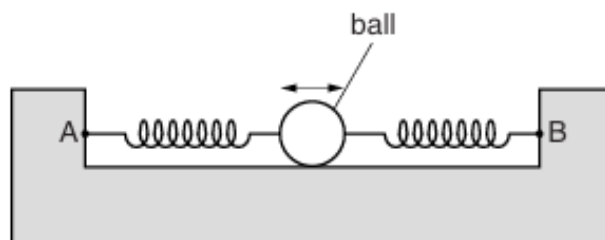


Fig. 4.1

The ball is free to oscillate horizontally along the line AB. During the oscillations, the springs remain stretched and do not exceed their limits of proportionality.

The variation of the acceleration a of the ball with its displacement x from its equilibrium position is shown in Fig. 4.2.

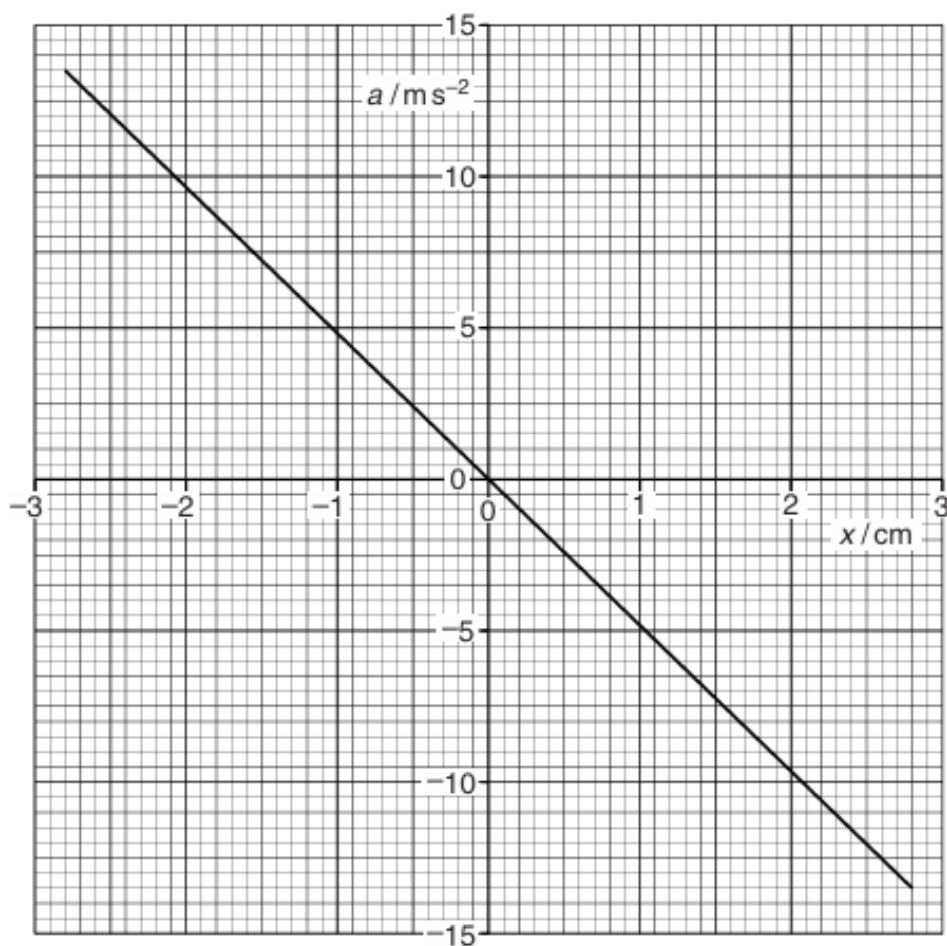


Fig. 4.2

- (a) State and explain the features of Fig. 4.2 that indicate that the motion of the ball is simple harmonic.

.....
.....
.....
.....
..... [4]

- (b) Use Fig. 4.2 to determine, for the oscillations of the ball,

- (i) the amplitude,

amplitude = cm [1]

- (ii) the frequency.

frequency = Hz [3]

- (c) The arrangement in Fig. 4.1 is now rotated through 90° so that the line AB is vertical. The ball now oscillates in a vertical plane.

Suggest one reason why the oscillations may no longer be simple harmonic.

.....
..... [1]

17. O/N/12/43/Q2

A small frictionless trolley is attached to a fixed point A by means of a spring. A second spring is used to attach the trolley to a variable frequency oscillator, as shown in Fig. 2.1.

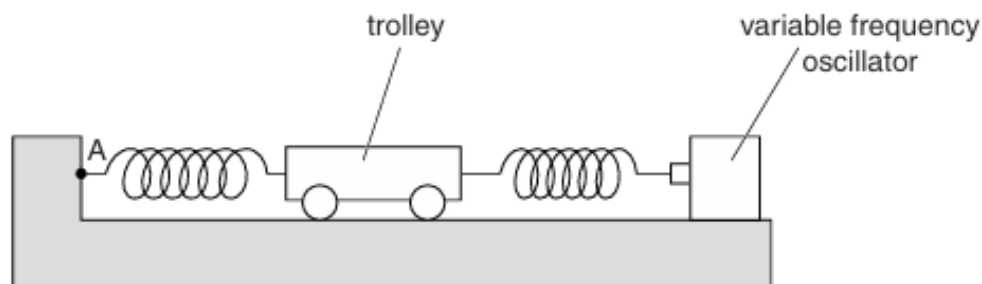


Fig. 2.1

Both springs remain extended within the limit of proportionality.

Initially, the oscillator is switched off. The trolley is displaced horizontally along the line joining the two springs and is then released.

The variation with time t of the velocity v of the trolley is shown in Fig. 2.2.

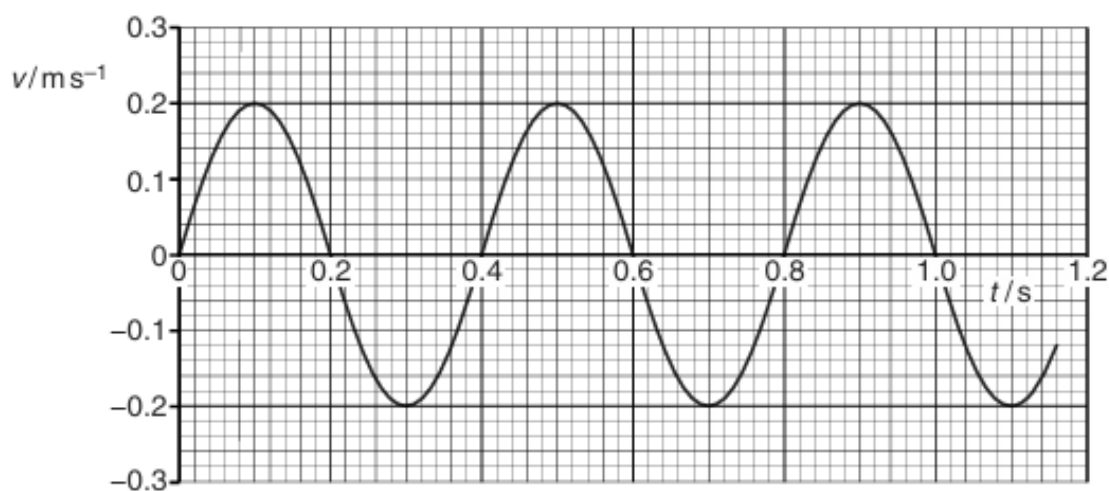


Fig. 2.2

(a) (i) Using Fig. 2.2, state two different times at which

1. the displacement of the trolley is zero,

time = s and time = s [1]

2. the acceleration in one direction is maximum.

time = s and time = s [1]

- (ii) Determine the frequency of oscillation of the trolley.

frequency = Hz [2]

- (iii) The variation with time of the displacement of the trolley is sinusoidal. The variation with time of the velocity of the trolley is also sinusoidal.

State the phase difference between the displacement and the velocity.

phase difference = [1]

- (b) The oscillator is now switched on. The amplitude of vibration of the oscillator is constant. The frequency f of vibration of the oscillator is varied. The trolley is forced to oscillate by means of vibrations of the oscillator. The variation with f of the amplitude a_0 of the oscillations of the trolley is shown in Fig. 2.3.

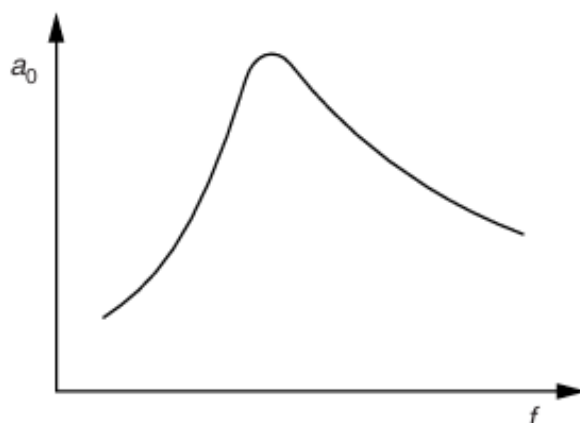


Fig. 2.3

By reference to your answer in (a), state the approximate frequency at which the amplitude is maximum.

frequency = Hz [1]

- (c) The amplitude of the oscillations in (b) may be reduced without changing significantly the frequency at which the amplitude is a maximum. State how this may be done and give a reason for your answer.
You may draw on Fig. 2.1 if you wish.

.....
.....

18. M/J/13/41/Q3

A ball is held between two fixed points A and B by means of two stretched springs, as shown in Fig. 3.1.

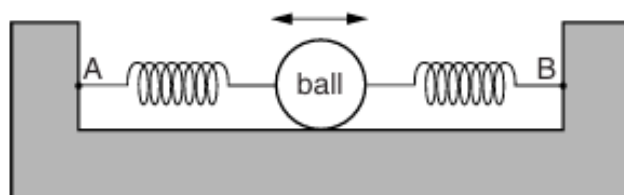


Fig. 3.1

The ball is free to oscillate along the straight line AB. The springs remain stretched and the motion of the ball is simple harmonic.

The variation with time t of the displacement x of the ball from its equilibrium position is shown in Fig. 3.2.

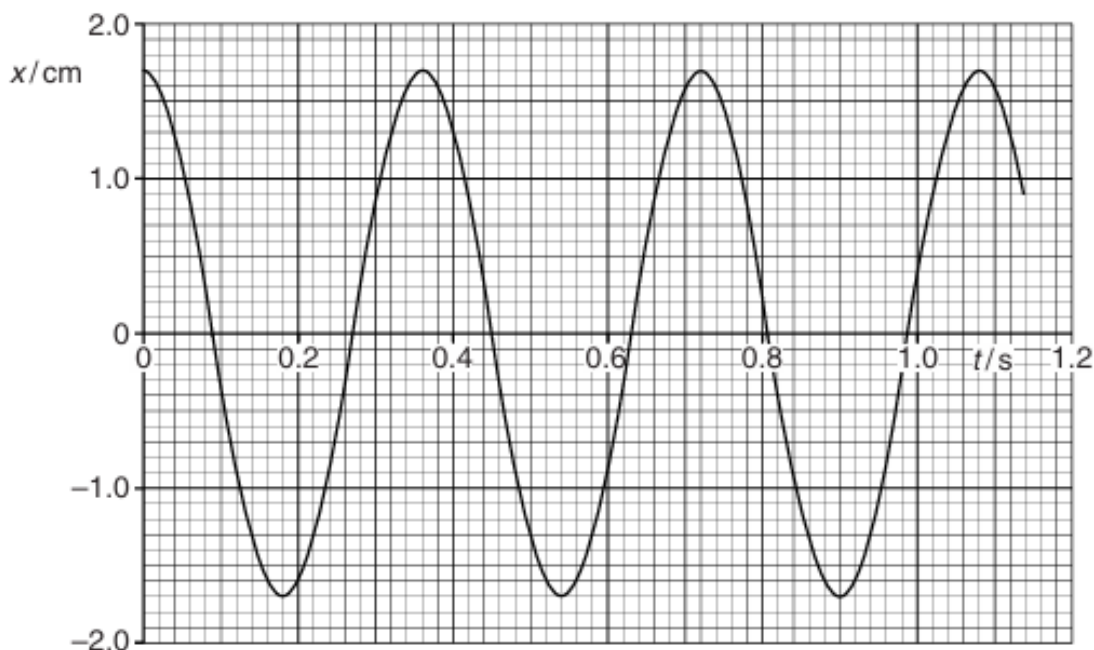


Fig. 3.2

(a) (i) Use Fig. 3.2 to determine, for the oscillations of the ball,

1. the amplitude,

amplitude = cm [1]

2. the frequency.

frequency = Hz [2]

- (ii) Show that the maximum acceleration of the ball is 5.2 ms^{-2} .

[2]

- (b) Use your answers in (a) to plot, on the axes of Fig. 3.3, the variation with displacement x of the acceleration a of the ball.

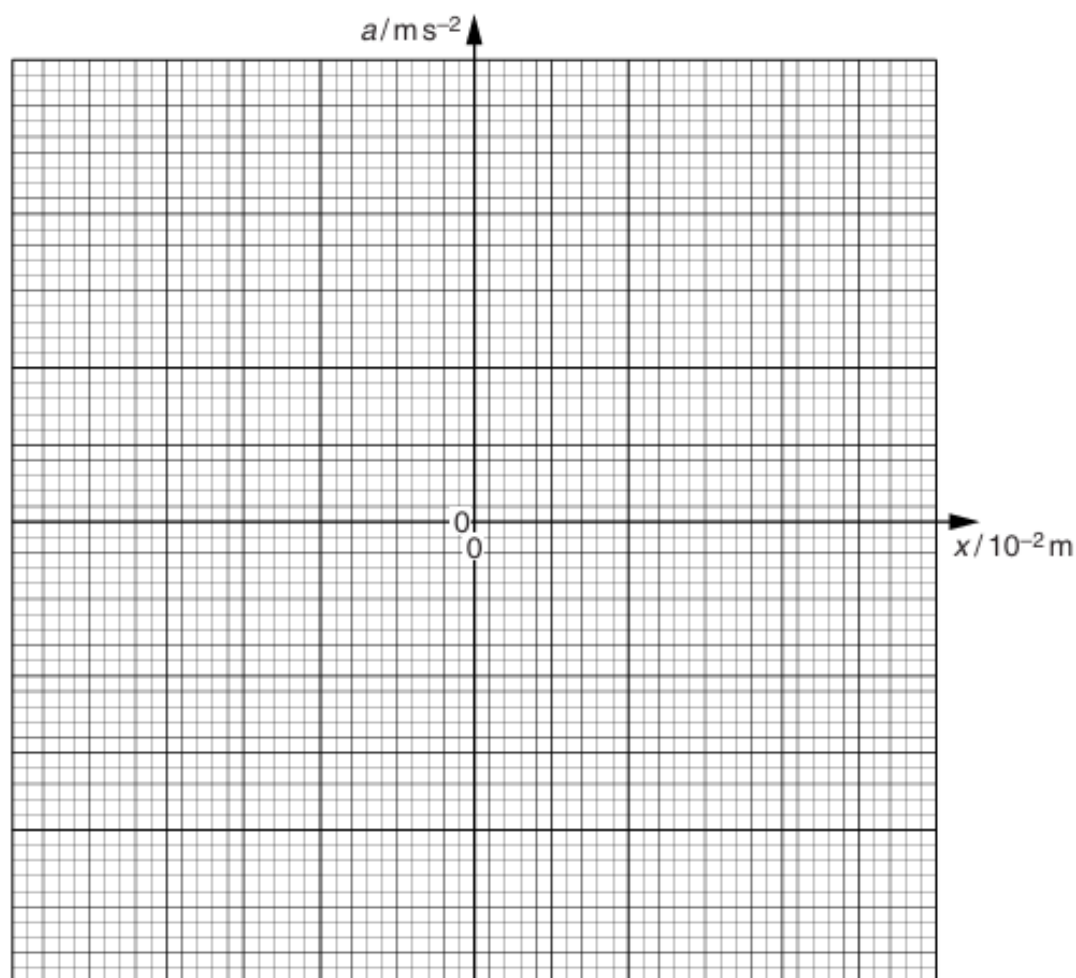


Fig. 3.3

[2]

19. M/J/13/42/Q3

A mass of 78 g is suspended from a fixed point by means of a spring, as illustrated in Fig. 3.1.

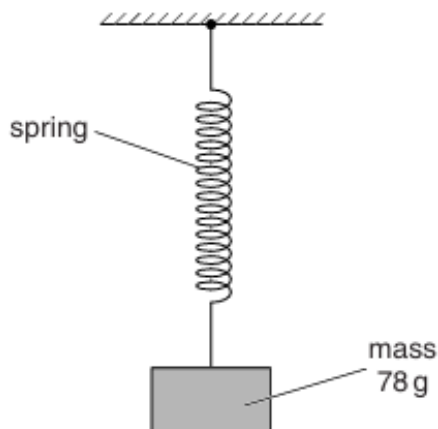


Fig. 3.1

The stationary mass is pulled vertically downwards through a distance of 2.1 cm and then released.

The mass is observed to perform simple harmonic motion with a period of 0.69 s.

- (a)** The mass is released at time $t = 0$.

For the oscillations of the mass,

- (i)** calculate the angular frequency ω ,

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii)** determine numerical equations for the variation with time t of

- 1.** the displacement x in cm,

.....
 [2]

- 2.** the speed v in ms^{-1} .

.....
 [2]

(b) Calculate the total energy of oscillation of the mass.

energy = J [2]

20. O/N/13/41/Q3

A metal ball is suspended from a fixed point by means of a string, as illustrated in Fig. 3.1.

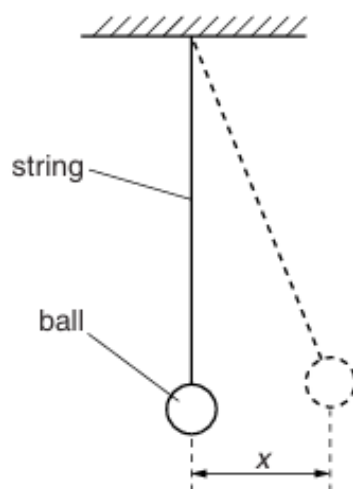


Fig. 3.1

The ball is given a small displacement and then released. The variation with time t of the displacement x of the ball is shown in Fig. 3.2.

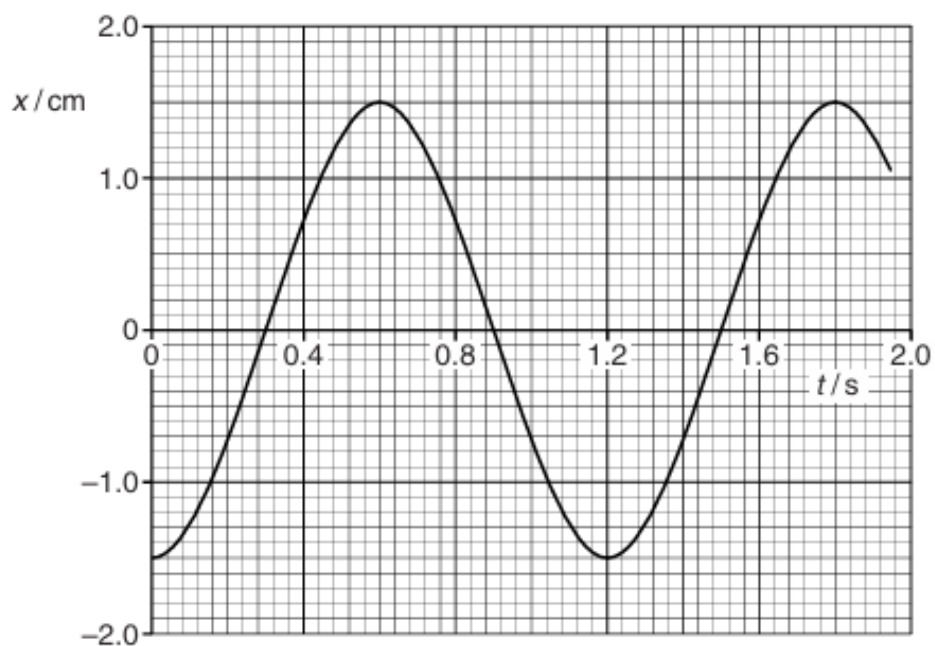


Fig. 3.2

- (a) (i) State two times at which the speed of the ball is a maximum.

time = s and time = s [1]

- (ii) Show that the maximum speed of the ball is approximately 0.08 ms^{-1} .

- (b) The variation with displacement x of the potential energy E_p of the oscillations of the ball is shown in Fig. 3.3.

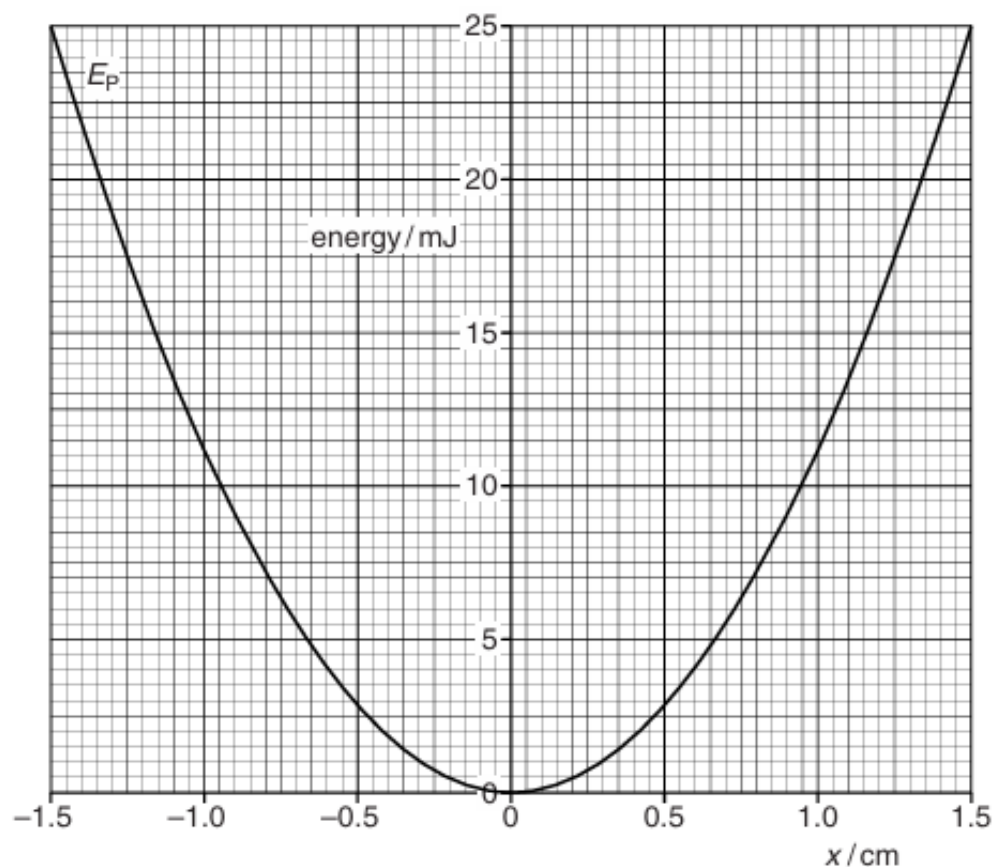


Fig. 3.3

- (i) On the axes of Fig. 3.3, sketch a graph to show the variation with displacement x of the kinetic energy of the ball. [2]
- (ii) The amplitude of the oscillations reduces over a long period of time. After many oscillations, the amplitude of the oscillations is 0.60 cm.

Use Fig. 3.3 to determine the total energy of the oscillations of the ball for oscillations of amplitude 0.60 cm. Explain your working.

energy = J [2]

21. M/J/14/41/Q4

A student investigates the energy changes of a mass oscillating on a vertical spring, as shown in Fig. 4.1.

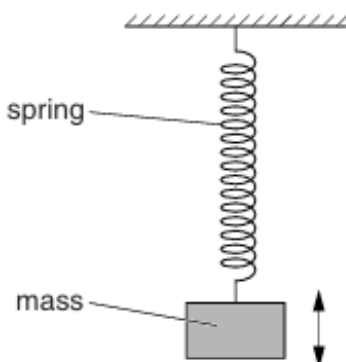


Fig. 4.1

The student draws a graph of the variation with displacement x of energy E of the oscillation, as shown in Fig. 4.2.

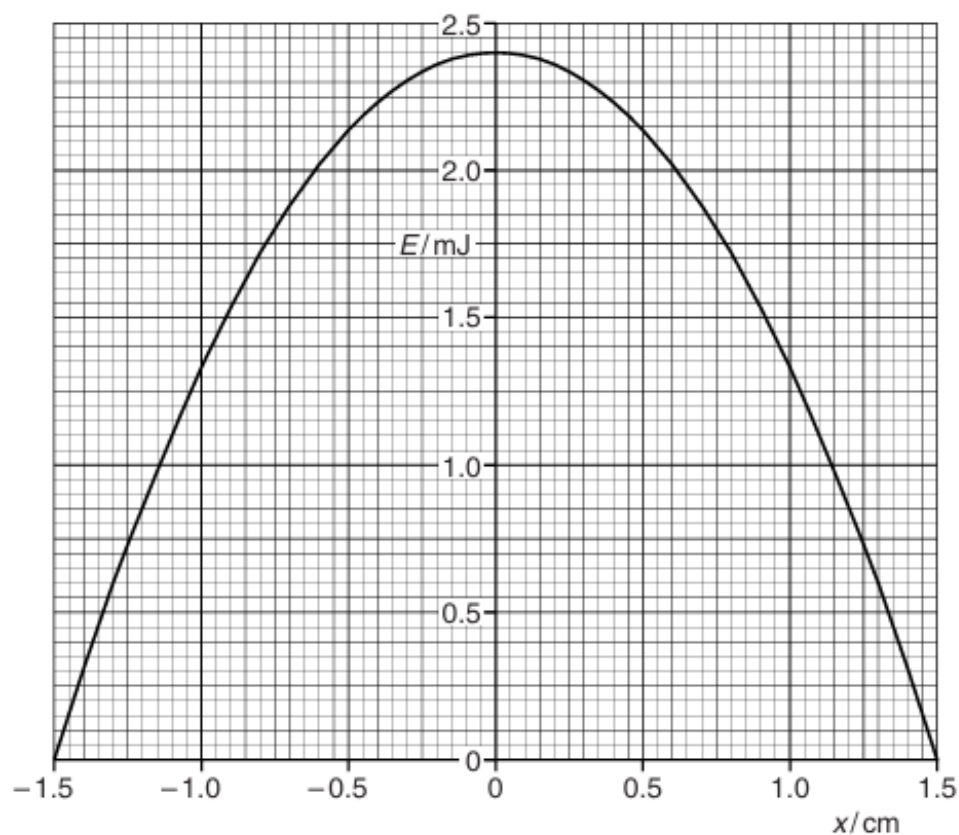


Fig. 4.2

- (a) State whether the energy E represents the total energy, the potential energy or the kinetic energy of the oscillations.

..... [1]

- (b) The student repeats the investigation but with a smaller amplitude. The maximum value of E is now found to be 1.8 mJ.

Use Fig. 4.2 to determine the change in the amplitude. Explain your working.

change in amplitude = cm [3]

22. O/N/14/41/Q4

- (a) State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

- (b) A small ball rests at point P on a curved track of radius r , as shown in Fig. 4.1.

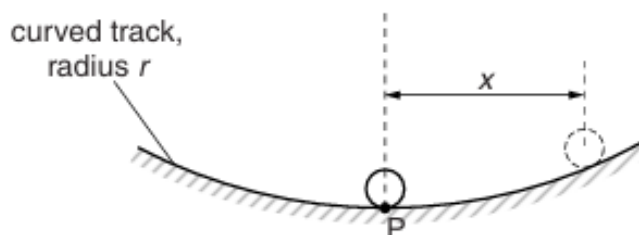


Fig. 4.1

The ball is moved a small distance to one side and is then released. The horizontal displacement x of the ball is related to its acceleration a towards P by the expression

$$a = -\frac{gx}{r}$$

where g is the acceleration of free fall.

- (i) Show that the ball undergoes simple harmonic motion.

.....

.....

.....

..... [2]

- (ii) The radius r of curvature of the track is 28 cm.

Determine the time interval τ between the ball passing point P and then returning to point P.

$\tau = \dots\dots\dots$ s [3]

(c) The variation with time t of the displacement x of the ball in (b) is shown in Fig. 4.2.

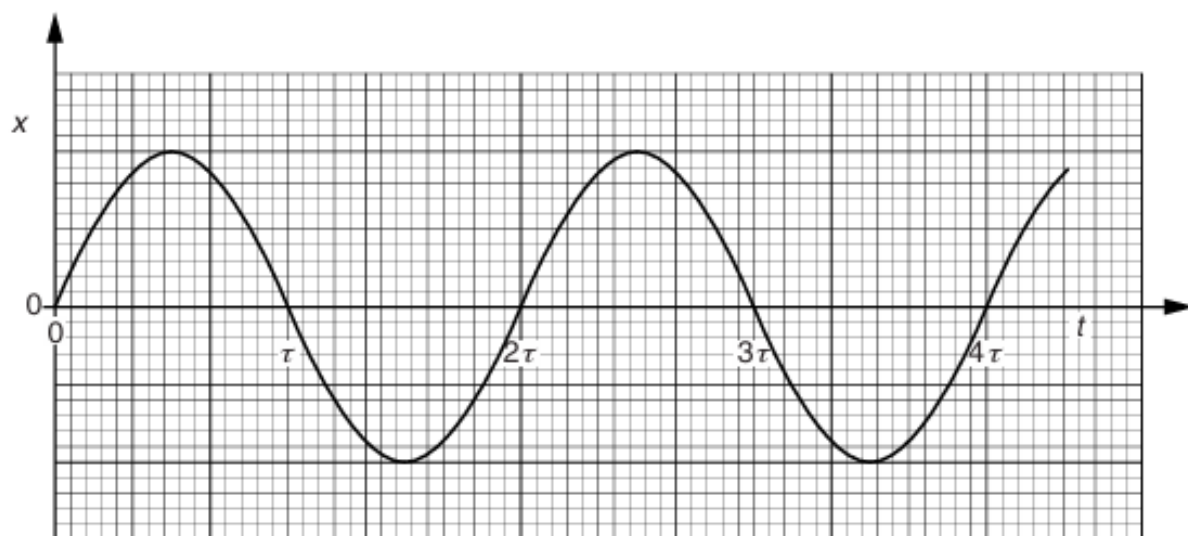


Fig. 4.2

Some moisture now forms on the track, causing the ball to come to rest after approximately 15 oscillations.

On the axes of Fig. 4.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. Assume the moisture forms at time $t = 0$.
[3]

23. O/N/14/43/Q4

- (a) State what is meant by *simple harmonic motion*.

.....

 [2]

- (b) A trolley is attached to two extended springs, as shown in Fig. 4.1.

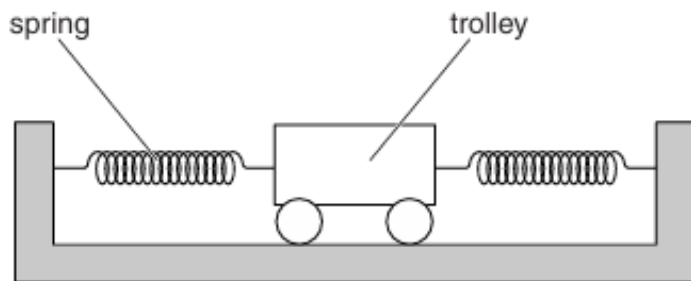


Fig. 4.1

The trolley is displaced along the line joining the two springs and is then released. At one point in the motion, a stopwatch is started. The variation with time t of the velocity v of the trolley is shown in Fig. 4.2.

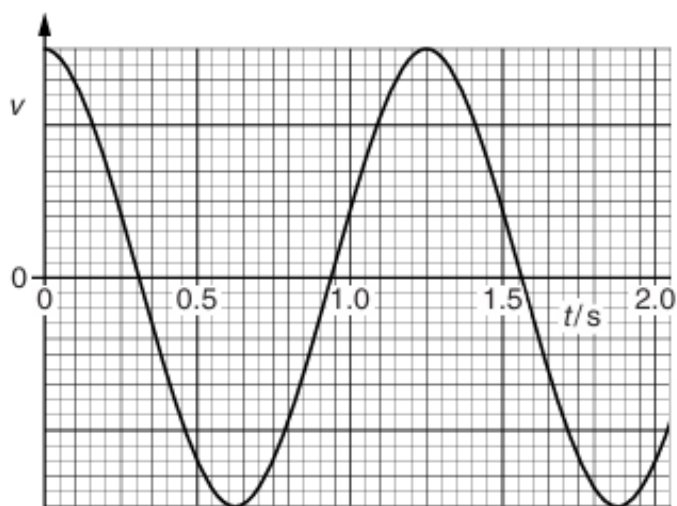


Fig. 4.2

The motion of the trolley is simple harmonic.

- (i) State one time at which the trolley is moving through the equilibrium position and also state the next time that it moves through this position.

.....s ands [1]

(ii) The amplitude of vibration of the trolley is 3.2 cm.

Determine

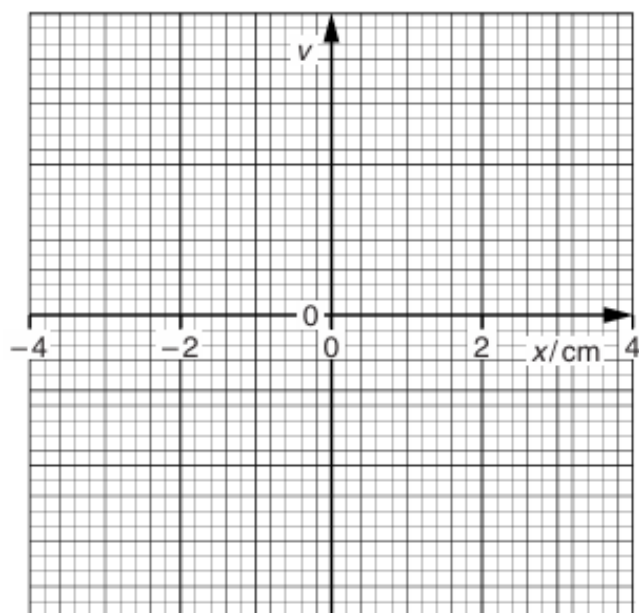
1. the maximum speed v_0 of the trolley,

$$v_0 = \dots\dots\dots \text{ cm s}^{-1} \quad [3]$$

2. the displacement of the trolley for a speed of $\frac{1}{2}v_0$.

$$\text{displacement} = \dots\dots\dots \text{ cm} \quad [2]$$

- (c) Use your answers in (b) to sketch, on the axes of Fig. 4.3, a graph to show the variation with displacement x of the velocity v of the trolley.



24. M/J/15/41/Q4

(a) For an oscillating body, state what is meant by

(i) forced frequency,

.....
..... [1]

(ii) natural frequency of vibration,

.....
..... [1]

(iii) resonance.

.....
.....
..... [2]

(b) State and explain one situation where resonance is useful.

.....
.....
.....
..... [2]

(c) In some situations, resonance should be avoided.

State one such situation and suggest how the effects of resonance are reduced.

.....
.....
.....
..... [2]

25. M/J/15/42/Q4

(a) State what is meant by *simple harmonic motion*.

.....

[2]

(b) The variation with time t of the displacement x of two oscillators P and Q is shown in Fig. 4.1.

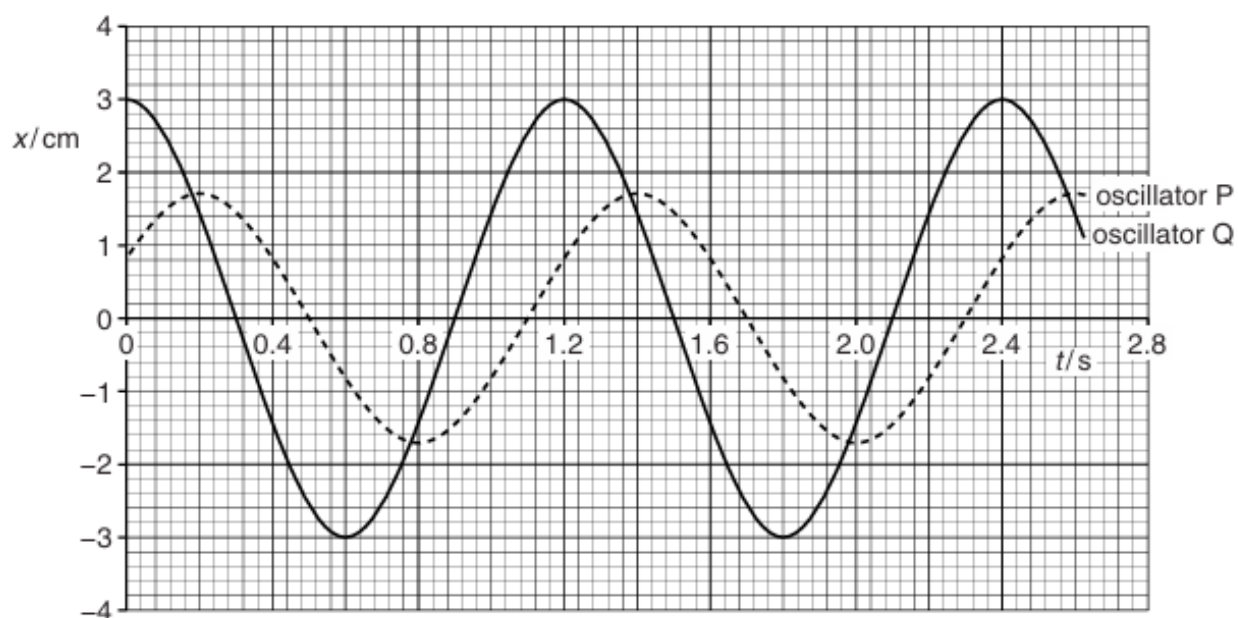


Fig. 4.1

The two oscillators each have the same mass.

Use Fig. 4.1 to determine

(i) the phase difference between the two oscillators,

phase difference = rad [1]

(ii) the maximum acceleration of oscillator Q,

maximum acceleration = m s^{-2} [2]

(iii) the ratio

$$\frac{\text{maximum kinetic energy of oscillations of Q}}{\text{maximum kinetic energy of oscillations of P}}$$

ratio = [2]

- (c) Use data from (b) to sketch, on the axes of Fig. 4.2, the variation with displacement x of the acceleration a of oscillator Q.

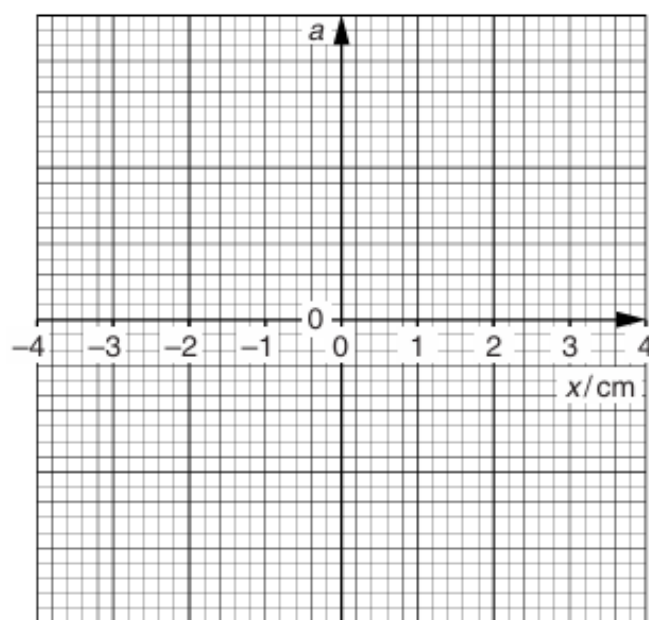


Fig. 4.2

[2]

26. O/N/15/41/Q4

- (a) Distinguish between free oscillations and forced oscillations.

free oscillations:

.....

forced oscillations:

.....

[2]

- (b) A trolley is held on a horizontal surface by means of two stretched springs, as shown in Fig. 4.1.

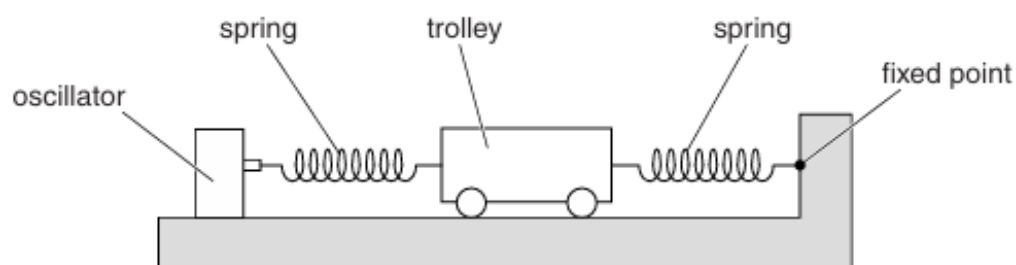


Fig. 4.1

One spring is attached to a fixed point. The other spring is attached to an oscillator that causes horizontal oscillations of the trolley.

The oscillator vibrates with a constant amplitude of vibration. The frequency of vibration of the oscillator is gradually increased from a very low value.

The variation with frequency f of the amplitude x_0 of vibration of the trolley is shown in Fig. 4.2.

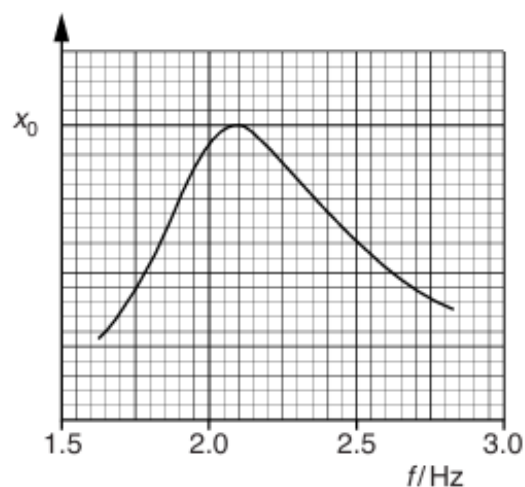


Fig. 4.2

Use Fig. 4.2 to state and explain

- (i) the value of the natural frequency of vibration of the trolley,

.....
.....
.....
.....[3]

- (ii) whether there are any frictional forces acting on the trolley.

.....
.....
.....[1]

- (c) The oscillator in (b) is now stopped.

The trolley is given a horizontal displacement of 4.7 cm along the line of the springs.
The trolley is then released.

Use information from Fig. 4.2 to estimate the maximum speed of the trolley.

speed = ms^{-1} [2]

27. O/N/15/43/Q4

(a) Define *simple harmonic motion*.

.....

 [2]

(b) A tube, sealed at one end, has a circular cross-sectional area A of $4.9 \times 10^{-4} \text{ m}^2$. Some sand is put in the tube so that the total mass M of the tube and its contents is 70 g . The tube floats upright in a liquid, as shown in Fig. 4.1.

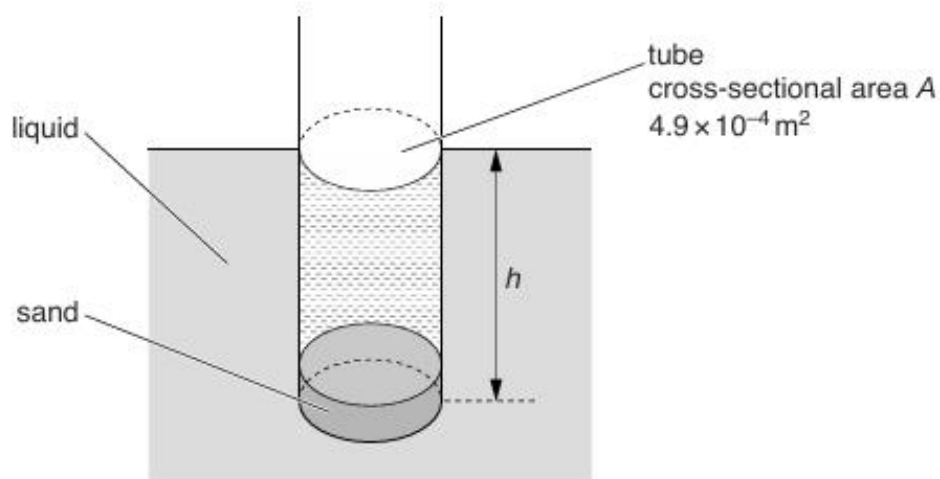


Fig. 4.1

The liquid has a density ρ of 0.79 g cm^{-3} .

By reference to the liquid pressure exerted on the base of the tube, show that the distance h of the base of the tube below the liquid surface is 18 cm . Explain your working.

[2]

- (c) The tube in (b) is displaced vertically and then released. The variation with time t of the distance h is shown in Fig. 4.2.

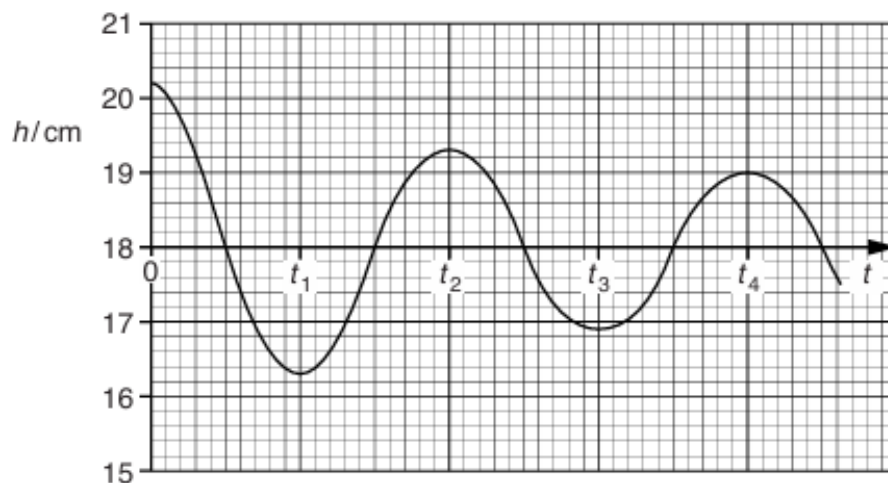


Fig. 4.2

The system oscillates with simple harmonic motion of angular frequency ω given by the expression

$$\omega^2 = \frac{\rho Ag}{M}$$

where g is the acceleration of free fall.

- (i) Use data from (b) to determine

1. the time t_1 ,

$$t_1 = \dots\dots\dots \text{s} \quad [3]$$

2. the time t_3 .

$$t_3 = \dots\dots\dots \text{s} \quad [1]$$

- (ii) Determine the loss in total energy of the oscillating system between time $t = 0$ and time $t = t_4$.

loss in energy =J [3]

28. F/M/16/42/Q4

An object of mass 80g oscillates with simple harmonic motion. The variation with time t of the displacement x of the object is shown in Fig. 4.1.

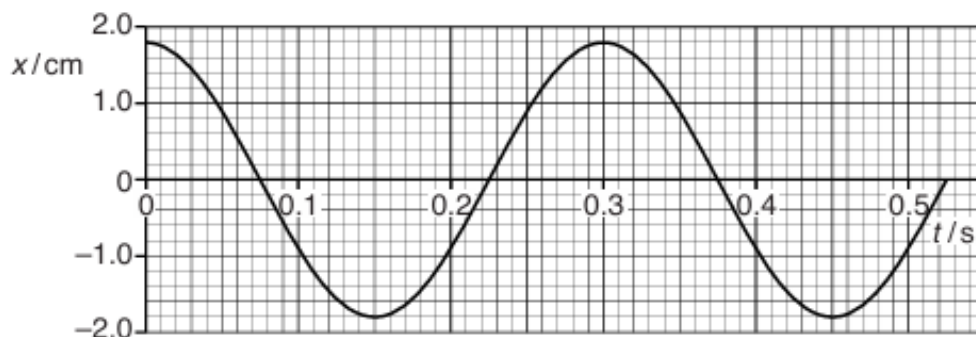


Fig. 4.1

- (a)** Use Fig. 4.1 to determine the amplitude and the period of the oscillations.

amplitude = cm

period = s

[1]

- (b)** Use Fig. 4.1 and your answers in **(a)** to calculate the kinetic energy of the object at time $t = 0.19$ s.

kinetic energy = J [3]

[Total: 4]

29. M/J/16/41/Q3

(a) State, by reference to displacement, what is meant by *simple harmonic motion*.

.....

.....

.....

..... [2]

(b) A mass is undergoing oscillations in a vertical plane.

The variation with displacement x of the acceleration a of the mass is shown in Fig. 3.1.

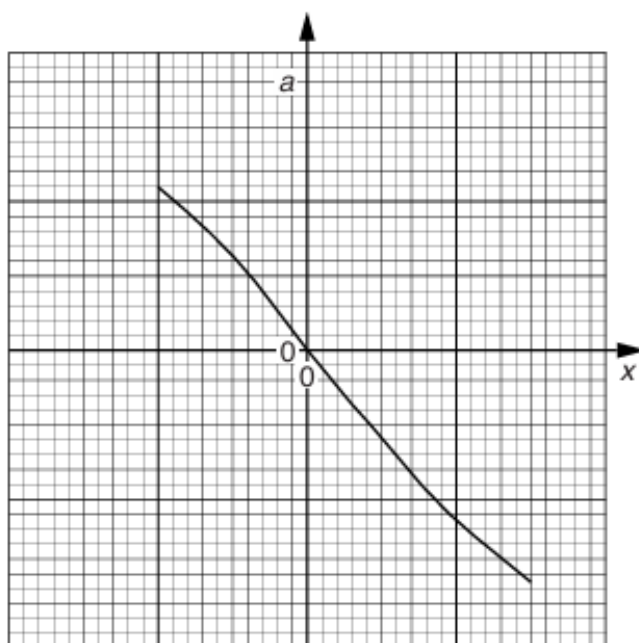


Fig. 3.1

State two reasons why the motion of the mass is not simple harmonic.

1.

.....

2.

.....

[2]

- (c) A block of wood is floating in a liquid, as shown in Fig. 3.2.

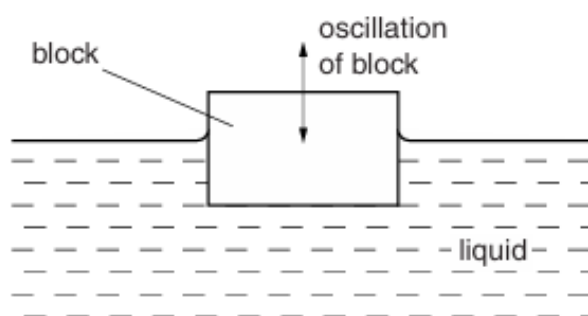


Fig. 3.2

The block is displaced vertically and then released.

The variation with time t of the displacement y of the block from its equilibrium position is shown in Fig. 3.3.

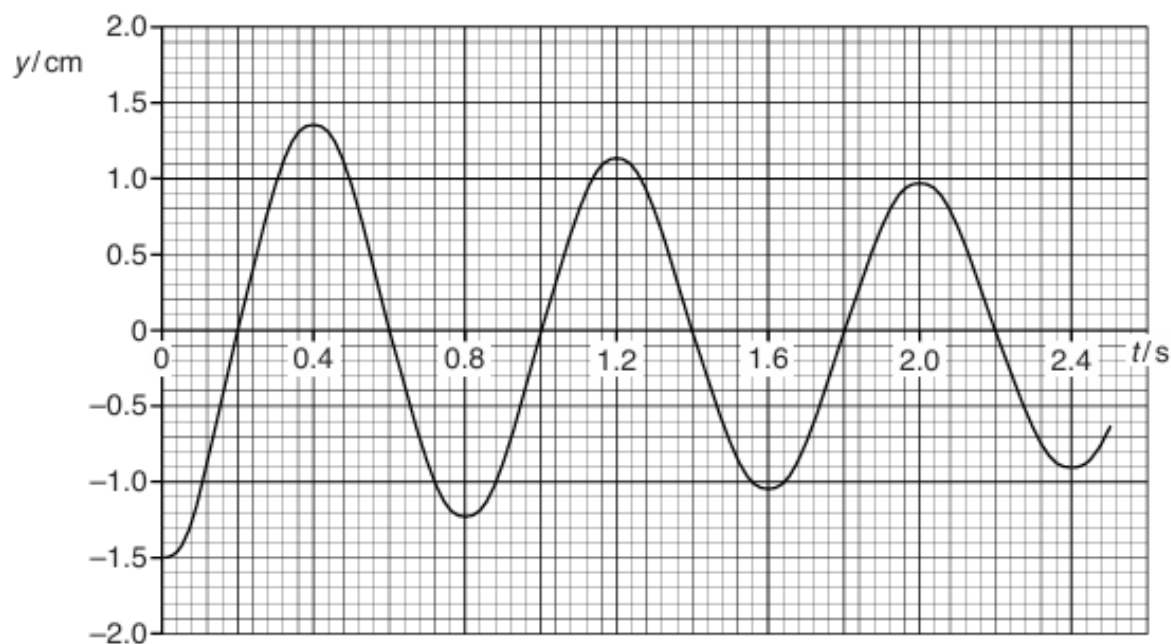


Fig. 3.3

Use data from Fig. 3.3 to determine

- (i) the angular frequency ω of the oscillations,

$$\omega = \dots\dots\dots \text{rads}^{-1} \quad [2]$$

- (ii) the maximum vertical acceleration of the block.

maximum acceleration = ms^{-2} [2]

- (iii) The block has mass 120 g.

The oscillations of the block are damped. Calculate the loss in energy of the oscillations of the block during the first three complete periods of its oscillations.

energy loss = J [3]

[Total: 11]

30. M/J/16/42/Q4

A metal block hangs vertically from one end of a spring. The other end of the spring is tied to a thread that passes over a pulley and is attached to a vibrator, as shown in Fig. 4.1.

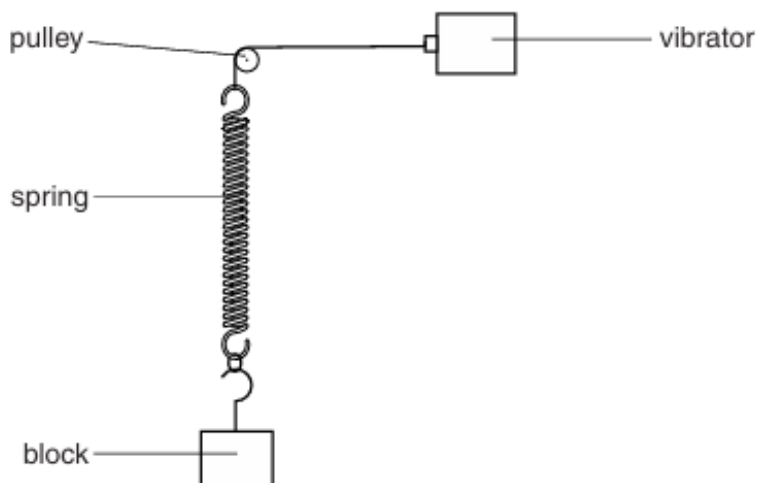


Fig. 4.1

- (a) The vibrator is switched off.
The metal block of mass 120 g is displaced vertically and then released. The variation with time t of the displacement y of the block from its equilibrium position is shown in Fig. 4.2.

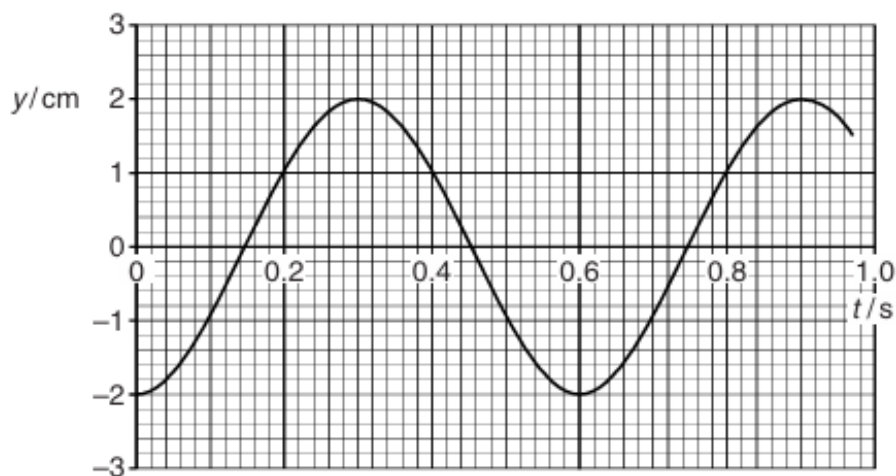


Fig. 4.2

For the vibrations of the block, calculate

- (i) the angular frequency ω ,

$\omega = \dots\dots\dots \text{rads}^{-1}$ [2]

(ii) the energy of the vibrations.

energy =J [2]

(b) The vibrator is now switched on.

The frequency of vibration is varied from $0.7f$ to $1.3f$ where f is the frequency of vibration of the block in (a).

For the block, complete Fig. 4.3 to show the variation with frequency of the amplitude of vibration. Label this line A. [3]

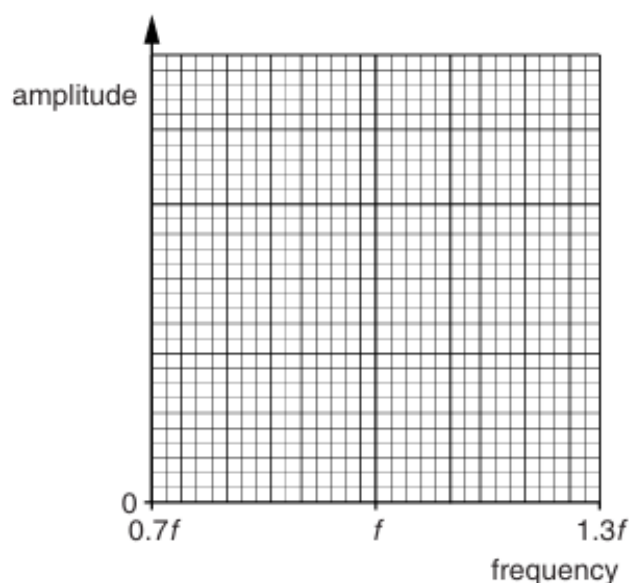


Fig. 4.3

(c) Some light feathers are now attached to the block in (b) to increase air resistance.

The frequency of vibration is once again varied from $0.7f$ to $1.3f$. The new amplitude of vibration is measured for each frequency.

On Fig. 4.3, draw a line to show the variation with frequency of the amplitude of vibration. Label this line B. [2]

[Total: 9]

31. O/N/16/41/Q3

To demonstrate simple harmonic motion, a student attaches a trolley to two similar stretched springs, as shown in Fig. 3.1.

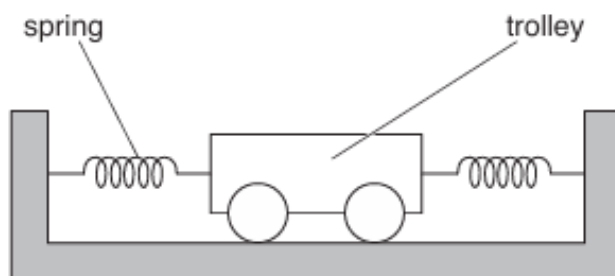


Fig. 3.1

The trolley has mass m of 810 g.

The trolley is displaced along the line of the two springs and then released. The subsequent acceleration a of the trolley is given by the expression

$$a = -\frac{2kx}{m}$$

where the spring constant k for each of the springs is 64 N m^{-1} and x is the displacement of the trolley.

(a) Show that the frequency of oscillation of the trolley is 2.0 Hz.

[3]

(b) The maximum displacement of the trolley is 1.6 cm.
Calculate the maximum speed of the trolley.

speed = ms^{-1} [2]

- (c) The mass of the trolley is increased. The initial displacement of the trolley remains unchanged.

Suggest the change, if any, that occurs in the frequency and in the maximum speed of the oscillations of the trolley.

frequency:

maximum speed:

[2]

[Total: 7]

32. O/N/16/42/Q4

A mass hangs vertically from a fixed point by means of a spring, as shown in Fig. 4.1.

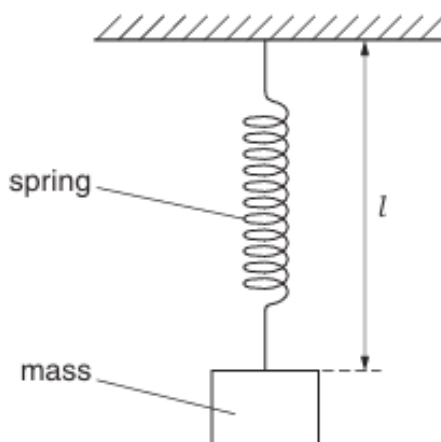


Fig. 4.1

The mass is displaced vertically and then released. The subsequent oscillations of the mass are simple harmonic.

The variation with time t of the length l of the spring is shown in Fig. 4.2.

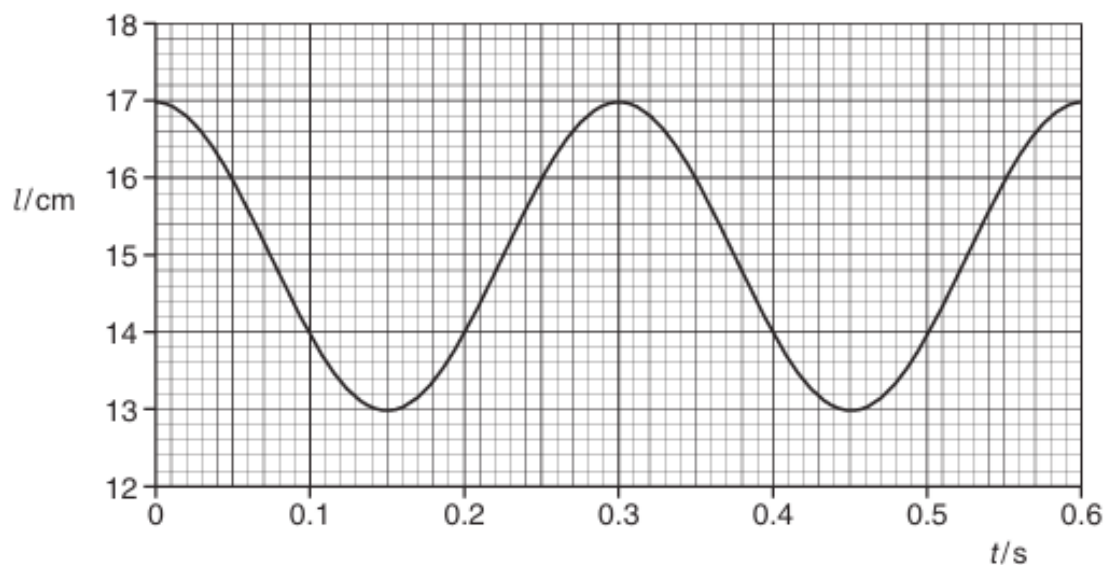


Fig. 4.2

(a) Use Fig. 4.2 to

(i) state two values of t at which the mass is moving downwards with maximum speed,

$t = \dots\dots\dots$ s and $t = \dots\dots\dots$ s [1]

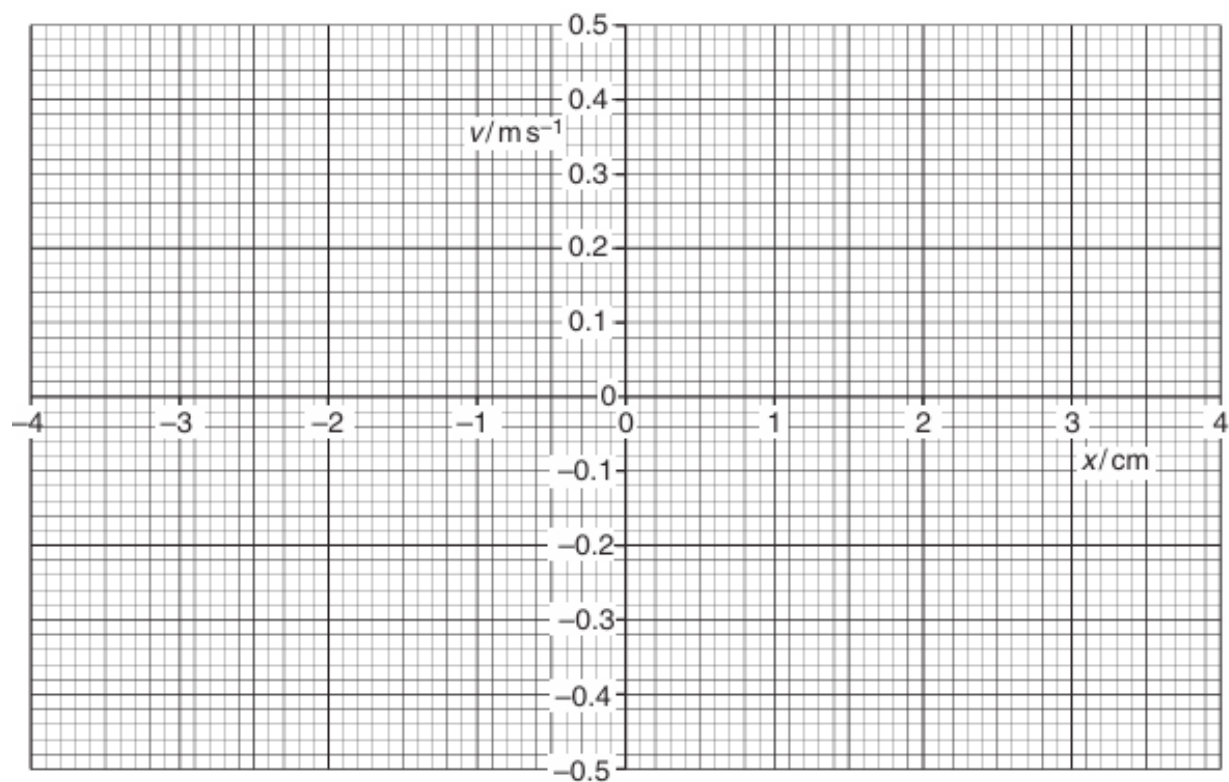
- (ii) determine, for these oscillations, the angular frequency ω ,

$$\omega = \dots\dots\dots \text{ rad s}^{-1} \quad [2]$$

- (iii) show that the maximum speed of the mass is 0.42 m s^{-1} .

[2]

- (b) Use data from Fig. 4.2 and (a)(iii) to sketch, on the axes of Fig. 4.3, the variation with displacement x from the equilibrium position of the velocity v of the mass.



33. F/M/17/42/Q3

A uniform beam is clamped at one end. A metal block of mass m is fixed to the other end of the beam causing it to bend, as shown in Fig. 3.1.

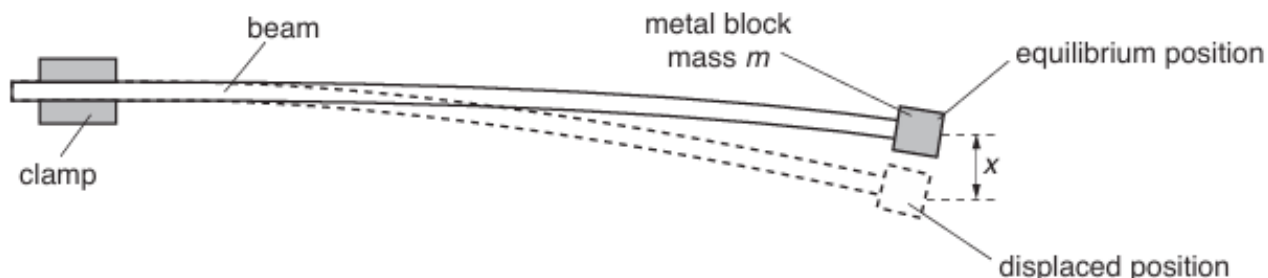


Fig. 3.1

The block is given a small vertical displacement and then released so that it oscillates with simple harmonic motion.

The acceleration a of the block is given by the expression

$$a = -\frac{k}{m}x$$

where k is a constant for the beam and x is the vertical displacement of the block from its equilibrium position.

- (a) Explain how it can be deduced from the expression that the block moves with simple harmonic motion.

.....

[2]

- (b) For the beam, $k = 4.0 \text{ kg s}^{-2}$. Show that the angular frequency ω of the oscillations is given by the expression

$$\omega = \frac{2.0}{\sqrt{m}}.$$

- (c) The initial amplitude of the oscillation of the block is 3.0 cm.

Use the expression in (b) to determine the maximum kinetic energy of the oscillations.

maximum kinetic energy = J [3]

- (d) Over a certain interval of time, the maximum kinetic energy of the oscillations in (c) is reduced by 50%. It may be assumed that there is negligible change in the angular frequency of the oscillations.

Determine the amplitude of oscillation.

amplitude = m [2]

- (e) Permanent magnets are now positioned so that the metal block oscillates between the poles, as shown in Fig. 3.2.

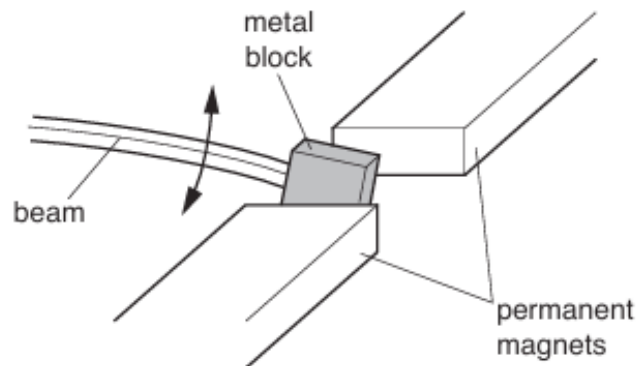


Fig.3.2

The block is made to oscillate with the same initial amplitude as in (c). Use energy conservation to explain why the energy of the oscillations decreases more rapidly than in (d).

.....

.....

.....

.....

.....

..... [3]

34. M/J/17/41/Q2

A bar magnet of mass 180 g is suspended from the free end of a spring, as illustrated in Fig. 2.1.

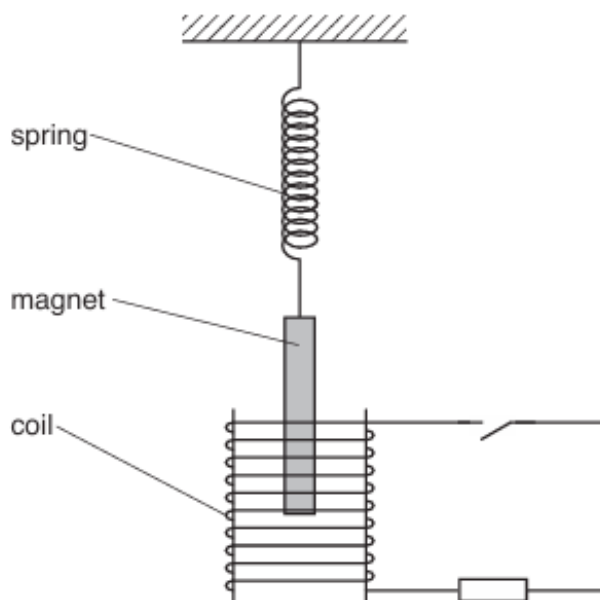


Fig. 2.1

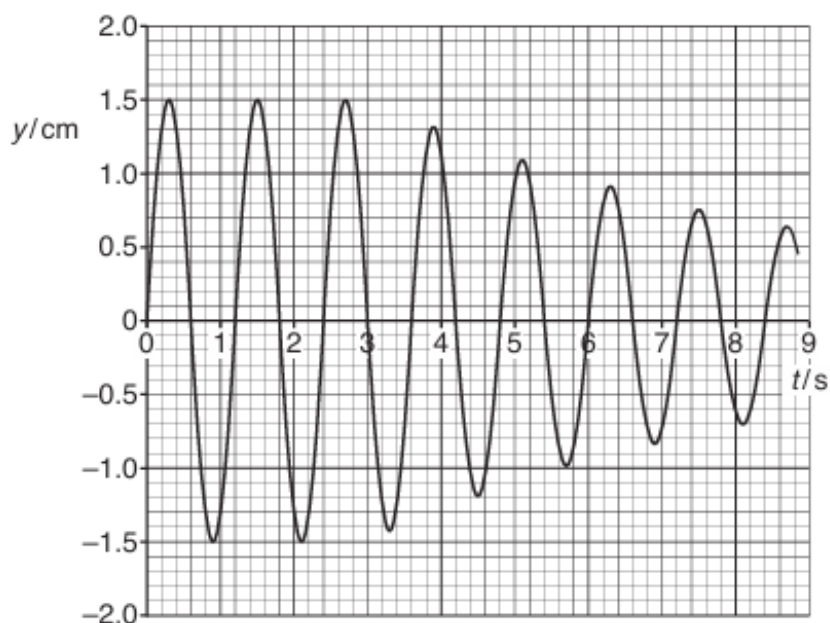
The magnet hangs so that one pole is near the centre of a coil of wire.

The coil is connected in series with a resistor and a switch. The switch is open.

The magnet is displaced vertically and then allowed to oscillate with one pole remaining inside the coil. The other pole remains outside the coil.

At time $t = 0$, the magnet is oscillating freely as it passes through its equilibrium position. At time $t = 3.0$ s, the switch in the circuit is closed.

The variation with time t of the vertical displacement y of the magnet is shown in Fig. 2.2.



- (a) Determine, to two significant figures, the frequency of oscillation of the magnet.

frequency = Hz [2]

- (b) State whether the closing of the switch gives rise to light, heavy or critical damping.

.....[1]

- (c) Calculate the change in the energy ΔE of oscillation of the magnet between time $t = 2.7$ s and time $t = 7.5$ s. Explain your working.

$\Delta E =$ J [6]

[Total: 9]

35. M/J/17/42/Q3

A bar magnet of mass 250 g is suspended from the free end of a spring, as illustrated in Fig. 3.1.

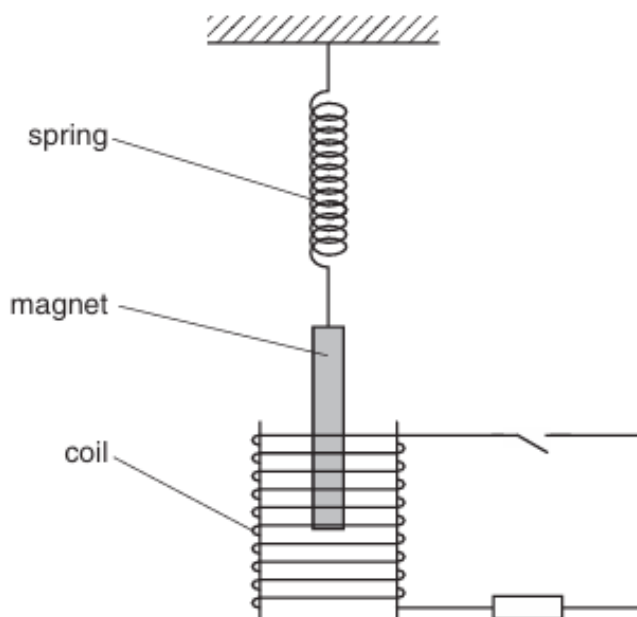


Fig. 3.1

The magnet hangs so that one pole is near the centre of a coil of wire.

The coil is connected in series with a resistor and a switch. The switch is open.

The magnet is displaced vertically and then allowed to oscillate with one pole remaining inside the coil. The other pole remains outside the coil.

At time $t = 0$, the magnet is oscillating freely as it passes through its equilibrium position. At time $t = 6.0$ s, the switch in the circuit is closed.

The variation with time t of the vertical displacement y of the magnet is shown in Fig. 3.2.

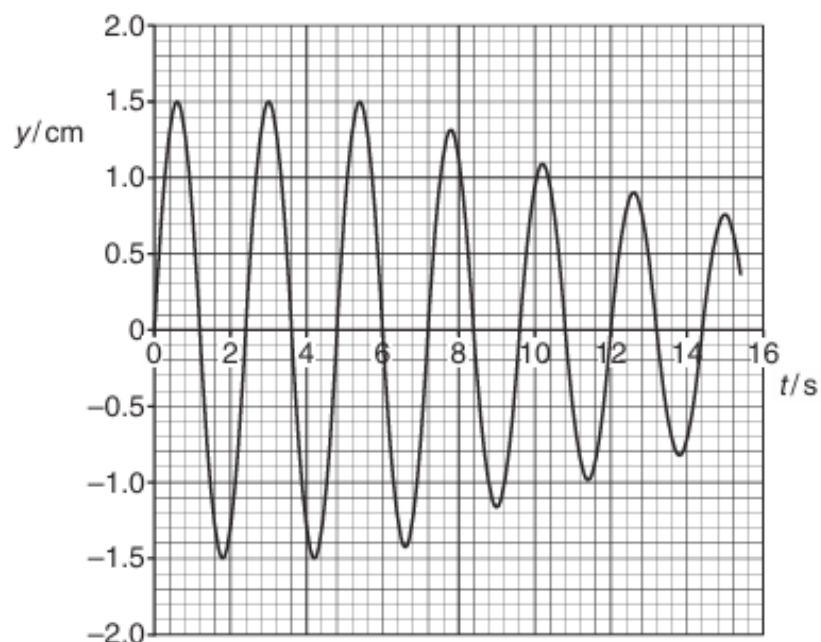


Fig. 3.2

(a) For the oscillating magnet, use data from Fig. 3.2 to calculate, to two significant figures,

(i) the frequency f ,

$f = \dots\dots\dots$ Hz [2]

(ii) the energy of the oscillations during the time $t = 0$ to time $t = 6.0$ s.

energy = $\dots\dots\dots$ J [3]

(b) (i) State Faraday's law of electromagnetic induction.

.....

.....

.....

.....[2]

(ii) Use Faraday's law and energy conservation to explain why the amplitude of the oscillations of the magnet reduces after time $t = 6.0\text{ s}$.

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

36. O/N/17/41/Q2

- (a) State, by reference to simple harmonic motion, what is meant by *angular frequency*.

.....
 [1]

- (b) A thin metal strip is clamped at one end so that it is horizontal. A load of mass M is attached to its free end. The load causes a displacement s of the end of the strip, as shown in Fig. 2.1.

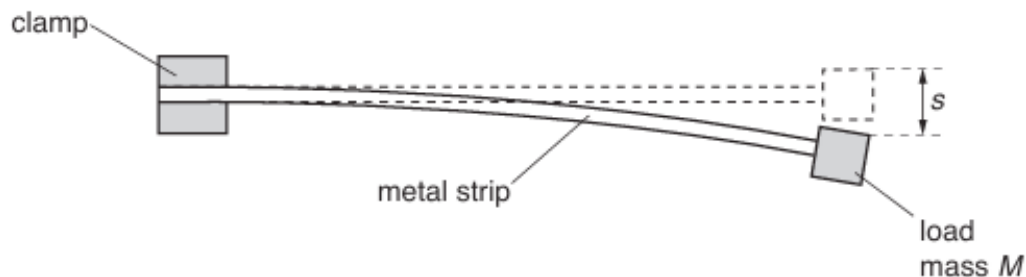


Fig. 2.1

The load is displaced vertically and then released. The load oscillates. The variation with the acceleration a of the displacement s of the load is shown in Fig. 2.2.

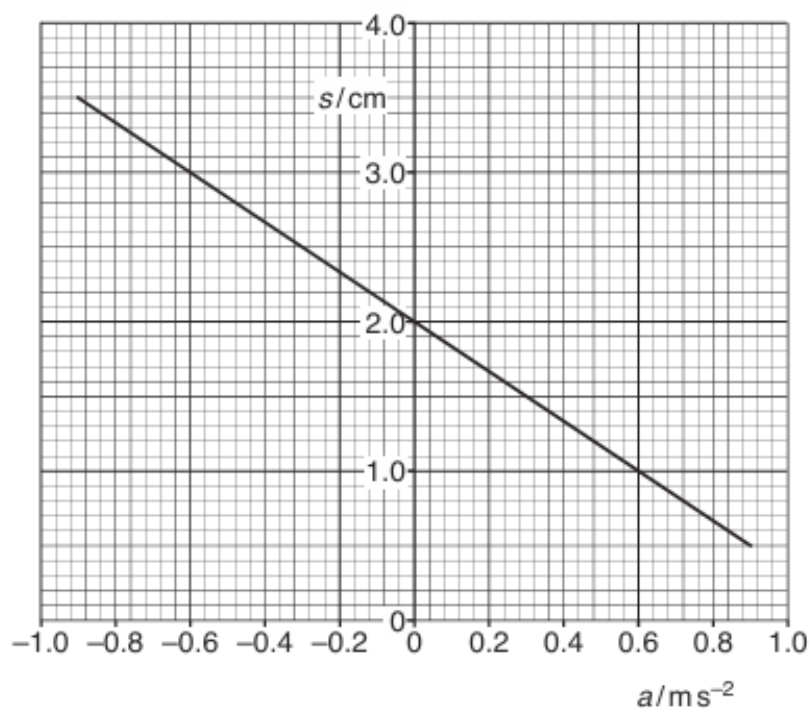


Fig. 2.2

(i) Use Fig. 2.2 to determine

1. the displacement of the load before it is made to oscillate,

displacement = cm

2. the amplitude of the oscillations of the load.

amplitude = cm
[2]

(ii) Show that the load is undergoing simple harmonic motion.

.....
.....
.....
..... [3]

(iii) Calculate the frequency of oscillation of the load.

frequency = Hz [3]

[Total: 9]

37. O/N/17/42/Q3

(a) (i) Define the *radian*.

.....

[2]

(ii) State, by reference to simple harmonic motion, what is meant by *angular frequency*.

.....
[1]

(b) A thin metal strip, clamped horizontally at one end, has a load of mass M attached to its free end, as shown in Fig. 3.1.

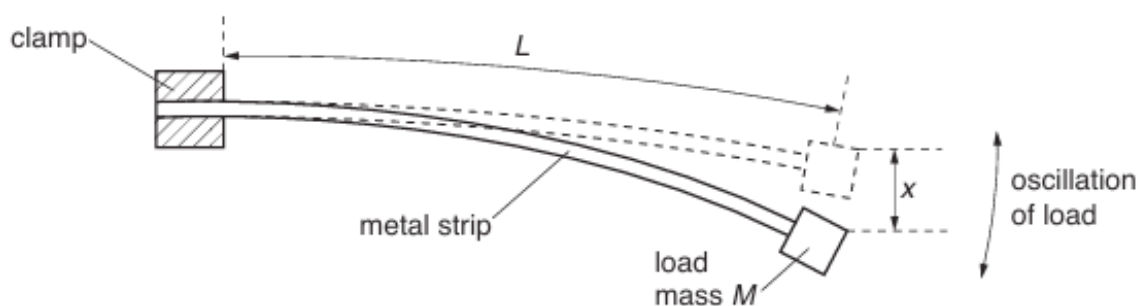


Fig. 3.1

The metal strip bends, as shown in Fig. 3.1.

When the free end of the strip is displaced vertically and then released, the mass oscillates in a vertical plane.

Theory predicts that the variation of the acceleration a of the oscillating load with the displacement x from its equilibrium position is given by

$$a = -\left(\frac{c}{ML^3}\right)x$$

where L is the effective length of the metal strip and c is a positive constant.

(i) Explain how the expression shows that the load is undergoing simple harmonic motion.

.....

[2]

- (ii) For a metal strip of length $L = 65\text{ cm}$ and a load of mass $M = 240\text{ g}$, the frequency of oscillation is 3.2 Hz .
Calculate the constant c .

$$c = \dots\dots\dots \text{ kg m}^3 \text{ s}^{-2} \text{ [3]}$$

[Total: 8]

38. F/M/18/42/Q4

- (a) Explain what is meant by the *natural frequency of vibration* of a system.

.....
.....
.....[1]

- (b) A block of metal is fixed to one end of a vertical spring. The other end of the spring is attached to an oscillator, as shown in Fig. 4.1.

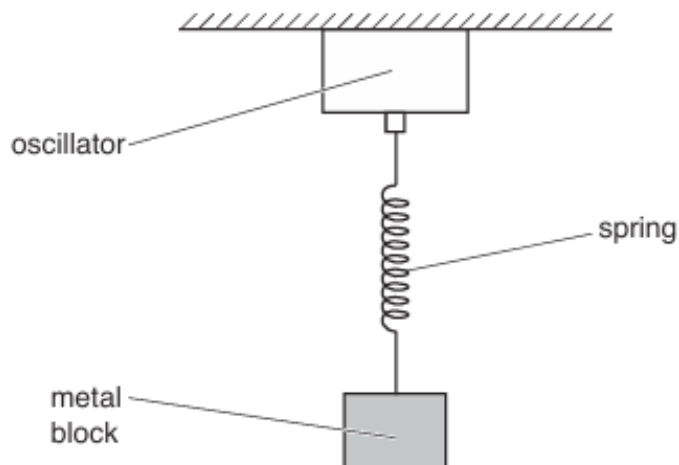


Fig. 4.1

The amplitude of oscillation of the oscillator is constant.

The variation of the amplitude x_0 of the oscillations of the block with frequency f of the oscillations is shown in Fig. 4.2.

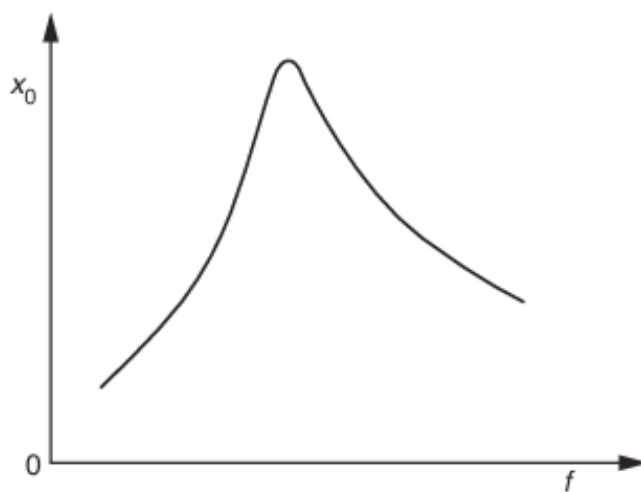


Fig. 4.2

(i) Name the effect shown in Fig. 4.2.

.....[1]

(ii) State and explain whether the block is undergoing damped oscillations.

.....

.....

.....[2]

(c) State **one** example in which the effect shown in Fig. 4.2 is useful.

.....

.....[1]

[Total: 5]

39. M/J/18/41/Q2

A metal plate is made to vibrate vertically by means of an oscillator, as shown in Fig. 2.1.

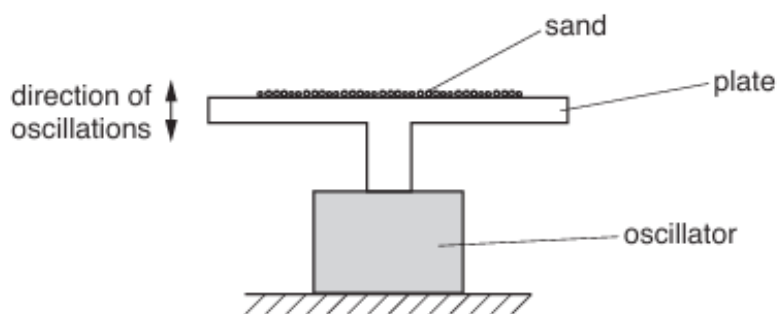


Fig. 2.1

Some sand is sprinkled on to the plate.

The variation with displacement y of the acceleration a of the sand on the plate is shown in Fig. 2.2.

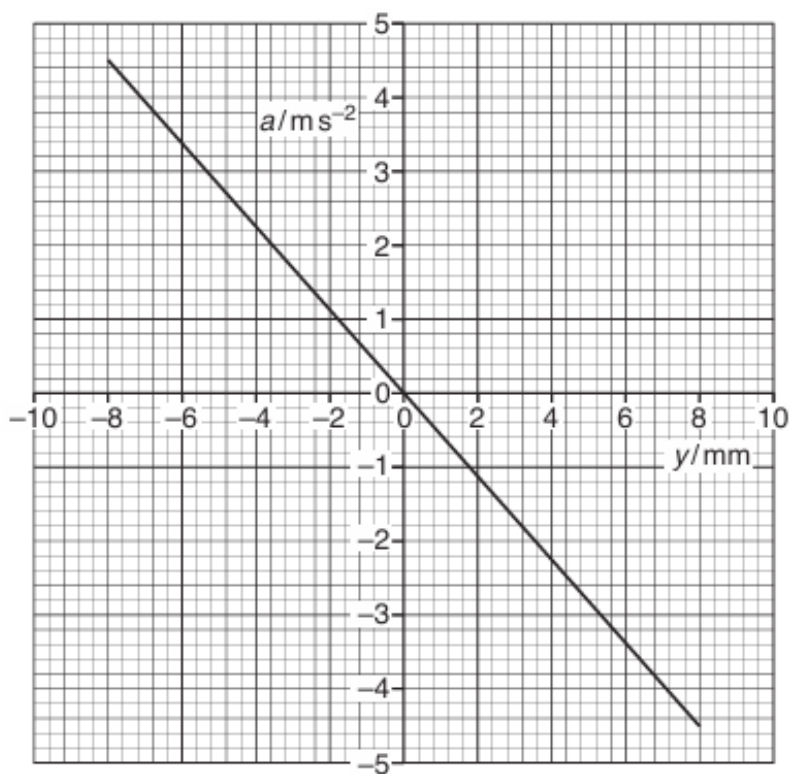


Fig. 2.2

- (a) (i) Use Fig. 2.2 to show how it can be deduced that the sand is undergoing simple harmonic motion.

.....

.....

- (ii) Calculate the frequency of oscillation of the sand.

frequency = Hz [2]

- (b) The amplitude of oscillation of the plate is gradually increased beyond 8 mm. The frequency is constant.

At one amplitude, the sand is seen to lose contact with the plate.

For the plate when the sand first loses contact with the plate,

- (i) state the position of the plate,

.....[1]

- (ii) calculate the amplitude of oscillation.

amplitude = mm [3]

[Total: 8]

40. M/J/18/42/Q4

(a) State two conditions necessary for a mass to be undergoing simple harmonic motion.

1.
2.

[2]

(b) A trolley of mass 950 g is held on a horizontal surface by means of two springs attached to fixed points P and Q, as shown in Fig. 4.1.

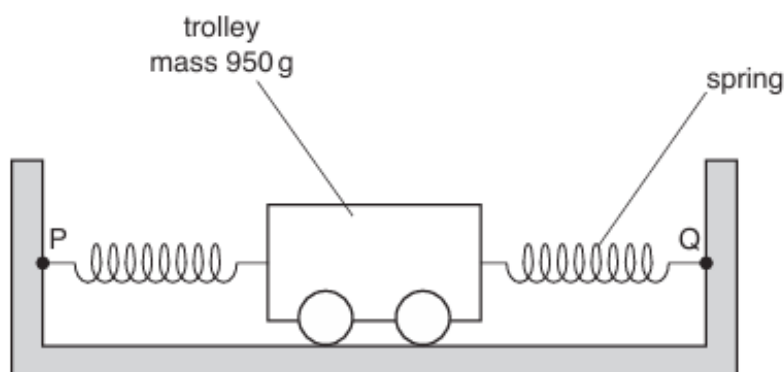
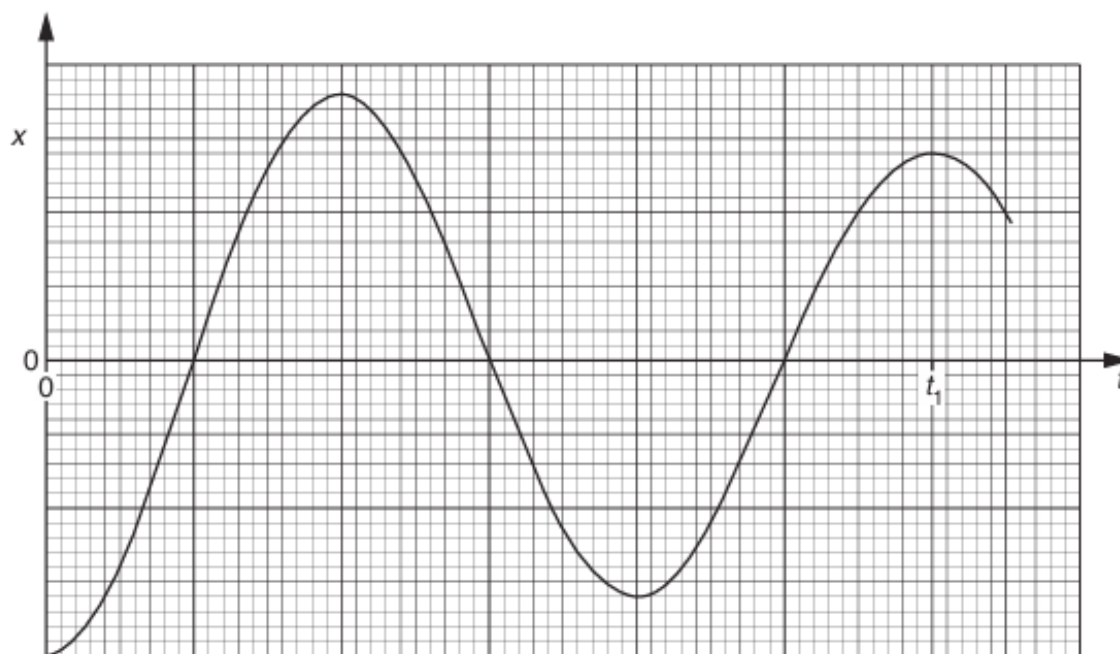


Fig. 4.1

The springs, each having a spring constant k of 230 N m^{-1} , are always extended.

The trolley is displaced along the line of the springs and then released.

The variation with time t of the displacement x of the trolley is shown in Fig. 4.2.



- (i) 1. State and explain whether the oscillations of the trolley are heavily damped, critically damped or lightly damped.

.....

2. Suggest the cause of the damping.

.....

[3]

- (ii) The acceleration a of the trolley of mass m may be assumed to be given by the expression

$$a = -\left(\frac{2k}{m}\right)x.$$

1. Calculate the angular frequency ω of the oscillations of the trolley.

$$\omega = \dots\dots\dots \text{rad s}^{-1} [3]$$

2. Determine the time t_1 shown on Fig. 4.2.

$$t_1 = \dots\dots\dots \text{s} [2]$$

[Total: 10]

41. O/N/18/41/Q3

A U-tube contains liquid, as shown in Fig. 3.1.

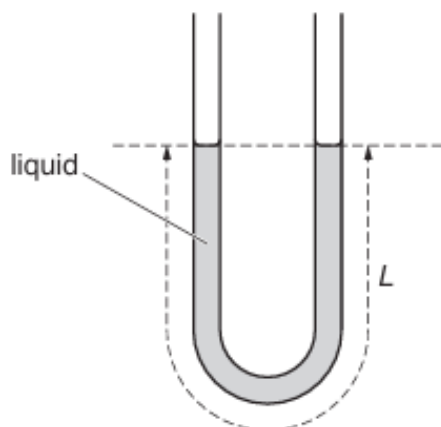


Fig. 3.1

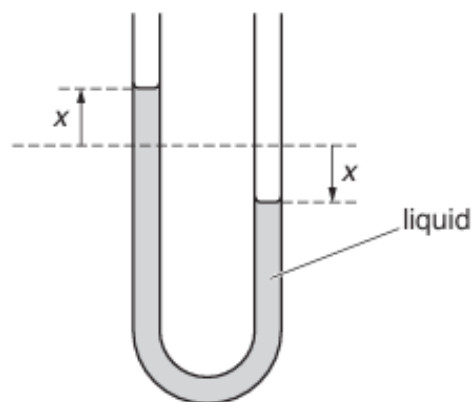


Fig. 3.2

The total length of the column of liquid in the tube is L .

The column of liquid is displaced so that the change in height of the liquid in each arm of the U-tube is x , as shown in Fig. 3.2.

The liquid in the U-tube then oscillates with simple harmonic motion such that the acceleration a of the column is given by the expression

$$a = -\left(\frac{2g}{L}\right)x$$

where g is the acceleration of free fall.

(a) Calculate the period T of oscillation of the liquid column for a column length L of 19.0 cm.

$T = \dots\dots\dots$ s [3]

(b) The variation with time t of the displacement x is shown in Fig. 3.3.

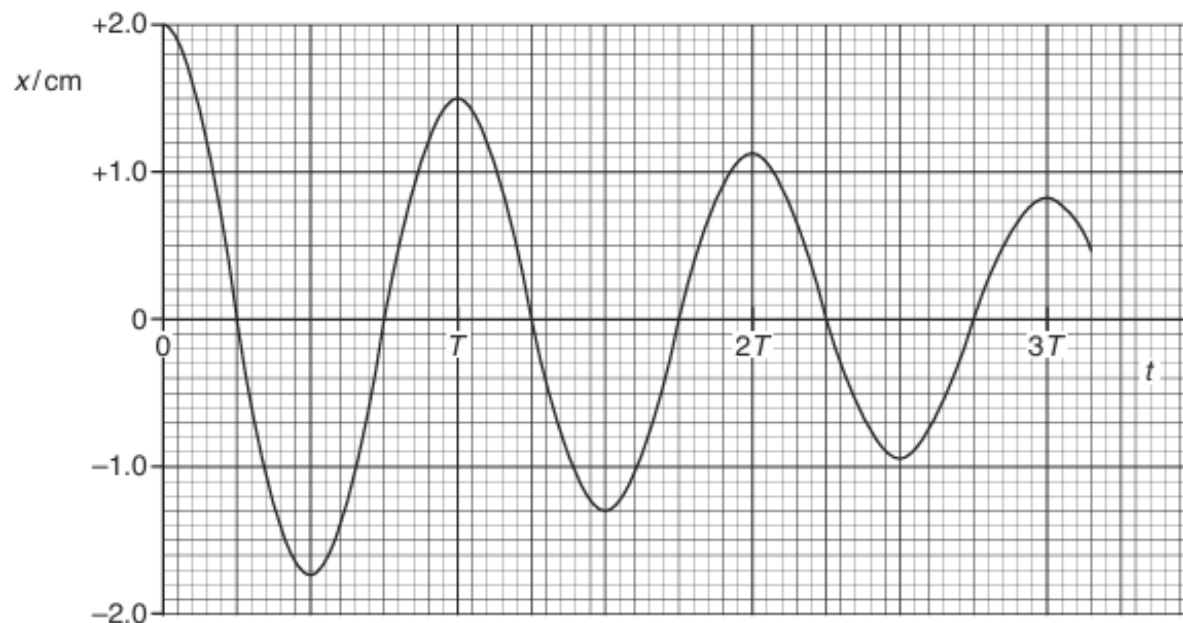


Fig. 3.3

The period of oscillation of the liquid column of mass 18.0 g is T .

The oscillations are damped.

(i) Suggest one cause of the damping.

.....
[1]

(ii) Calculate the loss in total energy of the oscillations during the first 2.5 periods of the oscillations.

energy loss = J [3]

[Total: 7]

42. O/N/18/42/Q4

A U-tube contains liquid, as shown in Fig. 4.1.

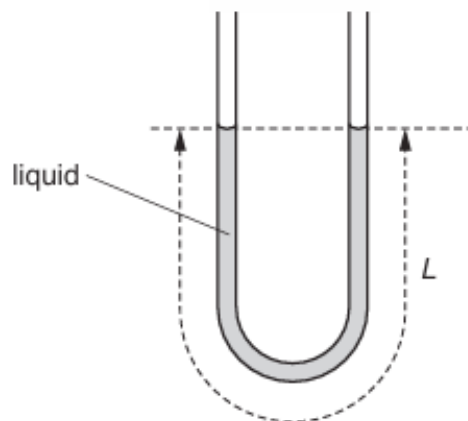


Fig. 4.1

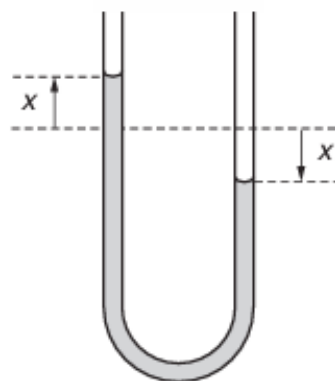


Fig. 4.2

The total length of the liquid column is L .

The column of liquid is displaced so that the change in height of the liquid level from the equilibrium position in each arm of the U-tube is x , as shown in Fig. 4.2.

The liquid in the U-tube then oscillates such that its acceleration a is given by the expression

$$a = -\left(\frac{2g}{L}\right)x$$

where g is the acceleration of free fall.

(a) Show that the liquid column undergoes simple harmonic motion.

(b) The variation with time t of the displacement x is shown in Fig. 4.3.

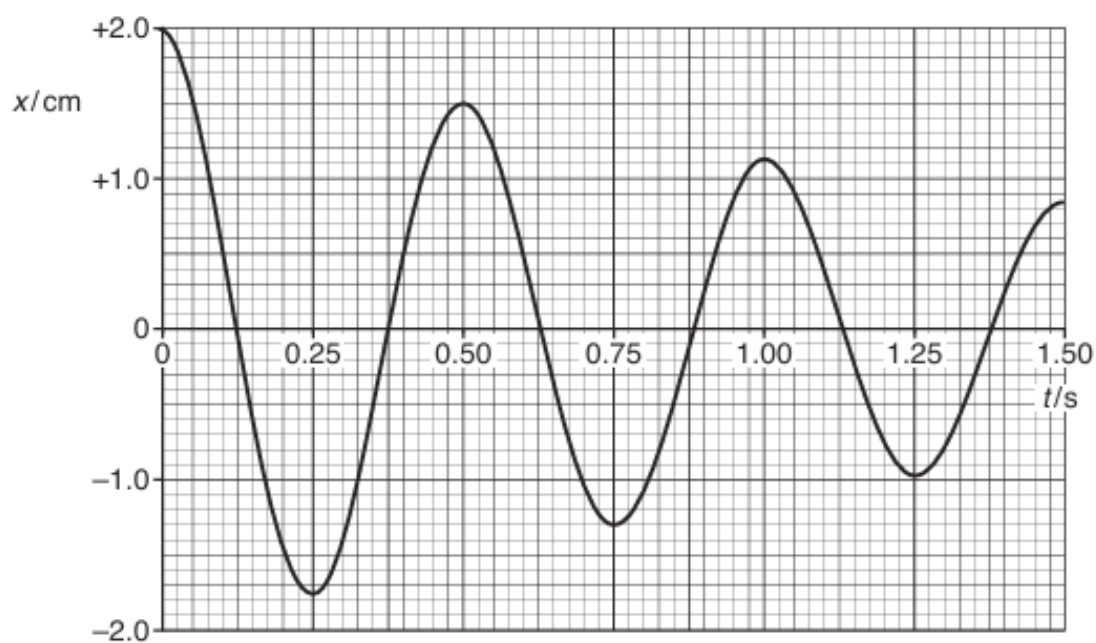


Fig. 4.3

Use data from Fig. 4.3 to determine the length L of the liquid column.

$L = \dots\dots\dots \text{ m [3]}$

(c) The oscillations shown in Fig. 4.3 are damped.

(i) Suggest one cause of this damping.

.....
.....[1]

(ii) Calculate the ratio

$$\frac{\text{total energy of oscillations after 1.5 complete oscillations}}{\text{total initial energy of oscillations}}$$

ratio = [2]

[Total: 8]

43. F/M/19/42/Q3

A cylindrical tube, sealed at one end, has cross-sectional area A and contains some sand. The total mass of the tube and the sand is M .

The tube floats upright in a liquid of density ρ , as illustrated in Fig. 3.1.

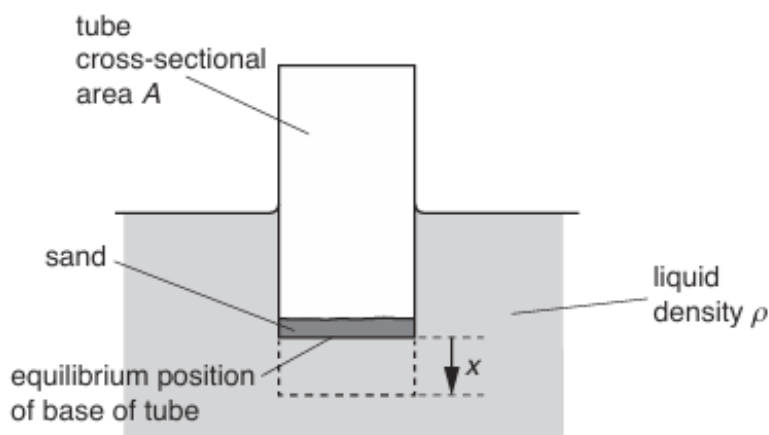


Fig. 3.1

The tube is pushed a short distance into the liquid and then released.

- (a) (i) State the two forces that act on the tube immediately after its release.

.....
 [1]

- (ii) State and explain the direction of the resultant force acting on the tube immediately after its release.

.....

 [2]

- (b) The acceleration a of the tube is given by the expression

$$a = -\left(\frac{A\rho g}{M}\right)x$$

where x is the vertical displacement of the tube from its equilibrium position.

Use the expression to explain why the tube undergoes simple harmonic oscillations in the liquid.

.....

 [2]

- (c) For a tube having cross-sectional area A of 4.5 cm^2 and a total mass M of 0.17 kg , the period of oscillation of the tube is 1.3 s .

- (i) Determine the angular frequency ω of the oscillations.

$$\omega = \dots\dots\dots \text{ rad s}^{-1} \text{ [2]}$$

- (ii) Use your answer in (i) and the expression in (b) to determine the density ρ of the liquid in which the tube is floating.

$$\rho = \dots\dots\dots \text{ kg m}^{-3} \text{ [3]}$$

[Total: 10]

44. M/J/19/41/Q3

A hollow tube, sealed at one end, has a cross-sectional area A of 24 cm^2 .
The tube contains sand so that the total mass M of the tube and sand is 0.23 kg .

The tube floats upright in a liquid of density ρ , as illustrated in Fig. 3.1.

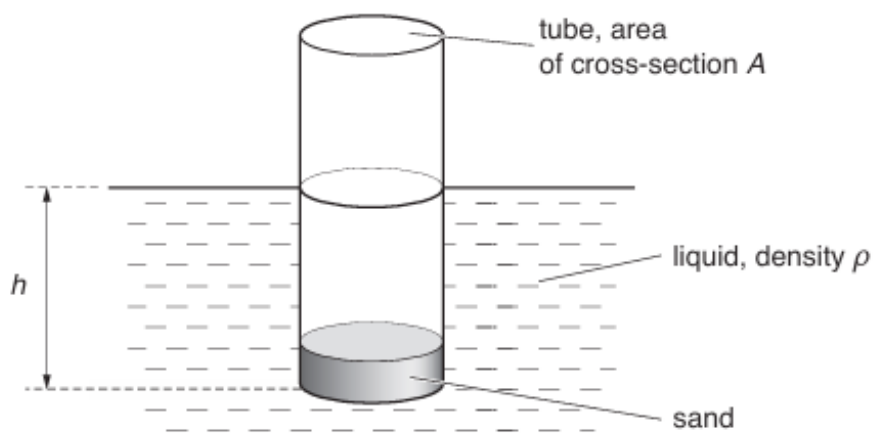
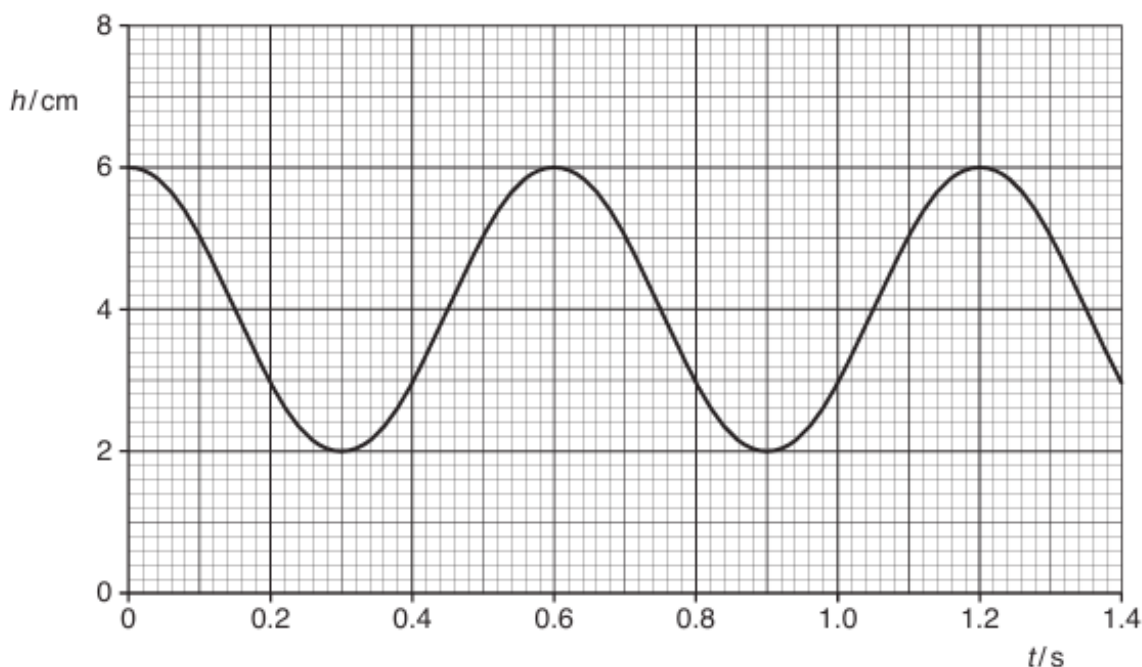


Fig. 3.1

The depth of the bottom of the tube below the liquid surface is h .

The tube is displaced vertically and then released. The variation with time t of the depth h is shown in Fig. 3.2.



(a) Determine:

(i) the amplitude, in metres, of the oscillations

amplitude = m [1]

(ii) the frequency of oscillation of the tube in the liquid

frequency =Hz [2]

(iii) the acceleration of the tube when h is a maximum.

acceleration =ms⁻² [2]

(b) The frequency f of oscillation of the tube is given by the expression

$$f = \frac{1}{2\pi} \sqrt{\left(\frac{A\rho g}{M}\right)}$$

where g is the acceleration of free fall.

Calculate the density ρ of the liquid in which the tube is floating.

$\rho =$ kgm⁻³ [2]

(c) The oscillations illustrated in Fig. 3.2 are undamped. In practice, the liquid does cause light damping.

On Fig. 3.2, draw a line to show light damping of the oscillations for time $t = 0$ to time $t = 1.4$ s.
[3]

45. M/J/19/42/Q3

A spring is hung vertically from a fixed point. A mass M is hung from the other end of the spring, as illustrated in Fig. 3.1.

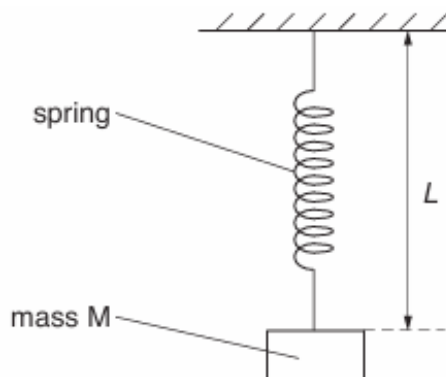
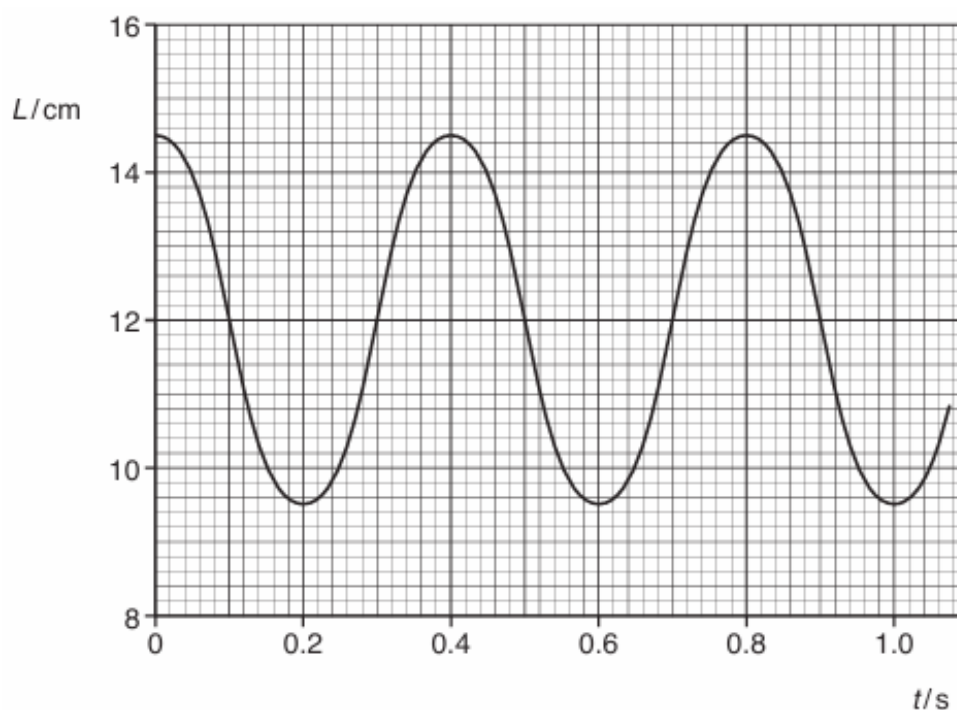


Fig. 3.1

The mass is displaced downwards and then released. The subsequent motion of the mass is simple harmonic.

The variation with time t of the length L of the spring is shown in Fig. 3.2.



(a) State:

(i) one time at which the mass is moving with maximum speed

time = s [1]

(ii) one time at which the spring has maximum elastic potential energy.

time = s [1]

(b) Use data from Fig. 3.2 to determine, for the motion of the mass:

(i) the angular frequency ω

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

(ii) the maximum speed

$$\text{maximum speed} = \dots\dots\dots \text{m s}^{-1} \quad [2]$$

(iii) the magnitude of the maximum acceleration.

$$\text{maximum acceleration} = \dots\dots\dots \text{m s}^{-2} \quad [2]$$

46. O/N/19/41/Q4

A mass is suspended vertically from a fixed point by means of a spring, as illustrated in Fig. 4.1.

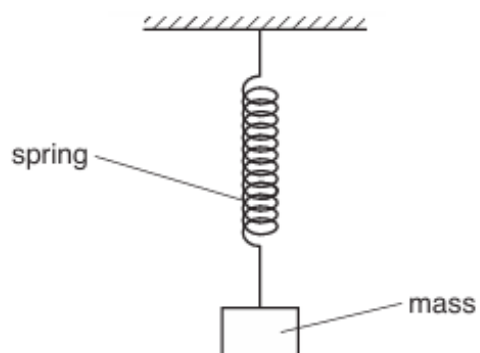


Fig. 4.1

The mass is oscillating vertically. The variation with displacement x of the acceleration a of the mass is shown in Fig. 4.2.

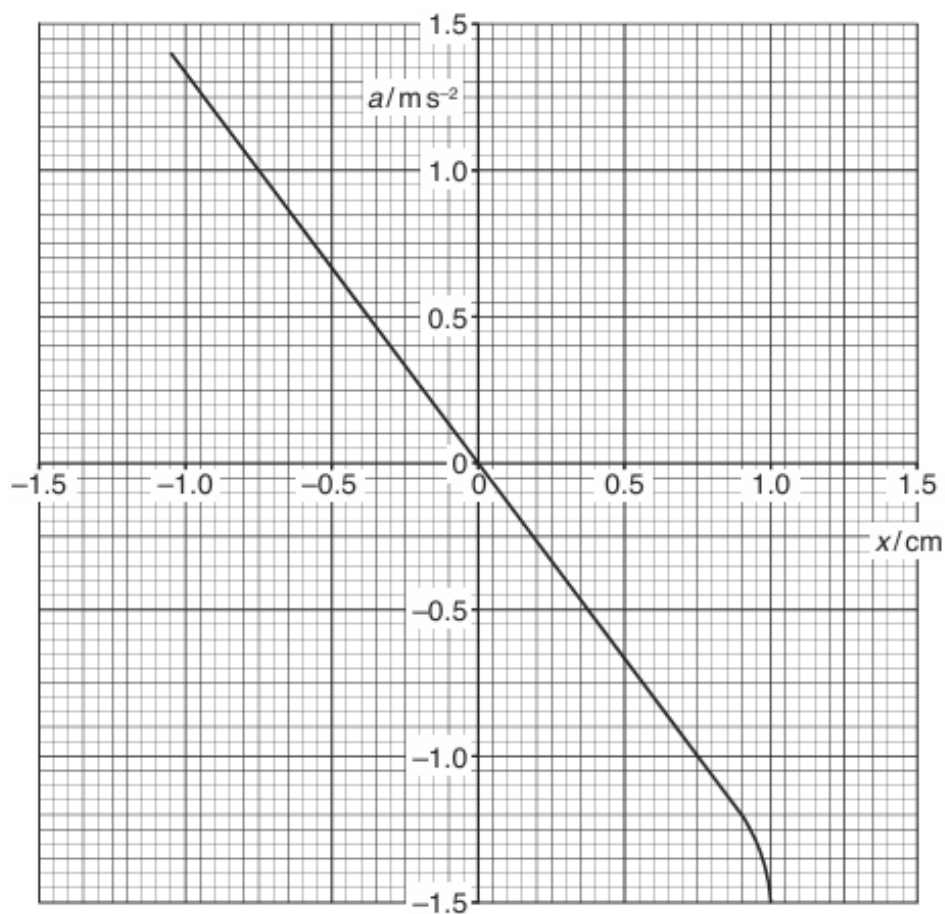


Fig. 4.2

- (a) (i) State what is meant by the *displacement* of the mass on the spring.

.....
 [1]

- (ii) Suggest how Fig. 4.2 shows that the mass is not performing simple harmonic motion.

.....
 [1]

- (b) (i) The amplitude of oscillation of the mass may be changed.

State the maximum amplitude x_0 for which the oscillations are simple harmonic.

$x_0 =$ cm [1]

- (ii) For the simple harmonic oscillations of the mass, use Fig. 4.2 to determine the frequency of the oscillations.

frequency = Hz [3]

- (c) The maximum speed of the mass when oscillating with simple harmonic motion of amplitude x_0 is v_0 .

On Fig. 4.3, show the variation with displacement x of the velocity v of the mass for displacements from $+x_0$ to $-x_0$.

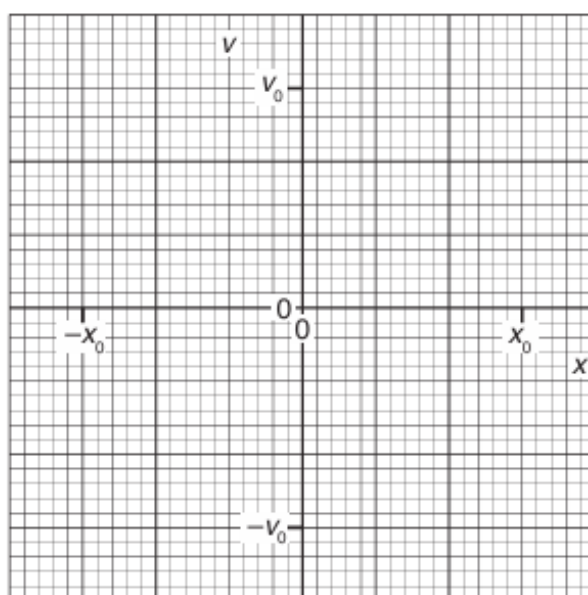


Fig. 4.3

[2]

[Total: 8]

47. O/N/19/42/Q4

A ball of mass M is held on a horizontal surface by two identical extended springs, as illustrated in Fig. 4.1.

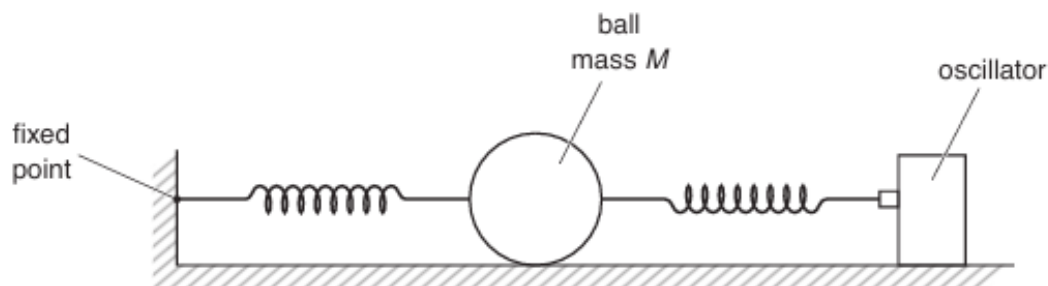


Fig. 4.1

One spring is attached to a fixed point. The other spring is attached to an oscillator.

The oscillator is switched off. The ball is displaced sideways along the axis of the springs and is then released. The variation with time t of the displacement x of the ball is shown in Fig. 4.2.

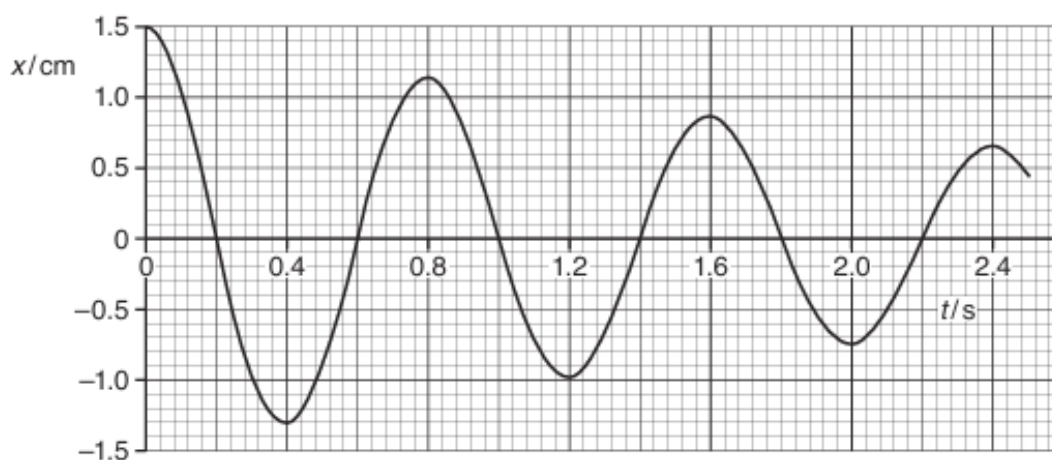


Fig. 4.2

(a) State:

(i) what is meant by *damping*

.....
 [1]

(ii) the evidence provided by Fig. 4.2 that the motion of the ball is damped.

.....
 [1]

- (b)** The acceleration a and the displacement x of the ball are related by the expression

$$a = -\left(\frac{2k}{M}\right)x$$

where k is the spring constant of one of the springs.

The mass M of the ball is 1.2 kg.

- (i)** Use data from Fig. 4.2 to determine the angular frequency ω of the oscillations of the ball.

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii)** Use your answer in **(i)** to determine the value of k .

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

48. F/M/20/42/Q3

- (a) A body undergoes simple harmonic motion.

The variation with displacement x of its velocity v is shown in Fig. 3.1.

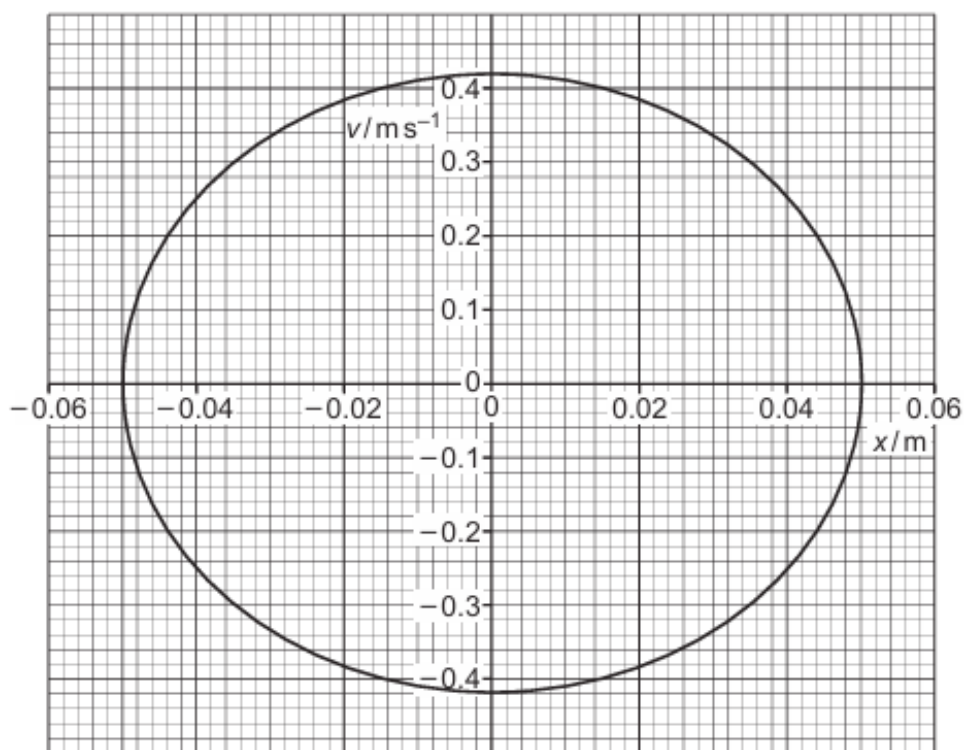


Fig. 3.1

- (i) State the amplitude x_0 of the oscillations.

$x_0 = \dots\dots\dots$ m [1]

- (ii) Calculate the period T of the oscillations.

$T = \dots\dots\dots$ s [3]

- (iii) On Fig. 3.1, label with a P a point where the body has **maximum** potential energy. [1]

- (b) A bar magnet is suspended from the free end of a spring, as shown in Fig. 3.2.

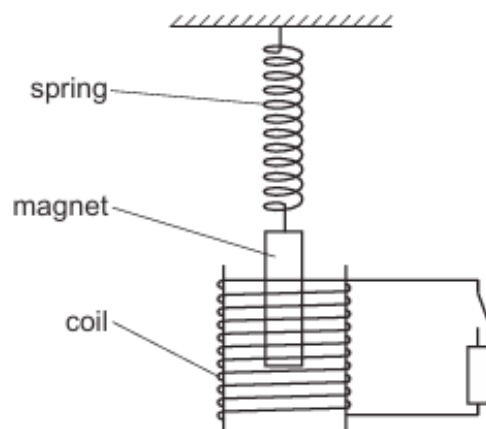


Fig. 3.2

One pole of the magnet is situated in a coil of wire. The coil is connected in series with a switch and a resistor. The switch is open.

The magnet is displaced vertically and then released. The magnet oscillates with simple harmonic motion.

- (i) State Faraday's law of electromagnetic induction.

.....

.....

.....

..... [2]

- (ii) The switch is now closed. Explain why the oscillations of the magnet are damped.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

49. M/J/20/41/Q3

The piston in the cylinder of a car engine moves in the cylinder with simple harmonic motion. The piston moves between a position of maximum height in the cylinder to a position of minimum height, as illustrated in Fig. 3.1.

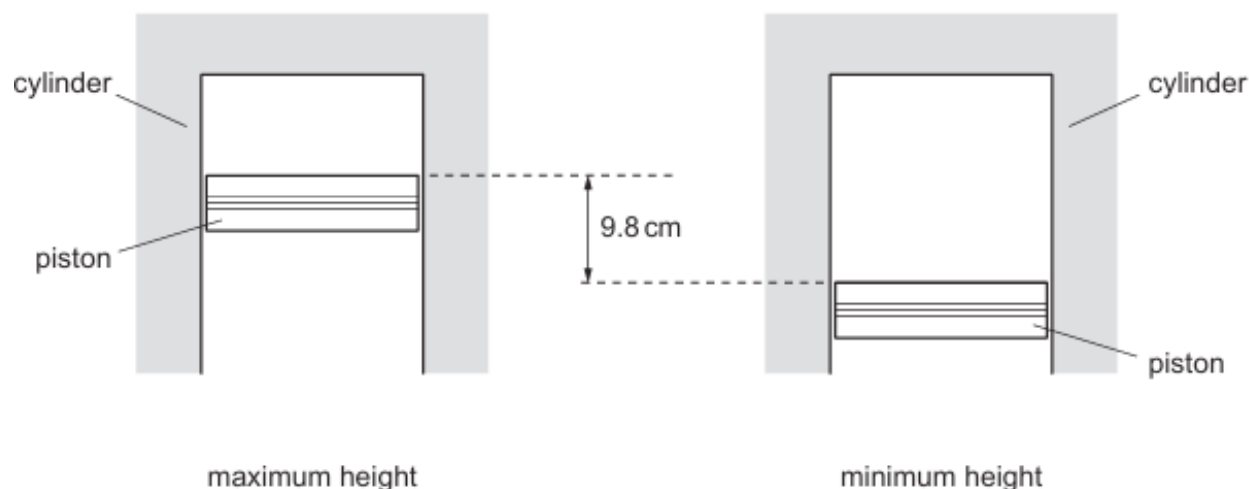


Fig. 3.1

The distance moved by the piston between the positions shown in Fig. 3.1 is 9.8 cm.

The mass of the piston is 640 g.

At one particular speed of the engine, the piston completes 2700 oscillations in 1.0 minute.

(a) For the oscillations of the piston in the cylinder, determine:

(i) the amplitude

amplitude = cm [1]

(ii) the frequency

frequency = Hz [1]

(iii) the maximum speed

maximum speed = ms^{-1} [2]

(iv) the speed when the top of the piston is 2.3 cm below its maximum height.

speed = ms^{-1} [2]

(b) The acceleration of the piston varies.

Determine the resultant force on the piston that gives rise to its maximum acceleration.

force = N [3]

[Total: 9]

50. M/J/20/42/Q4

A dish is made from a section of a hollow glass sphere.

The dish, fixed to a horizontal table, contains a small solid ball of mass 45 g, as shown in Fig. 4.1.

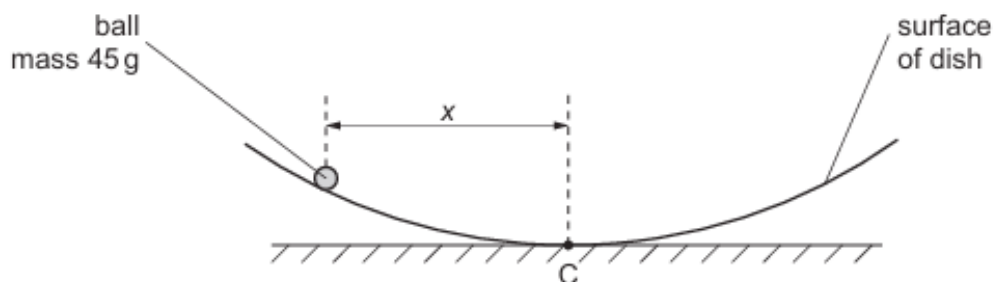


Fig. 4.1

The horizontal displacement of the ball from the centre C of the dish is x .

Initially, the ball is held at rest with distance $x = 3.0$ cm.

The ball is then released. The variation with time t of the horizontal displacement x of the ball from point C is shown in Fig. 4.2.

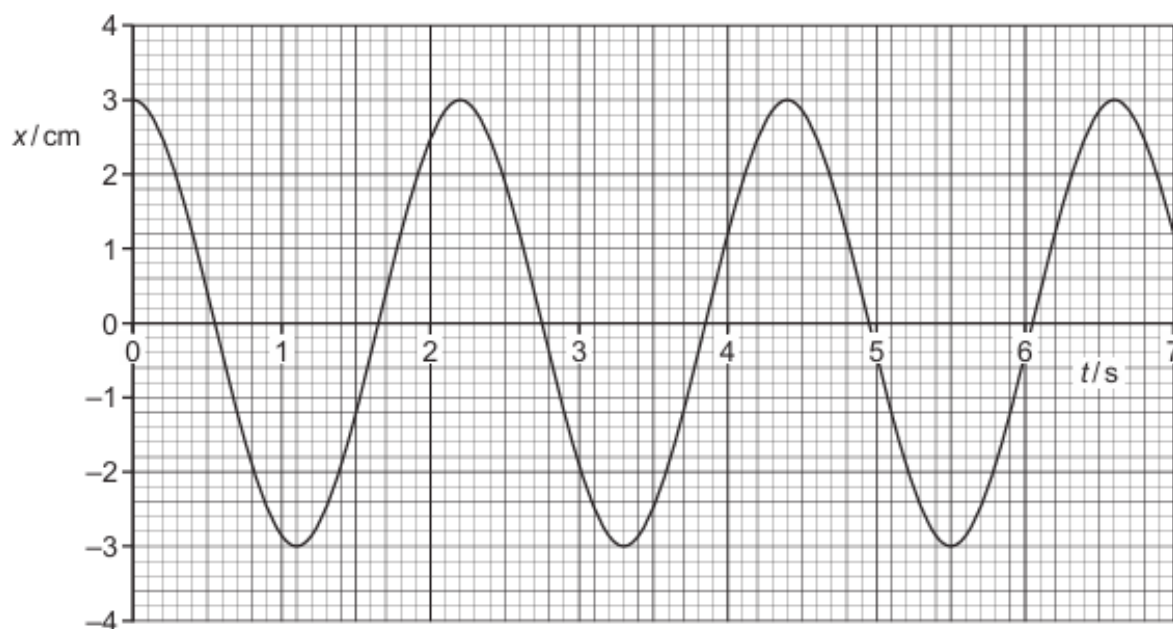


Fig. 4.2

The motion of the ball in the dish is simple harmonic with its acceleration a given by the expression

$$a = -\left(\frac{g}{R}\right)x$$

where g is the acceleration of free fall and R is a constant that depends on the dimensions of the dish and the ball.

- (a) Use Fig. 4.2 to show that the angular frequency ω of oscillation of the ball in the dish is 2.9rad s^{-1} .

[1]

- (b) Use the information in (a) to:

- (i) determine R

$R = \dots\dots\dots \text{ m}$ [2]

- (ii) calculate the speed of the ball as it passes over the centre C of the dish.

speed = $\dots\dots\dots \text{ m s}^{-1}$ [2]

- (c) Some moisture collects on the surface of the dish so that the motion of the ball becomes lightly damped.

On the axes of Fig. 4.2, draw a line to show the lightly damped motion of the ball for the first 5.0 s after the release of the ball. [3]

[Total: 8]

51. O/N/20/41/Q3

A pendulum consists of a metal sphere P suspended from a fixed point by means of a thread, as illustrated in Fig. 3.1.

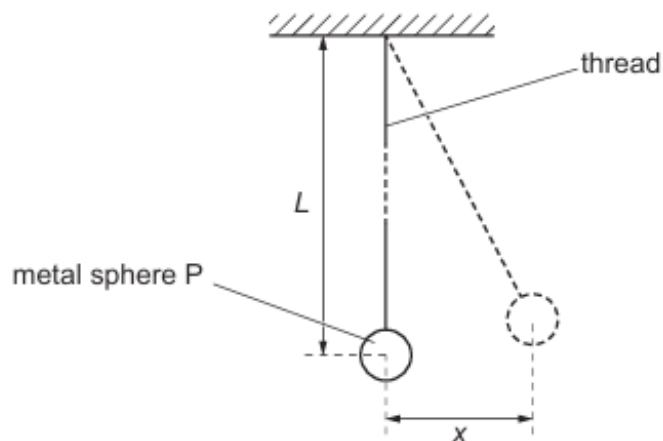


Fig. 3.1

The centre of gravity of sphere P is a distance L from the fixed point.

The sphere is pulled to one side and then released so that it oscillates. The sphere may be assumed to oscillate with simple harmonic motion.

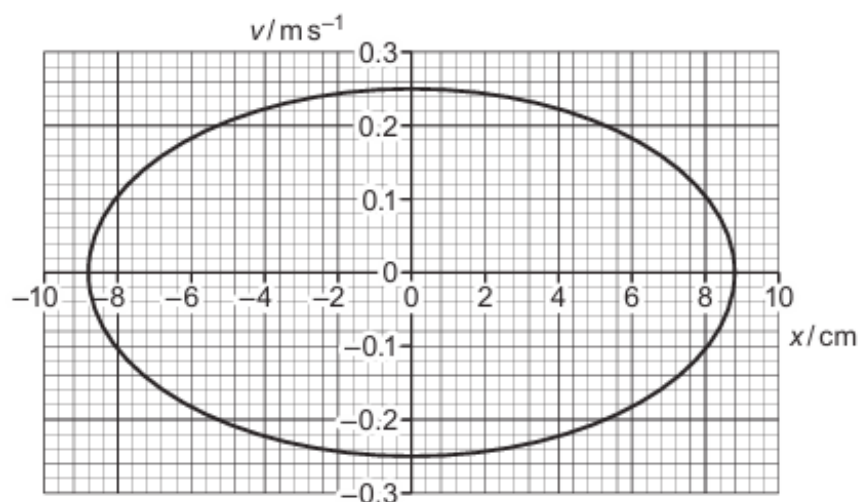
(a) State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

(b) The variation of the velocity v of sphere P with the displacement x from its mean position is shown in Fig. 3.2.



Use Fig. 3.2 to determine the frequency f of the oscillations of sphere P.

$$f = \dots\dots\dots \text{ Hz [3]}$$

- (c) The period T of the oscillations of sphere P is given by the expression

$$T = 2\pi\sqrt{\left(\frac{L}{g}\right)}$$

where g is the acceleration of free fall.

Use your answer in (b) to determine the length L .

$$L = \dots\dots\dots \text{ m [2]}$$

- (d) Another pendulum consists of a sphere Q suspended by a thread. Spheres P and Q are identical. The thread attached to sphere Q is longer than the thread attached to sphere P.

Sphere Q is displaced and then released. The oscillations of sphere Q have the same amplitude as the oscillations of sphere P.

On Fig. 3.2, sketch the variation of the velocity v with displacement x for sphere Q. [2]

[Total: 9]

52. O/N/20/42/Q3

A simple pendulum consists of a metal sphere suspended from a fixed point by means of a thread, as illustrated in Fig. 3.1.

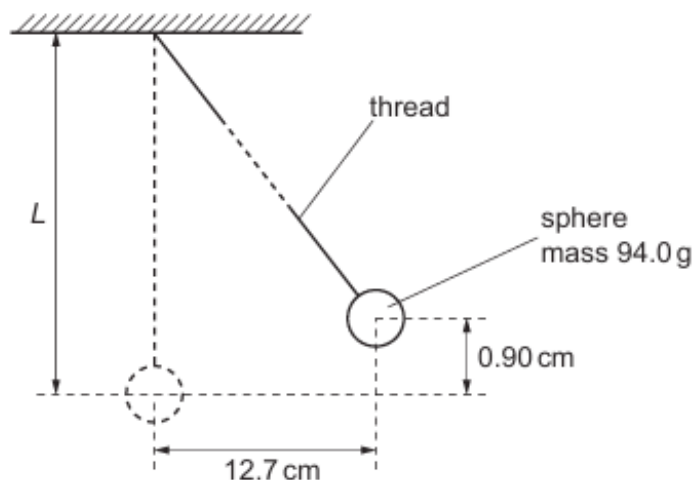


Fig. 3.1 (not to scale)

The sphere of mass 94.0 g is displaced to one side through a horizontal distance of 12.7 cm. The centre of gravity of the sphere rises vertically by 0.90 cm.

The sphere is released so that it oscillates. The sphere may be assumed to oscillate with simple harmonic motion.

- (a) State what is meant by *simple harmonic motion*.

.....

 [2]

- (b) (i) State the kinetic energy of the sphere when the sphere returns to the displaced position shown in Fig. 3.1.

kinetic energy = J [1]

- (ii) Calculate the total energy E_T of the oscillations.

E_T = J [2]

- (iii) Use your answer in (ii) to show that the angular frequency ω of the oscillations of the pendulum is 3.3 rad s^{-1} .

[2]

- (c) The period T of oscillation of the pendulum is given by the expression

$$T = 2\pi\sqrt{\left(\frac{L}{g}\right)}$$

where g is the acceleration of free fall and L is the length of the pendulum.

Use data from (b) to determine L .

$L = \dots\dots\dots \text{m}$ [3]

[Total: 10]

53. F/M/21/42/Q4

- (a) The defining equation of simple harmonic motion is

$$a = -\omega^2 x.$$

State the significance of the minus (–) sign in the equation.

.....
..... [1]

- (b) A trolley rests on a bench. Two identical stretched springs are attached to the trolley as shown in Fig. 4.1. The other end of each spring is attached to a fixed support.

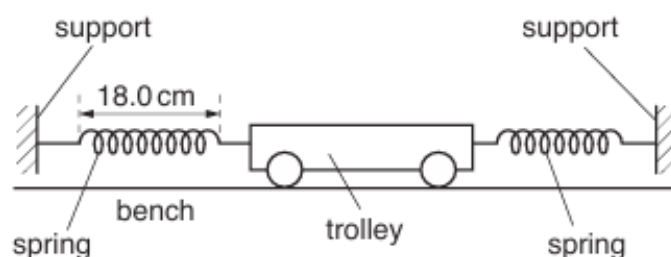


Fig. 4.1

The unstretched length of each spring is 12.0 cm. The spring constant of each spring is 8.0 N m^{-1} . When the trolley is in equilibrium the length of each spring is 18.0 cm.

The trolley is displaced 4.8 cm to one side and then released. Assume that resistive forces on the trolley are negligible.

- (i) Show that the resultant force on the trolley at the moment of release is 0.77 N.

- (ii) The mass of the trolley is 250g.

Calculate the maximum acceleration a of the trolley.

$$a = \dots\dots\dots \text{ms}^{-2} \quad [1]$$

- (iii) Use your answer in (ii) to determine the period T of the subsequent oscillation.

$$T = \dots\dots\dots \text{s} \quad [3]$$

- (iv) The experiment is repeated with an initial displacement of the trolley of 2.4 cm.

State and explain the effect, if any, this change has on the period of the oscillation of the trolley.

.....
.....
..... [2]

[Total: 9]

54. M/J/21/41/Q3

(a) State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

(b) A trolley of mass m is held on a horizontal surface by means of two springs. One spring is attached to a fixed point P. The other spring is connected to an oscillator, as shown in Fig. 3.1.

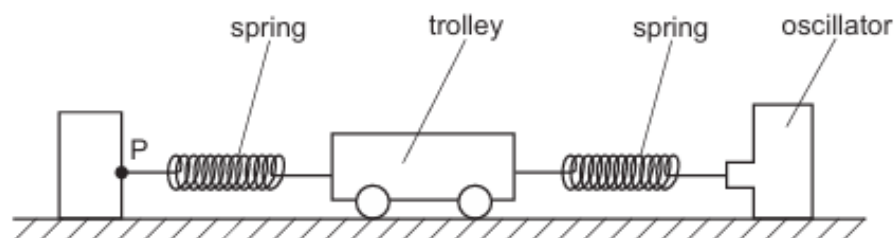


Fig. 3.1

The springs, each having spring constant k of 130 N m^{-1} , are always extended.

The oscillator is switched off. The trolley is displaced along the line of the springs and then released. The resulting oscillations of the trolley are simple harmonic.

The acceleration a of the trolley is given by the expression

$$a = -\left(\frac{2k}{m}\right)x$$

where x is the displacement of the trolley from its equilibrium position.

The mass of the trolley is 840 g .

Calculate the frequency f of oscillation of the trolley.

$f = \dots\dots\dots \text{ Hz}$ [3]

- (c) The oscillator in (b) is switched on. The frequency of oscillation of the oscillator is varied, keeping its amplitude of oscillation constant.

The amplitude of oscillation of the trolley is seen to vary. The amplitude is a maximum at the frequency calculated in (b).

- (i) State the name of the effect giving rise to this maximum.

..... [1]

- (ii) At any given frequency, the amplitude of oscillation of the trolley is constant.

Explain how this indicates that there are resistive forces opposing the motion of the trolley.

.....
.....
..... [2]

[Total: 8]

55. M/J/21/42/Q3

A U-shaped tube contains some liquid. The liquid column in each half of the tube has length L , as shown in Fig. 3.1.

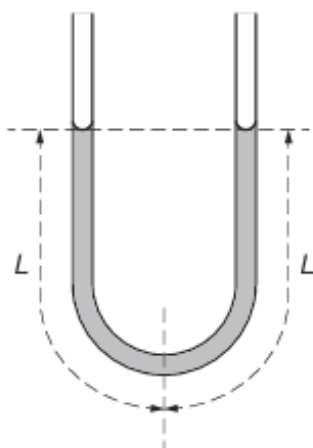


Fig. 3.1

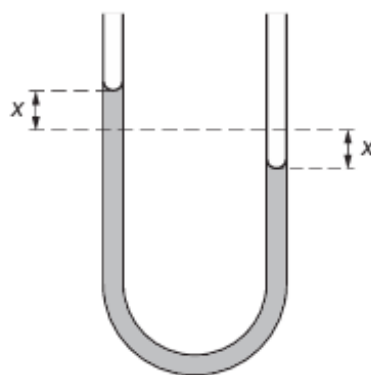


Fig. 3.2

The liquid columns are displaced vertically. The liquid then oscillates in the tube. The liquid levels are displaced from the equilibrium positions as shown in Fig. 3.2.

The acceleration a of the liquid in the tube is related to the displacement x by the expression

$$a = -\left(\frac{g}{L}\right)x$$

where g is the acceleration of free fall.

- (a) Explain how the expression shows that the liquid in the tube is undergoing simple harmonic motion.

.....

.....

.....

.....

..... [3]

- (b) The length L of each liquid column is 18 cm.

Determine the period T of the oscillations.

$T =$ s [3]

- (c) The oscillations of the liquid in the tube are damped.
In any one complete cycle of the oscillations, the amplitude decreases by 6.0% of its value at the beginning of the oscillation.

Determine the ratio

$$\frac{\text{energy of oscillations after 3 cycles}}{\text{initial energy of oscillations}}.$$

ratio = [3]

[Total: 9]

56. O/N/21/41/Q4

A trolley on a track is attached by springs to fixed blocks X and Y, as shown in Fig. 4.1. The track contains many small holes through which air is blown vertically upwards. This results in the trolley resting on a cushion of air rather than being in direct contact with the track.

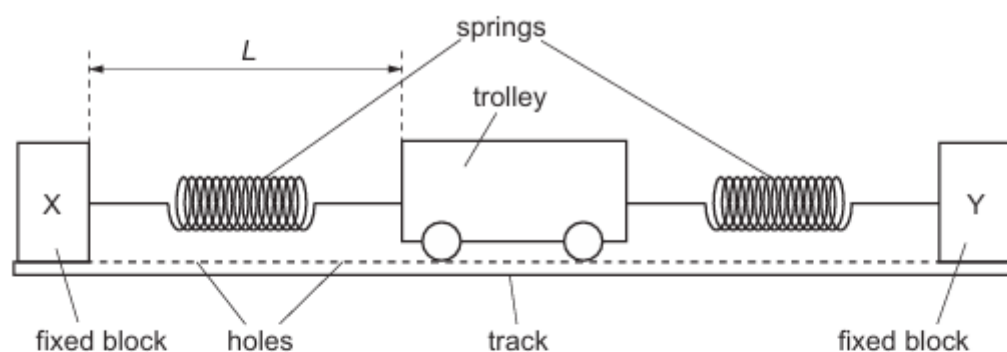


Fig. 4.1

The trolley is pulled to one side of its equilibrium position and then released so that it oscillates initially with simple harmonic motion. After a short time, the air blower is switched off. The variation with time t of the distance L of the trolley from block X is shown in Fig. 4.2.

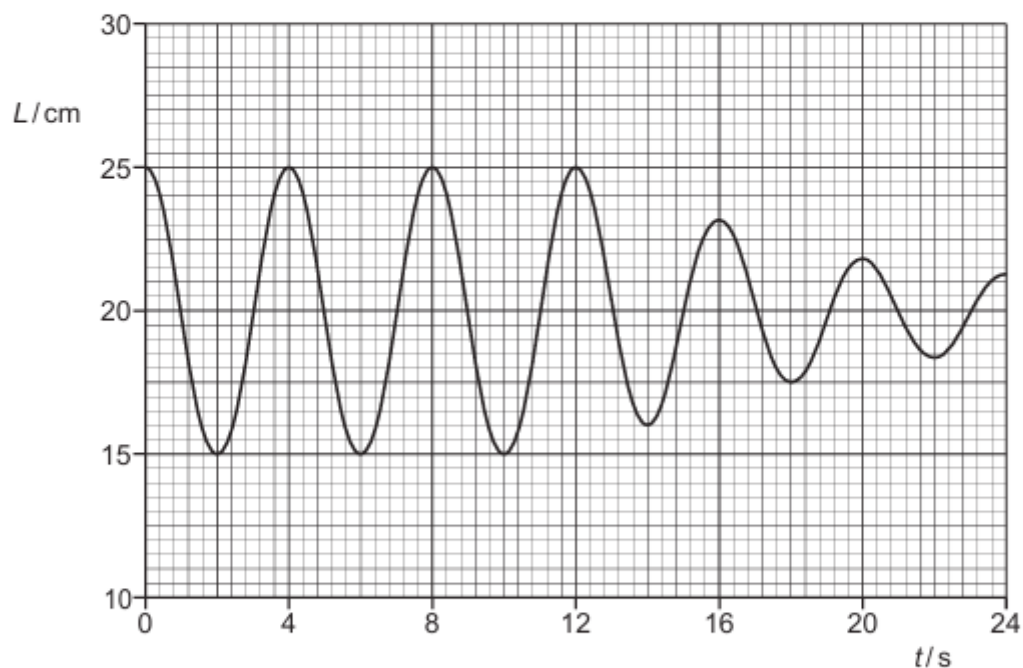


Fig. 4.2

(a) Use Fig. 4.2 to determine:

(i) the initial amplitude of the oscillations

amplitude = cm [1]

- (ii) the angular frequency ω of the oscillations

$$\omega = \dots\dots\dots \text{ rad s}^{-1} \quad [2]$$

- (iii) the maximum speed v_0 , in cm s^{-1} , of the oscillating trolley.

$$v_0 = \dots\dots\dots \text{ cm s}^{-1} \quad [2]$$

- (b) Apart from the quantities in (a), describe what may be deduced from Fig. 4.2 about the motion of the trolley between time $t = 0$ and time $t = 24$ s. No calculations are required.

.....

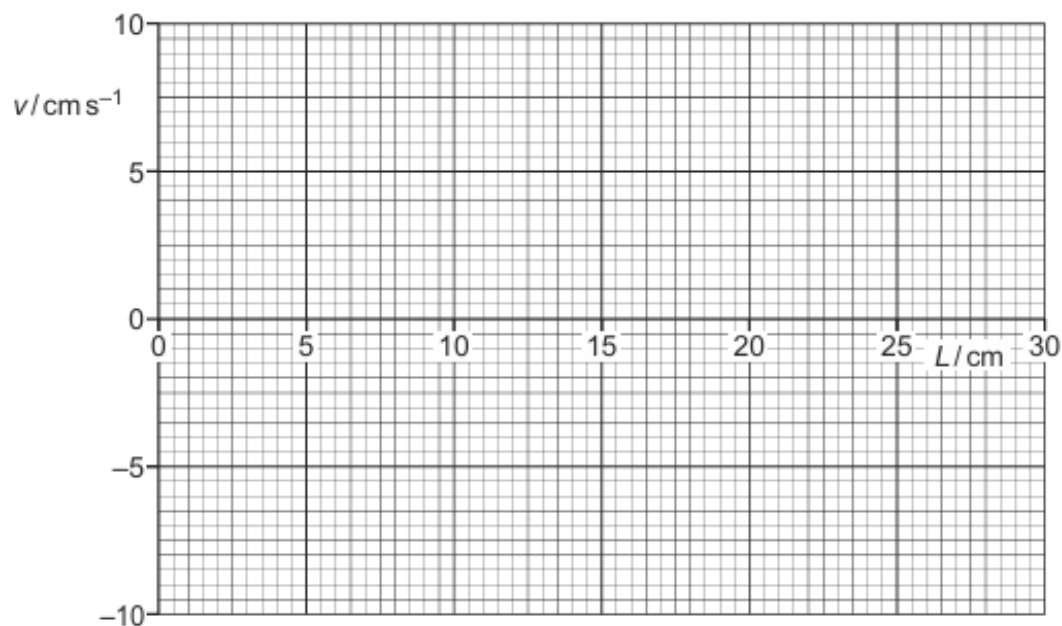
.....

.....

.....

..... [3]

- (c) On Fig. 4.3, sketch the variation with L of the velocity v of the trolley for its first complete oscillation.



57. O/N/21/42/Q4

A trolley on a smooth surface is attached by springs to fixed blocks as shown in Fig. 4.1.

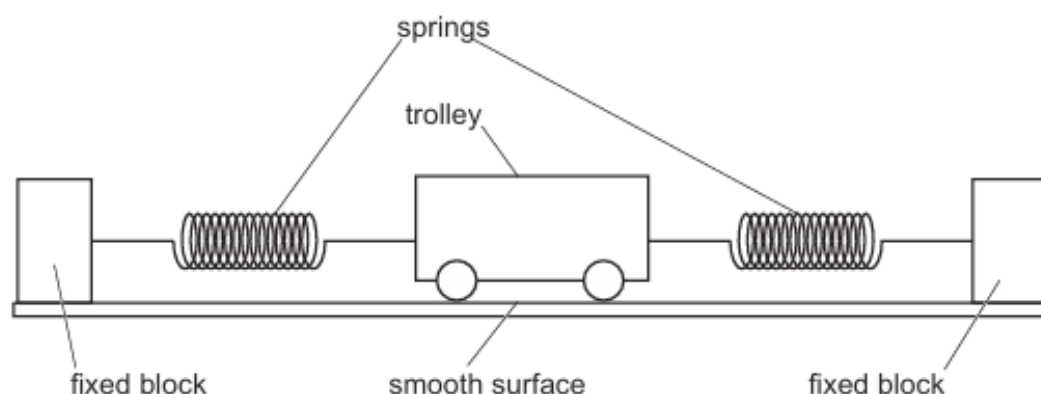


Fig. 4.1

The trolley oscillates horizontally about its equilibrium position with an amplitude of 12 cm. Fig. 4.2 shows the variation of the acceleration a of the trolley with displacement x from its equilibrium position. Friction between the trolley and the surface can be assumed to be negligible.

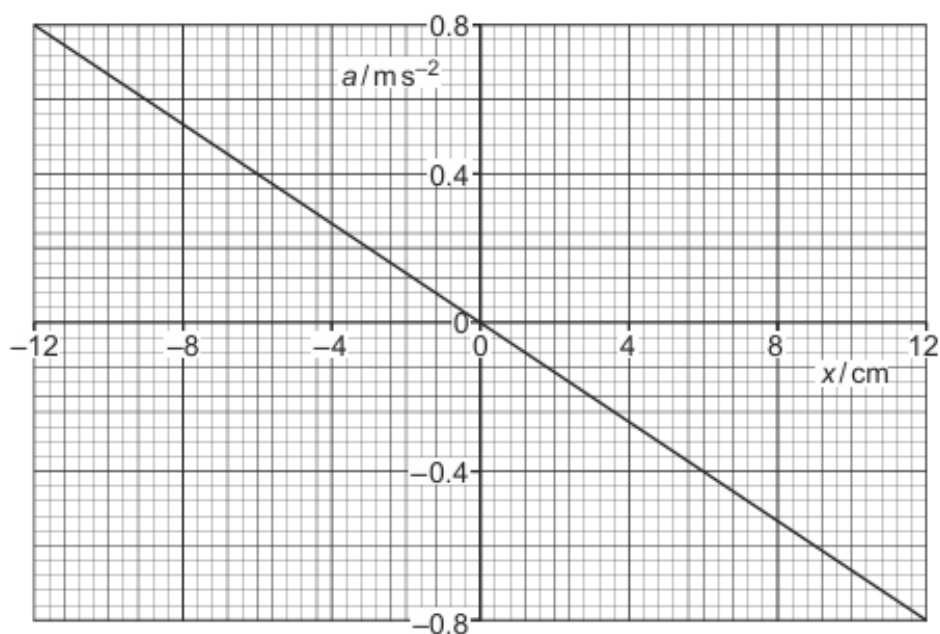


Fig. 4.2

- (a) Describe the features of the line in Fig. 4.2 that demonstrate that the motion of the trolley is simple harmonic.

.....

.....

..... [2]

(b) Use Fig. 4.2 to determine the period T of the oscillations of the trolley.

$T = \dots\dots\dots$ s [3]

(c) (i) On the line of the graph of Fig. 4.2, label with the letter P one point where the kinetic energy of the trolley is zero. [1]

(ii) On the line of the graph of Fig. 4.2, label with the letter Q an approximate position of one point where the kinetic energy of the trolley is equal to the potential energy stored in the springs. [1]

[Total: 7]

58. F/M/22/42/Q3

A small wooden block (cuboid) of mass m floats in water, as shown in Fig. 3.1.

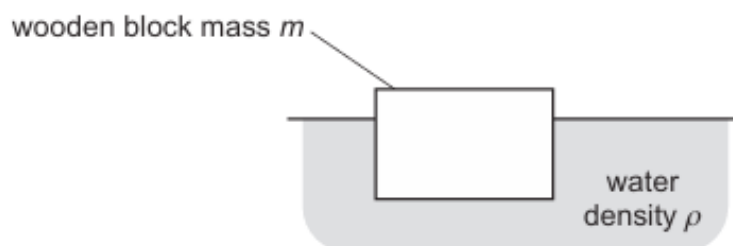


Fig. 3.1

The top face of the block is horizontal and has area A . The density of the water is ρ .

- (a) State the names of the two forces acting on the block when it is stationary.

..... [1]

- (b) The block is now displaced downwards as shown in Fig. 3.2 so that the surface of the water is higher up the block.

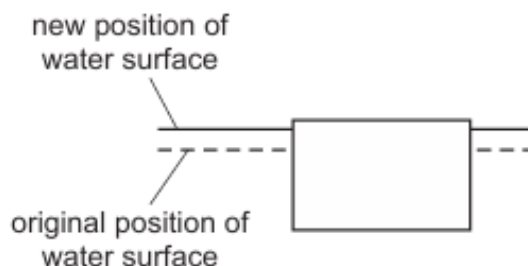


Fig. 3.2

State and explain the direction of the resultant force acting on the wooden block in this position.

.....
..... [1]

- (c) The block in (b) is now released so that it oscillates vertically.

The resultant force F acting on the block is given by

$$F = -A\rho g x$$

where g is the gravitational field strength and x is the vertical displacement of the block from the equilibrium position.

- (i) Explain why the oscillations of the block are simple harmonic.

.....
.....
..... [2]

- (ii) Show that the angular frequency ω of the oscillations is given by

$$\omega = \sqrt{\frac{A\rho g}{m}}.$$

[2]

- (d) The block is now placed in a liquid with a greater density. The block is displaced and released so that it oscillates vertically. The variation with displacement x of the acceleration a of the block is measured for the first half oscillation, as shown in Fig. 3.3.

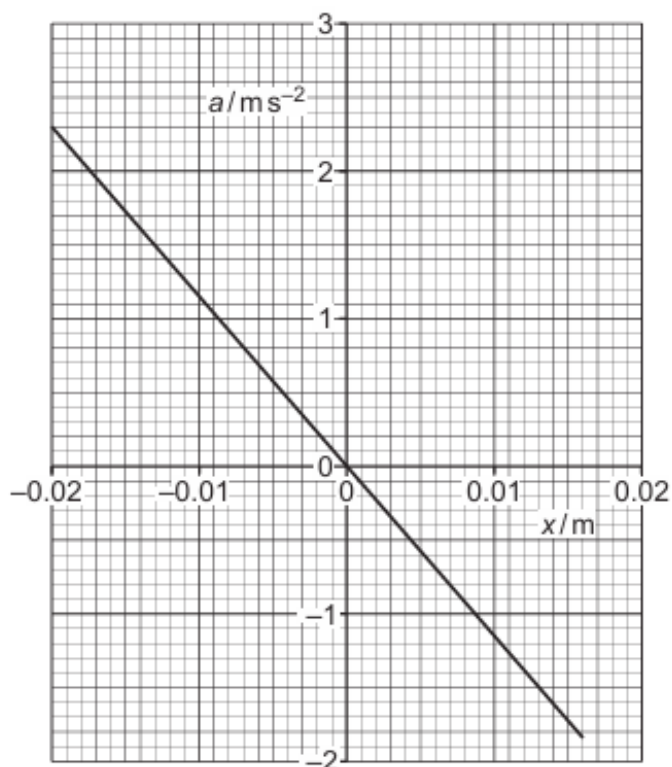


Fig. 3.3

- (i) Explain why the maximum negative displacement of the block is not equal to its maximum positive displacement.

.....

 [1]

- (ii) The mass of the block is 0.57 kg.

Use Fig. 3.3 to determine the decrease ΔE in energy of the oscillation for the first half oscillation.

$E = \dots\dots\dots$ J [3]

59. M/J/22/41/Q4

A pendulum consists of a bob (small metal sphere) attached to the end of a piece of string. The other end of the string is attached to a fixed point. The bob oscillates with small oscillations about its equilibrium position, as shown in Fig. 4.1.

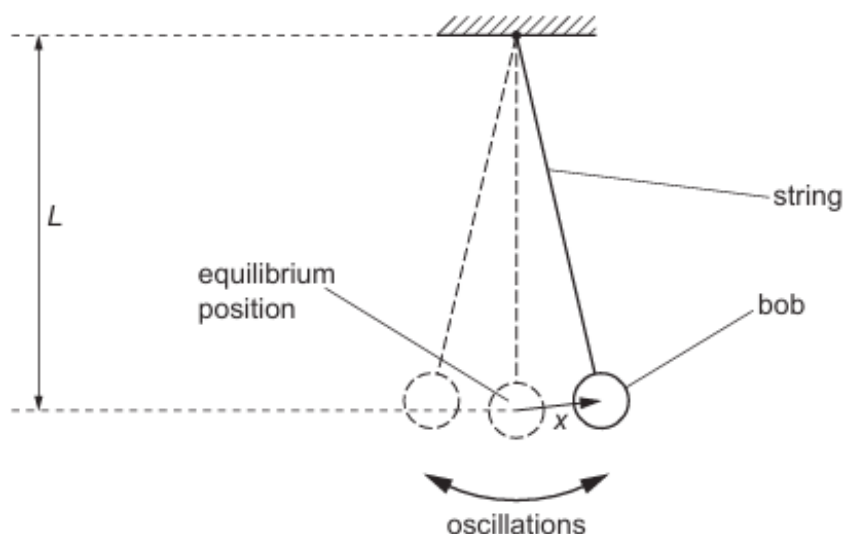


Fig. 4.1 (not to scale)

The length L of the pendulum, measured from the fixed point to the centre of the bob, is 1.24 m.

The acceleration a of the bob varies with its displacement x from the equilibrium position as shown in Fig. 4.2.

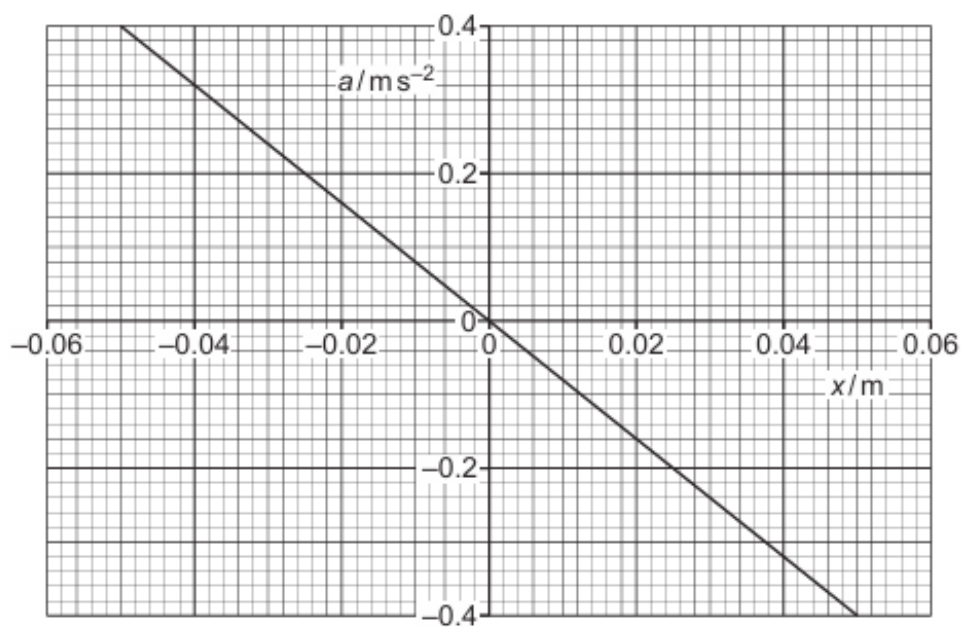


Fig. 4.2

- (a) State how Fig. 4.2 shows that the motion of the pendulum is simple harmonic.

.....

 [2]

- (b) (i) Use Fig. 4.2 to determine the angular frequency ω of the oscillations.

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) The angular frequency ω is related to the length L of the pendulum by

$$\omega = \sqrt{\frac{k}{L}}$$

where k is a constant.

Use your answer in (b)(i) to determine k . Give a unit with your answer.

$$k = \dots\dots\dots \text{unit} \dots\dots\dots [2]$$

- (c) While the pendulum is oscillating, the length of the string is increased in such a way that the total energy of the oscillations remains constant.

Suggest and explain the qualitative effect of this change on the amplitude of the oscillations.

.....

 [2]

[Total: 8]

60. M/J/22/42/Q4

- (a) State what is meant by resonance.

.....

 [2]

- (b) Fig. 4.1 shows a heavy pendulum and a light pendulum, both suspended from the same piece of string. This string is secured at each end to fixed points.

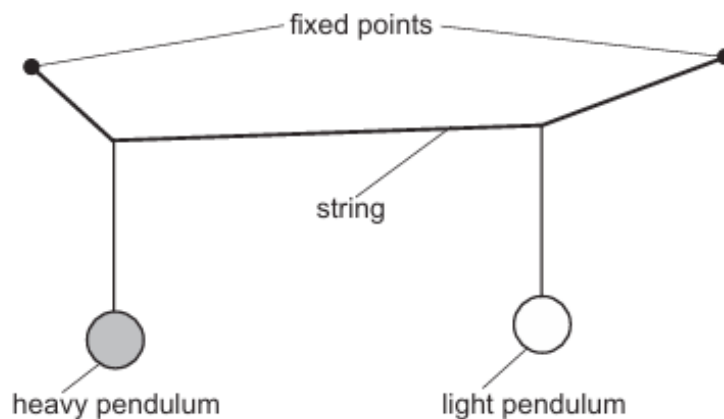
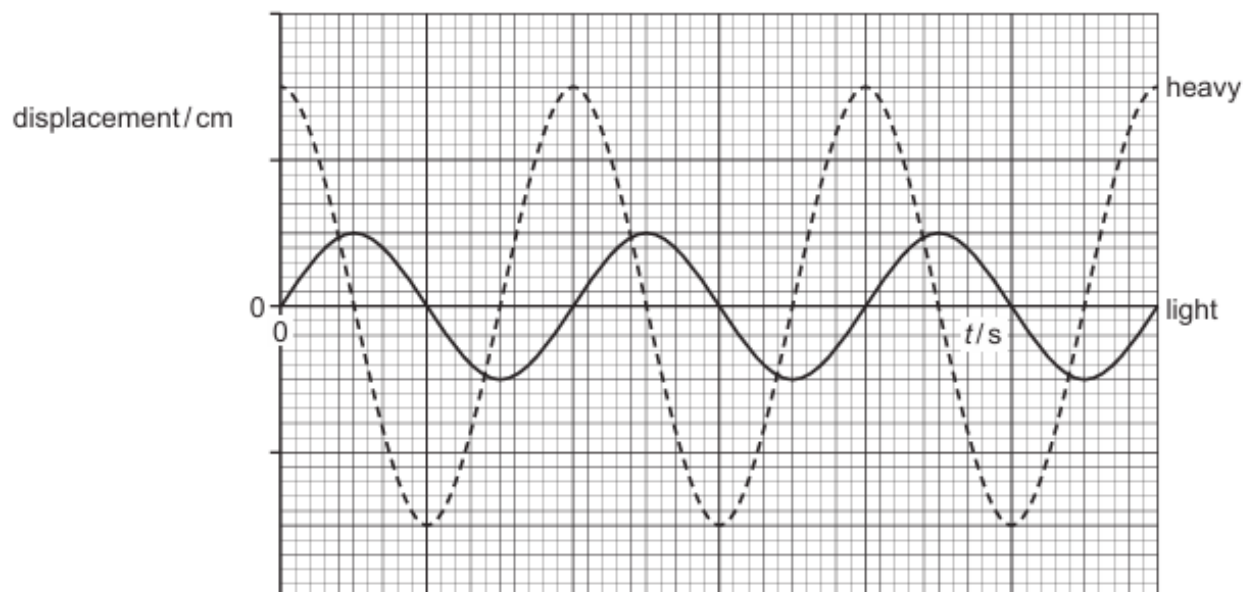


Fig. 4.1

Both pendulums have the same natural frequency. The heavy pendulum is set oscillating perpendicular to the plane of the diagram. As it oscillates, it causes the light pendulum to oscillate.

Fig. 4.2 shows the variation with time t of the displacements of the two pendulums for three oscillations.



The variation with t of the displacement x of the light pendulum is given by

$$x = 0.25 \sin 5.0\pi t$$

where x is in centimetres and t is in seconds.

- (i) Calculate the period T of the oscillations.

$$T = \dots\dots\dots \text{ s } [2]$$

- (ii) On Fig. 4.2, label both of the axes with the correct scales. Use the space below for any additional working that you need.

[2]

- (iii) Determine the magnitude of the phase difference ϕ between the oscillations of the light and heavy pendulums. Give a unit with your answer.

$$\phi = \dots\dots\dots \text{ unit } \dots\dots\dots [2]$$

[Total: 8]

61. O/N/22/41/Q3

An object is suspended from a spring that is attached to a fixed point as shown in Fig. 3.1.

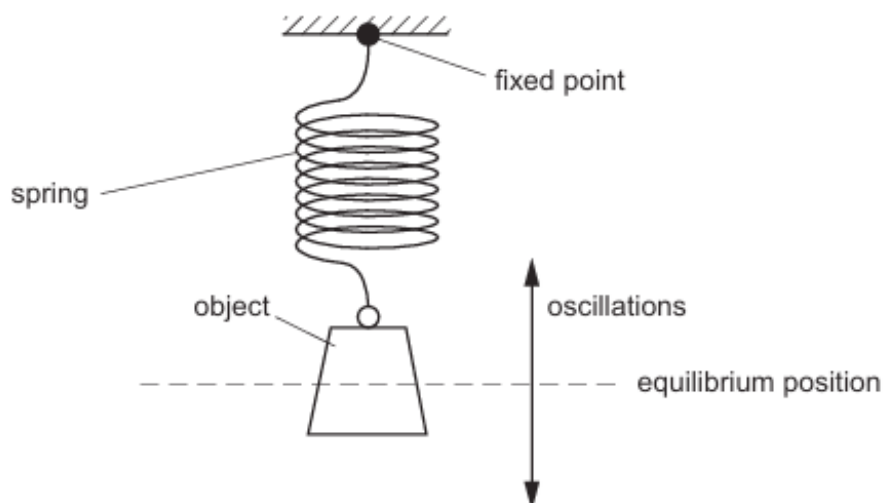


Fig. 3.1

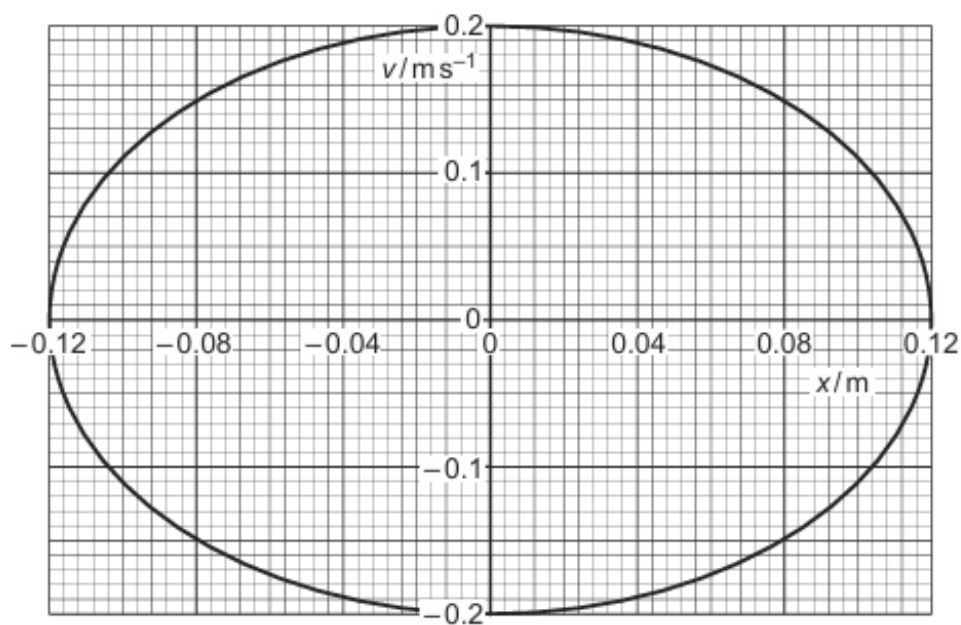
The object oscillates vertically with simple harmonic motion about its equilibrium position.

- (a) State the defining equation for simple harmonic motion. Identify the meaning of each of the symbols used to represent physical quantities.

.....

 [2]

- (b) The variation with displacement x from the equilibrium position of the velocity v of the object is shown in Fig. 3.2.



The variation with x of the potential energy E_p of the oscillations of the object is shown in Fig. 3.3.

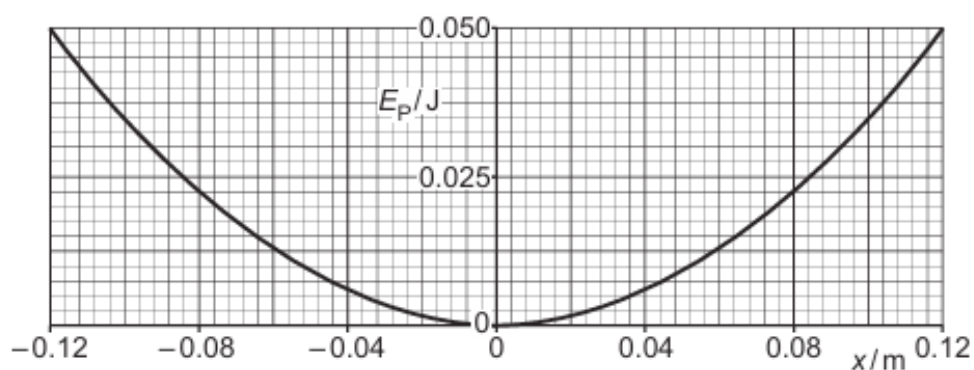


Fig. 3.3

Use Fig. 3.2 and Fig. 3.3 to:

- (i) determine the amplitude x_0 of the oscillations

$$x_0 = \dots\dots\dots \text{ m [1]}$$

- (ii) show that the angular frequency of the oscillations is 1.7 rad s^{-1}

[2]

- (iii) determine the mass M of the object.

$$M = \dots\dots\dots \text{ kg [2]}$$

(c) The oscillations of the object are now lightly damped.

(i) State what is meant by damping.

.....
.....
..... [2]

(ii) Assume that the damping does not change the angular frequency of the oscillations.

On Fig. 3.2, sketch the variation with x of v when the amplitude of the oscillations is 0.060 m. [2]

[Total: 11]

62. O/N/22/42/Q4

Fig. 4.1 shows the variation with time t of the height h above the ground of an object of mass 36 kg that is undergoing vertical simple harmonic motion.

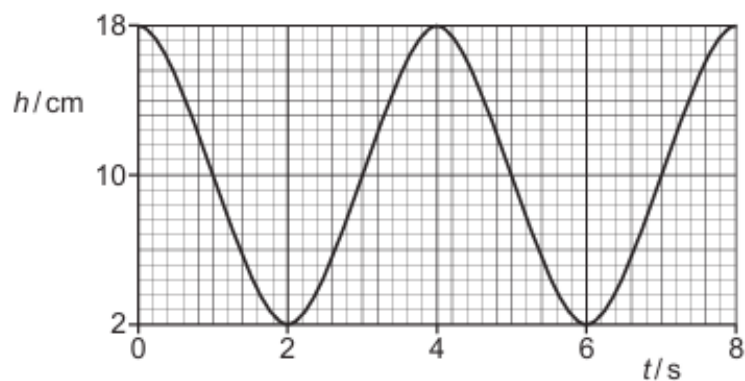


Fig. 4.1

(a) For the oscillations of the object:

(i) determine the amplitude x_0 , in cm

$$x_0 = \dots\dots\dots \text{ cm [1]}$$

(ii) show that the angular frequency ω is 1.6 rad s^{-1}

[2]

(iii) determine the total energy E .

$$E = \dots\dots\dots \text{ J [3]}$$

(b) On Fig. 4.2, sketch the variation with h of the kinetic energy E_K of the object.

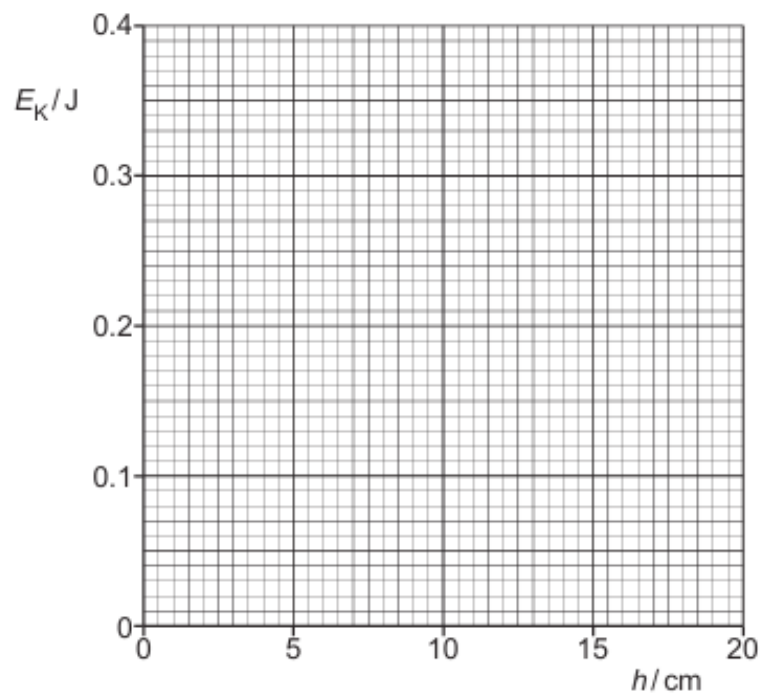


Fig. 4.2

[4]

[Total: 10]

63. F/M/23/42/Q3

An object is suspended from a vertical spring as shown in Fig. 3.1.

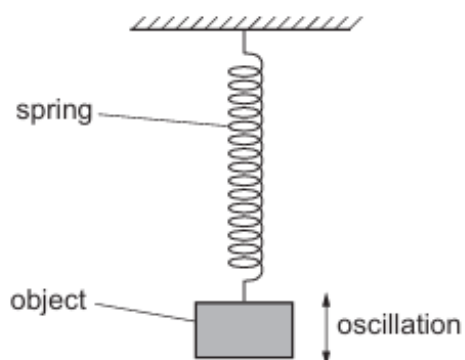
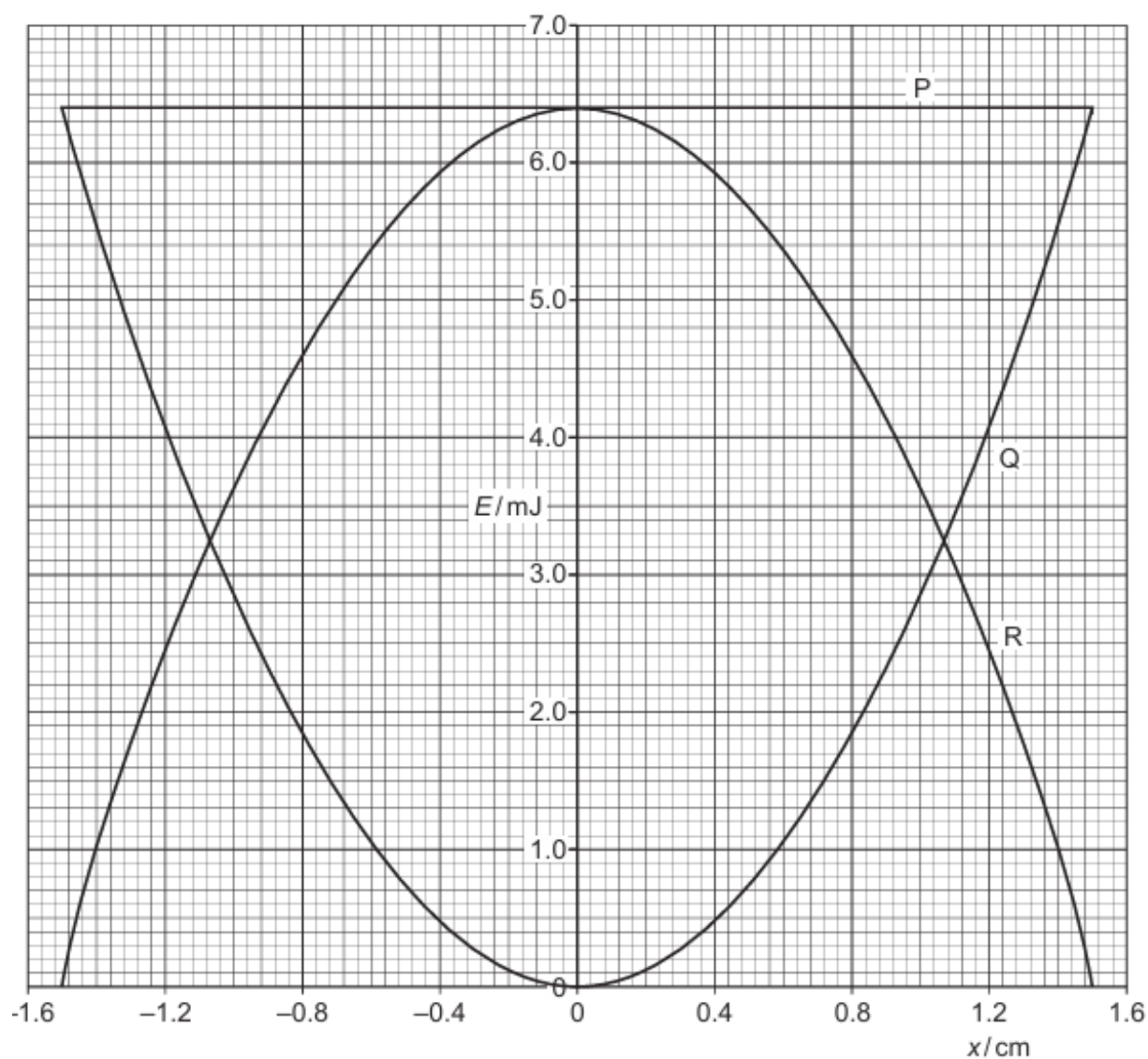


Fig. 3.1

The object is displaced vertically and then released so that it oscillates, undergoing simple harmonic motion.

Fig. 3.2 shows the variation with displacement x of the energy E of the oscillations.



The kinetic energy, the potential energy and the total energy of the oscillations are each represented by one of the lines P, Q and R.

- (a) State the energy that is represented by each of the lines P, Q and R.

P

Q

R

[2]

- (b) The object has a mass of 130 g.
Determine the period of the oscillations.

period = s [4]

- (c) (i) State the cause of damping.

.....

..... [1]

- (ii) A light card is attached to the object. The object is displaced with the same initial amplitude and then released. During each complete oscillation the total energy of the system decreases by 8.0% of the total energy at the start of that oscillation.

Determine the decrease in total energy, in mJ, of the system by the end of the first 6 complete oscillations.

energy lost = mJ [2]

64. M/J/23/42/Q4

A small steel sphere is oscillating vertically on the end of a spring, as shown in Fig. 4.1.

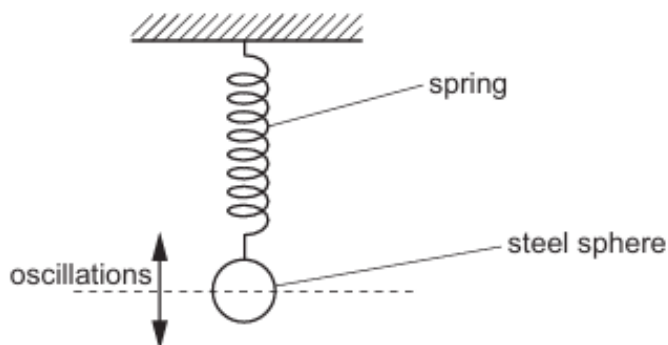


Fig. 4.1

The velocity v of the sphere varies with displacement x from its equilibrium position according to

$$v = \pm 9.7 \sqrt{(11.6 - x^2)}$$

where v is in cm s^{-1} and x is in cm.

- (a) (i) Calculate the frequency of the oscillations.

frequency = Hz [2]

- (ii) Show that the amplitude of the oscillations is 3.4 cm.

[1]

- (iii) Calculate the maximum acceleration a_0 of the sphere.

$a_0 = \dots \text{ms}^{-2}$ [2]

- (b) On Fig. 4.2, sketch the variation with x of the acceleration a of the sphere.

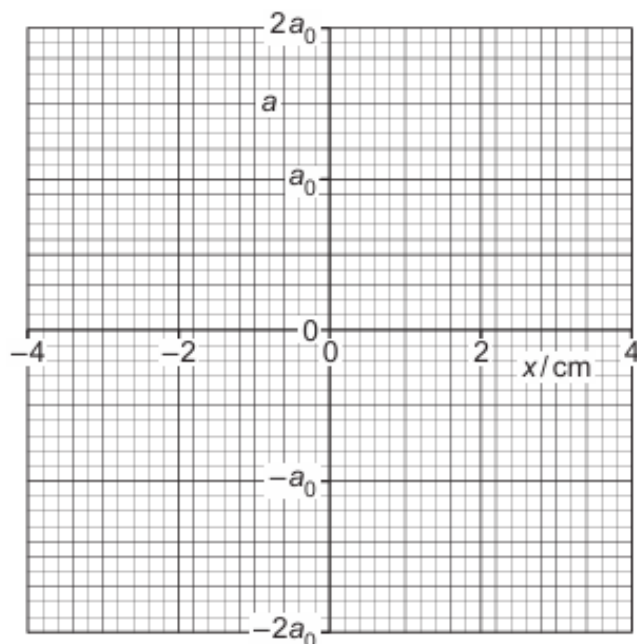


Fig. 4.2

[3]

- (c) Describe, without calculation, the interchange between the potential energy and the kinetic energy of the oscillations.

.....

.....

.....

.....

..... [3]

[Total: 11]

65. O/N/23/41/Q4

A heavy metal sphere of mass 0.81 kg is suspended from a string. The sphere is undergoing small oscillations from side to side, as shown in Fig. 4.1.

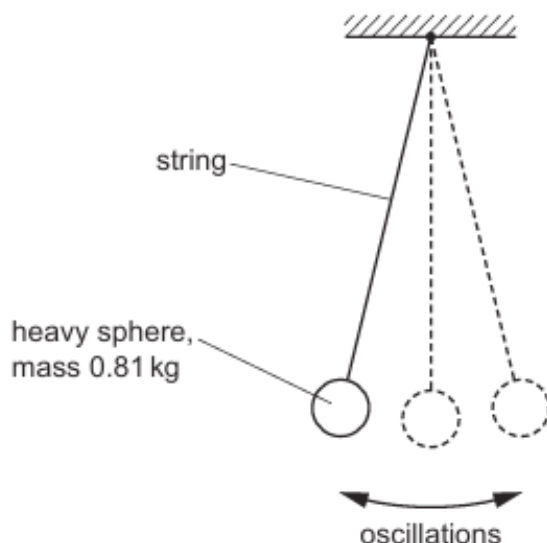


Fig. 4.1

The oscillations of the sphere may be considered to be simple harmonic with amplitude 0.036 m and period 3.0 s.

(a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) Calculate:

(i) the angular frequency of the oscillations

angular frequency = rad s^{-1} [2]

(ii) the total energy of the oscillations.

total energy = J [2]

- (c) The suspended sphere is now lowered into water. The sphere is given a sideways displacement of $+0.036\text{ m}$ from its equilibrium position and is then released at time $t = 0$. The water causes the motion of the sphere to be critically damped.

On Fig. 4.2, sketch the variation of the displacement x of the sphere from its equilibrium position with t from $t = 0$ to $t = 6.0\text{ s}$.

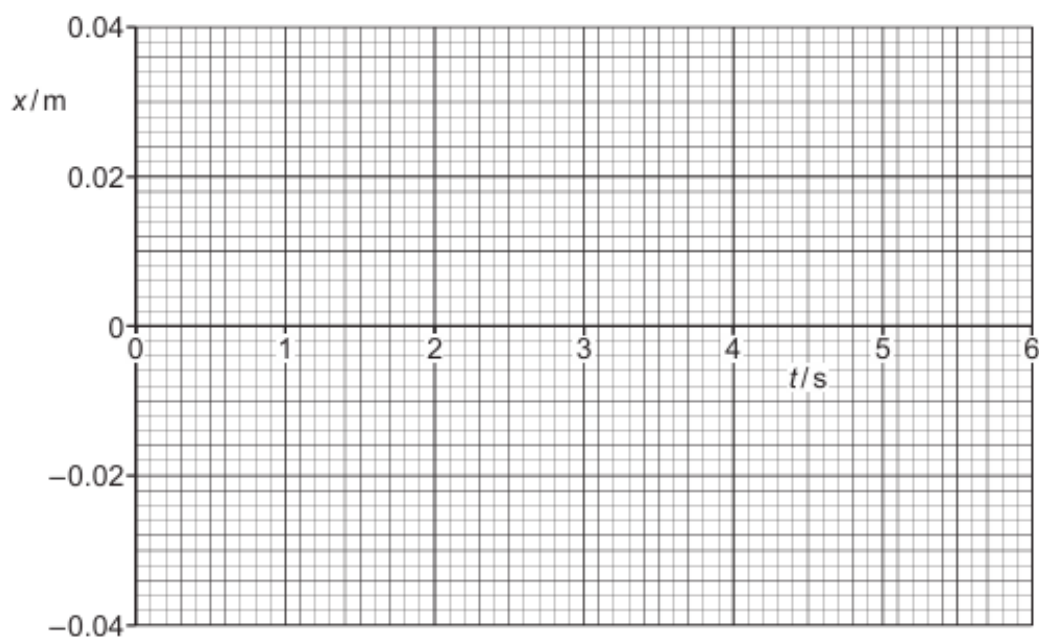


Fig. 4.2

[3]

[Total: 9]

66. O/N/23/42/Q4

An electron in a metal rod moves randomly about a mean position. When an alternating voltage is applied to the ends of the rod, the mean position can be considered to oscillate with simple harmonic motion along the axis of the rod. Fig. 4.1 shows the variation with time t of the displacement x of the mean position from a fixed point on the axis of the rod.

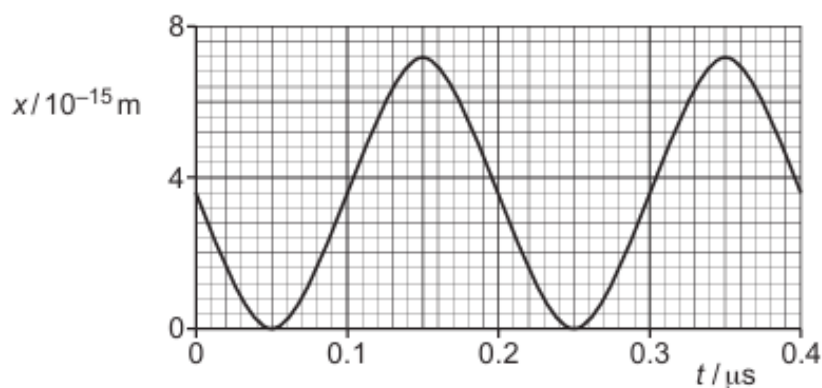


Fig. 4.1

- (a) (i) Determine the amplitude of the oscillations.

amplitude = m [1]

- (ii) Determine the angular frequency of the oscillations.

angular frequency = rad s^{-1} [1]

- (iii) Use your answers in (a)(i) and (a)(ii) to show that the maximum drift speed v_0 of the electron is $1.1 \times 10^{-7} \text{ m s}^{-1}$.

- (b) The rod has a cross-sectional area of 4.3 cm^2 and contains a number density of conduction electrons (charge carriers) of $8.5 \times 10^{28}\text{ m}^{-3}$.

All of the conduction electrons in the rod may be assumed to be oscillating in phase with, and with the same amplitude as, the oscillation shown in Fig. 4.1.

- (i) Use the information in (a)(iii) to calculate the magnitude I_0 of the maximum current in the rod.

$$I_0 = \dots\dots\dots\text{A} \quad [2]$$

- (ii) On Fig. 4.2, sketch the variation of the current I in the rod with time t between $t = 0$ and $t = 0.40\text{ }\mu\text{s}$.

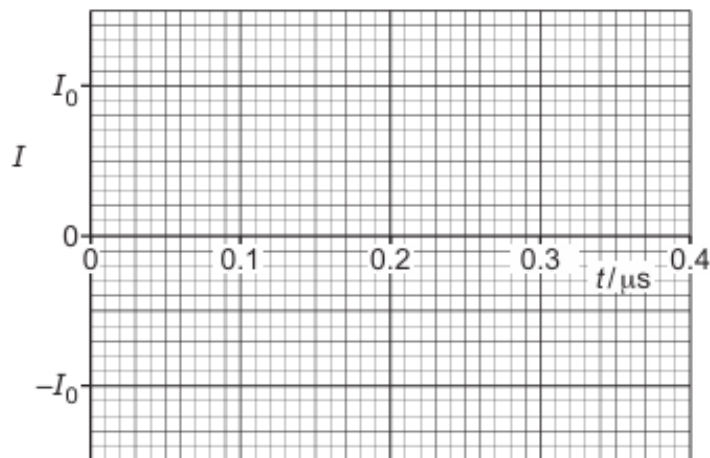


Fig. 4.2

[2]

- (iii) Use your answers in (a)(ii) and (b)(i) to determine an expression for I in terms of t , where I is in A and t is in s.

$$I = \dots\dots\dots \quad [2]$$

- (iv) Determine the root-mean-square (r.m.s.) current in the rod.

$$\text{r.m.s. current} = \dots\dots\dots\text{A} \quad [1]$$

ANSWERS

1.

- (a) (i) 0.8 cm B1
- (ii) (max.) kinetic energy = 2.56 mJ C1
 $v_{\text{(MAX)}} = \omega a$ C1
(max.) kinetic energy = $\frac{1}{2}m\omega^2 a^2$ or $\frac{1}{2}m\omega^2 (a^2 - x^2)$ C1
 $2.56 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times (0.8 \times 10^{-2})^2$ M1
 $\omega = 24.8 \text{ rad s}^{-1}$ C1
 $f = \omega/2\pi$ M1
= 4.0 Hz (3.95 Hz) A0
- (b) (i) line parallel to x-axis at 2.56 mJ B1
- (ii) 1 4.0 Hz B1
- 2 0.50 cm (allow $\pm 0.03 \text{ cm}$) B1

2.

- (a) (i) amplitude = 0.5 cm A1
- (ii) period = 0.8 s A1
- (b) (i) $\omega = 2\pi / T$ C1
= 7.85 rad s⁻¹
correct use of $v = \omega \sqrt{(x_0^2 - x^2)}$ B1
= $7.85 \times \sqrt{\{(0.5 \times 10^{-2})^2 - \{0.2 \times 10^{-2}\}^2\}}$
= 3.6 cm s⁻¹ A1
(if tangent drawn or clearly implied (B1)
3.6 $\pm$ 0.3 cm s⁻¹ (A2)
but allow 1 mark for $> \pm 0.3$ but $\leq \pm 0.6 \text{ cm s}^{-1}$)
- (ii) $d = 15.8 \text{ cm}$ A1
- (c) (i) (continuous) loss of energy / reduction in
amplitude (from the oscillating system) B1
caused by force acting in opposite direction to the motion / friction /
viscous forces B1
- (ii) same period / small increase in period B1
line displacement always less than that on Fig.3.2 (ignore first T/4) M1
peak progressively smaller A1

3.

- (a) acceleration / force (directly) proportional to displacement M1
 and *either* directed towards fixed point
 or acceleration & displacement in opposite directions A1

- (b) (i) maximum / minimum height / 8 mm above cloth / 14 mm below cloth B1

(ii) 1. $a = 11 \text{ mm}$ A1

2. $\omega = 2\pi f$ C1

$= 2\pi \times 4.5$

$= 28.3 \text{ rad s}^{-1}$ (do not allow 1 s.f.) A1

(c) (i) $v = \omega a$ C1

$= 28.3 \times 11 \times 10^{-3}$

$= 0.31 \text{ m s}^{-1}$ (do not allow 1 s.f.) A1

(ii) $v = \omega \sqrt{a^2 - y^2}$

$y = 3 \text{ mm}$ C1

$= 28.3 \times 10^{-3} \sqrt{11^2 - 3^2}$ C1

$= 0.30 \text{ m s}^{-1}$ (allow 1 s.f.) A1

4.

(a) (i) $(\theta =) \omega t$ (allow any subject if all terms given) B1

(ii) $(SQ =) r \sin \omega t$ (allow any subject if all terms given) B1

(b) this is the solution of the equation $a = -\omega^2 x$ M1

$a = -\omega^2 x$ is the (defining) equation of s.h.m. A1

(c) (i) $f = \omega / 2\pi$ C1

$= 4.7 / 2\pi$

$= 0.75 \text{ Hz}$ A1

(ii) $v = r\omega$ (r must be identified) C1

$= 4.7 \times 12$

$= 56 \text{ cm s}^{-1}$ A1

5.

- (a) (i) amplitude = 0.2 mm A1
- (ii) period = 1.2 ms C1
frequency = 830 Hz A1
- (b) (i) any two of zero, 0.6 ms and 1.2 ms A1
- (ii) any two of 0.3 ms, 0.9 ms, 1.5 ms A1
- (c) either $v = \omega x_0 = 2\pi f x_0$
 $= 2\pi \times 830 \times 0.2 \times 10^{-3} = 1.05 \text{ m s}^{-1}$
 or slope of graph = 1.0 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$) C1
 $E_K = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 2.5 \times 10^{-3} \times 1.05^2$ C1
 $= 1.4 \times 10^{-3} \text{ J}$ A1
- (d) (i) large / maximum amplitude of vibration B1
 when impressed frequency equals natural frequency of vibration B1
- (ii) e.g. metal panels on machinery vibrate / oscillate (M1)
 motor in machine impresses frequency on panel (A1)
 e.g. car suspension system vibrates / oscillates (M1)
 going over bumps would give large amplitude vibrations (A1)
 any feasible example, M1 + A1

6.

- (a) straight line through origin B1
 negative gradient B1
- (b) $a = -\omega^2 x$ and $\omega = 2\pi f$ C1
 $750 = (2\pi f)^2 \times 0.3 \times 10^{-3}$ C1
 $f = 250 \text{ Hz}$ A1
- (c) straight line between (-0.3, +190) and (+0.3, -190) A2
 (allow 1 mark for end of line incorrect by one grid square or line does not extend to $\pm 0.3 \text{ mm}$)

7.

- (a) (i) to-and-fro / backward and forward motion (between two limits) B1
- (ii) no energy loss or gain / no external force acting / constant energy B1
- (iii) acceleration directed towards a fixed point B1
acceleration proportional to distance from the fixed point / displacement B1
- (b) acceleration is constant (magnitude) M1
so cannot be s.h.m. A1

8.

- (a) (i) reduction in energy (of the oscillations) (B1)
reduction in amplitude / energy of oscillations (B1)
due to force (always) opposing motion / resistive forces (B1)
any two of the above, max 2
- (ii) amplitude is decreasing (very) gradually / oscillations would continue (for a long time) / many oscillations M1
light damping A1
- (b) (i) frequency = $1 / 0.3$
= 3.3 Hz A1
allow points taken from time axis giving $f = 3.45$ Hz
- (ii) energy = $\frac{1}{2}mv^2$ and $v = \omega a$ C1
= $\frac{1}{2} \times 0.065 \times (2\pi/0.3)^2 \times (1.5 \times 10^{-2})^2$ M1
= 3.2 mJ A0
- (c) amplitude reduces exponentially / does not decrease linearly M1
so will be not be 0.7 cm A1

9.

- (a) (i) resonance B1
- (ii) amplitude 16 mm and frequency 4.6 Hz A1
- (b) (i) $a = (-)\omega^2 x$ and $\omega = 2\pi f$ C1
 $a = 4\pi^2 \times 4.6^2 \times 16 \times 10^{-3}$ C1
 $= 13.4 \text{ ms}^{-2}$ A1
- (ii) $F = ma$ C1
 $= 150 \times 10^{-3} \times 13.4$
 $= 2.0 \text{ N}$ A1
- (c) line always 'below' given line and never zero M1
peak is at 4.6 Hz (or slightly less) and flatter A1

10.

- (a) (i) 8.0 cm A1
- (ii) $2\pi f = 220$ C1
 $f = 35$ (condone unit) A1
- (iii) line drawn mid-way between AB and CD (allow $\pm 2 \text{ mm}$) B1
- (iv) $v = \omega a$ C1
 $= 220 \times 4.0$
 $= 880 \text{ cm s}^{-1}$ A1
- (b) (i) 1. line drawn 3 cm above AB (allow $\pm 2 \text{ mm}$) B1
2. arrow pointing upwards B1
- (ii) 1. line drawn 3 cm above AB (allow $\pm 2 \text{ mm}$) B1
2. arrow pointing downwards B1
- (iii) $v = \omega \sqrt{a^2 - x^2}$
 $= 220 \times \sqrt{4.0^2 - 2.0^2}$ C1
 $= 760 \text{ cm s}^{-1}$ A1
(incorrect value for x , 0/2 marks)

11.		
(a)	acceleration / force proportional to displacement from a fixed point	M1
	acceleration / force (always) directed towards that fixed point / in opposite direction to displacement	A1
(b) (i)	$A\rho g / m$ is a constant and so acceleration proportional to x	B1
	negative sign shows acceleration towards a fixed point / in opposite direction to displacement	B1
(ii)	$\omega^2 = (A\rho g / m)$	C1
	$\omega = 2\pi f$	C1
	$(2 \times \pi \times 1.5)^2 = (\{4.5 \times 10^{-4} \times 1.0 \times 10^3 \times 9.81\} / m)$	C1
	$m = 50 \text{ g}$	A1
12.		
(a) (i)	<u>amplitude</u> remains constant	B1
	(ii) <u>amplitude</u> decreases gradually	M1
	light damping	A1
(iii)	period = 0.80 s	C1
	frequency = 1.25 Hz (<i>period not 0.8 s, then 0/2</i>)	A1
(b) (i)	(induced) e.m.f. is proportional to	M1
	rate of change/cutting of (magnetic) flux (linkage)	A1
(ii)	a current is induced in the coil	M1
	as magnet moves in coil	A1
	current in resistor gives rise to a heating effect	M1
	thermal energy is derived from energy of oscillation of the magnet	A1
13.		
(a)	acceleration proportional to displacement/distance from fixed point	M1
	and in opposite directions/directed towards fixed point	A1
(b)	energy = $\frac{1}{2}m\omega^2x_0^2$ and $\omega = 2\pi f$	C1
	= $\frac{1}{2} \times 5.8 \times 10^{-3} \times (2\pi \times 4.5)^2 \times (3.0 \times 10^{-3})^2$	C1
	= $2.1 \times 10^{-5} \text{ J}$	A1
(c) (i)	at maximum displacement	M1
	above rest position	A1
(ii)	acceleration = $(-)\omega^2x_0$ and acceleration = 9.81 or g	C1
	$9.81 = (2\pi \times 4.5)^2 \times x_0$	
	$x_0 = 1.2 \times 10^{-2} \text{ m}$	A1

14.

(a) $a = (-)\omega^2 x$ and $\omega = 2\pi/T$ C1
 $T = 0.60 \text{ s}$ C1
 $a = (4\pi^2 \times 2.0 \times 10^{-2}) / (0.6)^2$
 $= 2.2 \text{ ms}^{-2}$ A1

(b) sinusoidal wave with all values positive B1
 all values positive, all peaks at E_K and energy = 0 at $t = 0$ B1
 period = 0.30 s B1

15.

(a) energy = $\frac{1}{2}m\omega^2 a^2$ and $\omega = 2\pi f$ C1
 $= \frac{1}{2} \times 37 \times 10^{-3} \times (2\pi \times 3.5)^2 \times (2.8 \times 10^{-2})^2$ M1
 $= 7.0 \times 10^{-3} \text{ J}$ A0
 (allow $2\pi \times 3.5$ shown as 7π)

Energy = $\frac{1}{2}mv^2$ and $v = r\omega$ (C1)
 Correct substitution (M1)
 Energy = $7.0 \times 10^{-3} \text{ J}$ (A0)

(b) $E_K = E_P$
 $\frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}m\omega^2 x^2$ or E_K or $E_P = 3.5 \text{ mJ}$ C1
 $x = a/\sqrt{2} = 2.8/\sqrt{2}$ or $E_K = \frac{1}{2}m\omega^2(a^2 - x^2)$ or $E_P = \frac{1}{2}m\omega^2 x^2$ C1
 $= 2.0 \text{ cm}$ A1
 (E_K or $E_P = 7.0 \text{ mJ}$ scores 0/3)

Allow: $k = 17.9$ (C1)
 $E = \frac{1}{2}kx^2$ (C1)
 $x = 2.0 \text{ cm}$ (A1)

(c) (i) graph: horizontal line, y-intercept = 7.0 mJ with end-points of line at +2.8 cm and -2.8 cm B1

(ii) graph: reasonable curve B1
 with maximum at (0, 7.0) end-points of line at (-2.8, 0) and (+2.8, 0) B1

(iii) graph: inverted version of (ii) M1
 with intersections at (-2.0, 3.5) and (+2.0, 3.5) A1
 (Allow marks in (iii), but not in (ii), if graphs K & P are not labelled)

(d) gravitational potential energy B1

- 16.**
- (a) straight line through origin M1
 shows acceleration proportional to displacement A1
 negative gradient M1
 shows acceleration and displacement in opposite directions A1
- (b) (i) 2.8 cm A1
- (ii) *either* gradient = ω^2 and $\omega = 2\pi f$ or $a = -\omega^2 x$ and $\omega = 2\pi f$ C1
 gradient = $13.5 / (2.8 \times 10^{-2}) = 482$
 $\omega = 22 \text{ rad s}^{-1}$ C1
 frequency = $(22/2\pi =) 3.5 \text{ Hz}$ A1
- (c) e.g. lower spring may not be extended
 e.g. upper spring may exceed limit of proportionality / elastic limit
 (*any sensible suggestion*) B1
- 17.**
- (a) (i) 1. 0.1 s, 0.3 s, 0.5 s, etc (*any two*) A1
2. either 0, 0.4 s, 0.8 s, 1.2 s
 or
 0.2 s, 0.6 s, 1.0 s (*any two*) A1
- (ii) period = 0.4 s C1
 frequency = $(1/0.4 =) 2.5 \text{ Hz}$ A1
- (iii) phase difference = 90° or $\frac{1}{2} \pi \text{ rad}$ B1
- (b) frequency = 2.4 – 2.5 Hz B1
- (c) e.g. attach sheet of card to trolley M1
 increases damping / frictional force A1
 e.g. reduce oscillator amplitude (M1)
 reduces power/energy input to system (A1)

18.

(a) (i) 1. amplitude = 1.7 cm A1

2. period = 0.36 cm C1
frequency = $1/0.36$
= 2.8 Hz A1

(ii) $a = (-)\omega^2 x$ and $\omega = 2\pi/T$ C1
acceleration = $(2\pi/0.36)^2 \times 1.7 \times 10^{-2}$ M1
= 5.2 m s^{-2} A0

(b) graph: straight line, through origin, with negative gradient M1
from $(-1.7 \times 10^{-2}, 5.2)$ to $(1.7 \times 10^{-2}, -5.2)$ A1
(if scale not reasonable, do not allow second mark)

(c) either kinetic energy = $\frac{1}{2}m\omega^2(x_0^2 - x^2)$ B1
or potential energy = $\frac{1}{2}m\omega^2 x^2$ and potential energy = kinetic energy C1
 $\frac{1}{2}m\omega^2(x_0^2 - x^2) = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$ or $\frac{1}{2}m\omega^2 x^2 = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$
 $x_0^2 = 2x^2$
 $x = x_0 / \sqrt{2} = 1.7 / \sqrt{2}$
= 1.2 cm A1

19.

(a) (i) $\omega = 2\pi / T$
= $2\pi / 0.69$ C1
= 9.1 rad s^{-1} A1
(allow use of $f = 1.5 \text{ Hz}$ to give $\omega = 9.4 \text{ rad s}^{-1}$)

(ii) 1. $x = 2.1 \cos 9.1t$ B1
2.1 and 9.1 numerical values B1
use of cos

2. $v_0 = 2.1 \times 10^{-2} \times 9.1$ (allow ecf of value of x_0 from (ii)1.) B1
= 0.19 m s^{-1} B1
 $v = v_0 \sin 9.1t$ (allow $\cos 9.1t$ if $\sin$ used in (ii)1.)

(b) energy = either $\frac{1}{2}mv_0^2$ or $\frac{1}{2}m\omega^2 x_0^2$ C1
= either $\frac{1}{2} \times 0.078 \times 0.19^2$ or $\frac{1}{2} \times 0.078 \times 9.1^2 \times (2.1 \times 10^{-2})^2$ A1
= $1.4 \times 10^{-3} \text{ J}$

20.

- (a) (i) any two from 0.3(0) s, 0.9(0) s, 1.50 s (*allow 2.1 s etc.*) B1
- (ii) *either* $v = \omega x$ and $\omega = 2\pi/T$ C1
 $v = (2\pi/1.2) \times 1.5 \times 10^{-2}$ M1
 $= 0.079 \text{ m s}^{-1}$ A0
or gradient drawn clearly at a correct position (C1)
working clear (M1)
to give $(0.08 \pm 0.01) \text{ m s}^{-1}$ (A0)
- (b) (i) sketch: curve from $(\pm 1.5, 0)$ passing through $(0, 25)$ M1
reasonable shape (*curved with both intersections between*
 $y = 12.0 \rightarrow 13.0$) A1
- (ii) at max. amplitude potential energy is total energy B1
total energy = 4.0 mJ B1

21.

- (a) kinetic (energy)/KE/ E_k B1
- (b) *either* change in energy = 0.60 mJ
or max E proportional to (amplitude)²/equivalent numerical working B1
new amplitude is 1.3 cm B1
change in amplitude = 0.2 cm B1

22.

- (a) acceleration/force proportional to displacement (from a fixed point) M1
either acceleration and displacement in opposite directions
or acceleration always directed towards a fixed point A1
- (b) (i) g and r are constant so a is proportional to x B1
negative sign shows a and x are in opposite directions B1
- (ii) $\omega^2 = g/r$ and $\omega = 2\pi/T$ C1
 $\omega^2 = 9.8/0.28$
 $= 35$ C1
 $T = 2\pi/\sqrt{35} = 1.06 \text{ s}$
time interval $\tau = 0.53 \text{ s}$ A1
- (c) sketch: time period constant (or increases very slightly) M1
drawn line always 'inside' given loops A1
successive decrease in peak height A1

- 23.
- (a) acceleration/force proportional to displacement (from a fixed point) M1
either acceleration and displacement in opposite directions A1
or acceleration always directed towards a fixed point
- (b) (i) zero & 0.625 s *or* 0.625 s & 1.25 s *or* 1.25 s & 1.875 s *or* 1.875 s & 2.5 s A1
- (ii) 1. $\omega = 2\pi/T$ and $v_0 = \omega x_0$ C1
 $\omega = 2\pi/1.25$
 $= 5.03 \text{ rad s}^{-1}$ C1
- $v_0 = 5.03 \times 3.2$
 $= 16.1 \text{ cm s}^{-1}$ (allow 2 s.f.) A1
2. $v = \omega\sqrt{(x_0^2 - x^2)}$
- either* $\frac{1}{2}\omega a = \omega\sqrt{(x_0^2 - x^2)}$ *or* $\frac{1}{2} \times 16.1 = 5.03\sqrt{(3.2^2 - x^2)}$ C1
 $x_0^2/4 = x_0^2 - x^2$ $2.58 = 3.2^2 - x^2$
 $x = 2.8 \text{ cm}$ $x = 2.8 \text{ cm}$ A1
- (c) sketch: loop with origin at its centre M1
correct intercepts & shape based on (b)(ii) A1
- 24.
- (a) (i) frequency at which object is made to vibrate/oscillate B1
- (ii) frequency at which object vibrates when free to do so B1
- (iii) maximum amplitude of vibration of oscillating body B1
when forced frequency equals natural frequency (of vibration) B1
- (b) e.g. vibration of quartz/piezoelectric crystal (*what is vibrating*) M1
either for accurate timing
or maximise amplitude of ultrasound waves (*why it is useful*) A1
- (c) e.g. vibrating metal panels (*what is vibrating*) M1
either place strengthening struts across the panel
or change shape/area of panel (*how it is reduced*) A1
- 25.
- (a) displacement (directly) proportional to acceleration/force M1
either displacement and acceleration in opposite directions
or acceleration (always) towards a (fixed) point A1

- (b) (i) $\frac{1}{3}\pi$ rad or 1.05 rad (*allow 60° if unit clear*) A1
- (ii) $a_0 = -\omega^2 x_0$
 $= (-) (2\pi/1.2)^2 \times 0.030$ C1
 $= (-) 0.82 \text{ m s}^{-2}$ A1
(special case: using oscillator P gives $x_0 = 1.7 \text{ cm}$ and $a_0 = 0.47 \text{ m s}^{-1}$ for 1/2)
- (iii) max. energy $\propto x_0^2$
ratio $= 3.0^2/1.7^2$ C1
 $= 3.1$ (*at least 2 s.f.*) A1
(if has inverse ratio but has stated max. energy $\propto x_0^2$ then allow 1/2)
- (c) graph: straight line through (0,0) with negative gradient M1
correct end-points $(-3.0, +0.82)$ and $(+3.0, -0.82)$ A1
- 26.
- (a) free: (body oscillates) without any loss of energy/no resistive forces/no external forces applied B1
forced: continuous energy input (required)/body is made to vibrate by an (external) periodic force/driving oscillator B1
- (b) (i) idea of resonance B1
maximum amplitude at natural frequency B1
frequency = 2.1 Hz (*allow 2.08 to 2.12 Hz*) B1
- (ii) peak not very sharp/amplitude not infinite so frictional forces are present B1
- (c) $v = \omega x_0$
 $= 2\pi \times 2.1 \times 4.7 \times 10^{-2}$ (*allow ECF from (b)(i)*) C1
 $= 0.62 \text{ m s}^{-1}$ A1
- 27.
- (a) acceleration/force proportional to distance from a fixed point or displacement M1
either acceleration/force and displacement in opposite directions
or acceleration/force (always) directed towards a fixed point/mean position/equilibrium position A1
- (b) $h\rho g = Mg/A$ B1
 $h \times 790 \times 4.9 \times 10^{-4} = 70 \times 10^{-3}$ leading to $h = 0.18 \text{ m}$ or 18 cm A1

(c) (i) 1. $\omega^2 = (790 \times 4.9 \times 10^{-4} \times 9.81) / (70 \times 10^{-3})$ C1
 $= 54.25$

$\omega = 7.37 \text{ (rad s}^{-1}\text{)}$
 period $(= 2\pi / \omega) = 0.85 \text{ s}$ C1

$t_1 = 0.43 \text{ s}$ A1

2. $t_3 = 1.28 \text{ s (allow 2 s.f.)}$ A1

(ii) energy of peak $= \frac{1}{2} M \omega^2 x_0^2$ B1

change $= \frac{1}{2} \times 70 \times 10^{-3} \times 54.25 \{ (2.2 \times 10^{-2})^2 - (1.0 \times 10^{-2})^2 \}$ C1
 $= 7.3 \times 10^{-4} \text{ J}$ A1

28.

(a) amplitude $= 1.8 \text{ cm}$ and period $= 0.30 \text{ s}$ A1

(b) $E_K = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ or $E_K = \frac{1}{2} m v^2$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$ C1
 $= \frac{1}{2} \times 0.080 \times (2\pi / 0.30)^2 \times [(1.8 \times 10^{-2})^2 - (1.2 \times 10^{-2})^2]$ C1
 $= 3.2 \times 10^{-3} \text{ J}$ A1

29.

(a) acceleration/force proportional to displacement (from fixed point) M1

acceleration/force and displacement in opposite directions A1

(b) maximum displacements/accelerations are different B1

graph is curved/not a straight line B1

(c) (i) $\omega = 2\pi / T$ and $T = 0.8 \text{ s}$ C1

$\omega = 7.9 \text{ rad s}^{-1}$ A1

(ii) $a = (-)\omega^2 x$
 $= 7.85^2 \times 1.5 \times 10^{-2}$ C1

$= 0.93 \text{ ms}^{-2}$ or 0.94 ms^{-2} A1

(iii) $\Delta E = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ C1

$= \frac{1}{2} \times 120 \times 10^{-3} \times 7.85^2 \times \{ (1.5 \times 10^{-2})^2 - (0.9 \times 10^{-2})^2 \}$ C1

$= 5.3 \times 10^{-4} \text{ J}$ A1

30.

(a) (i) $T = 0.60 \text{ s}$ and $\omega = 2\pi / T$

C1

$$\omega = 10(10.47) \text{ rad s}^{-1}$$

A1

(ii) energy $= \frac{1}{2}m\omega^2 x_0^2$ or $\frac{1}{2}mv^2$ and $v = \omega x_0$

C1

$$= \frac{1}{2} \times 120 \times 10^{-3} \times (10.5)^2 \times (2.0 \times 10^{-2})^2$$

$$= 2.6 \times 10^{-3} \text{ J}$$

A1

(b) sketch: smooth curve in correct directions

B1

peak at f

M1

amplitude never zero and line extends from $0.7f$ to $1.3f$

A1

(c) sketch: peaked line always below a peaked line A

M1

peak not as sharp and at (or slightly less than) frequency of peak in line A

A1

31.

(a) $2k/m = \omega^2$

M1

$$\omega = 2\pi f$$

M1

$$(2 \times 64 / 0.810) = (2\pi \times f)^2 \text{ leading to } f = 2.0 \text{ Hz}$$

A1

(b) $v_0 = \omega x_0$ or $v_0 = 2\pi f x_0$
or

$$v = \omega(x_0^2 - x^2)^{1/2} \text{ and } x = 0$$

C1

$$v_0 = 2\pi \times 2.0 \times 1.6 \times 10^{-2}$$

$$= 0.20 \text{ m s}^{-1}$$

A1

(c) frequency: reduced/decreased

B1

maximum speed: reduced/decreased

B1

32.

(a) (i) 0.225 s and 0.525 s A1

(ii) period or $T = 0.30$ s and $\omega = 2\pi / T$ C1

$$\omega = 2\pi / 0.30$$

$$\omega = 21 \text{ rad s}^{-1} \quad \text{A1}$$

(iii) speed = ωx_0 or $\omega(x_0^2 - x^2)^{1/2}$ and $x = 0$ C1

$$= 20.9 \times 2.0 \times 10^{-2} = 0.42 \text{ ms}^{-1} \quad \text{A1}$$

or

use of tangent method:

correct tangent shown on Fig. 4.2

(C1)

working e.g. $\Delta y / \Delta x$ leading to maximum speed in range (0.38–0.46) ms^{-1}

(A1)

(b) sketch: reasonably shaped continuous oval/circle surrounding (0,0) B1

curve passes through (0, 0.42) and (0, –0.42) B1

curve passes through (2.0, 0) and (–2.0, 0) B1

33.

3(a)	m is constant or k/m is constant <u>and</u> so acceleration / a proportional to displacement / x	B1
	negative sign shows that acceleration / a is in opposite direction to displacement / x	B1
	or negative sign shows acceleration / a is towards fixed point	
3(b)	evidence of comparison to expression to $a = -\omega^2 x$	B1
	$\omega^2 = k/m$ or $\omega^2 = 4.0/m$ hence $\omega = 2.0/\sqrt{m}$	A1
3(c)	$E_K = \frac{1}{2} m \omega^2 x_0^2$ or $E_K = \frac{1}{2} m v^2$ <u>and</u> $v = \omega x_0$	C1
	$= \frac{1}{2} m (4.0/m) (3.0 \times 10^{-2})^2$	C1
	$= 1.8 \times 10^{-3} \text{ J}$	A1
3(d)	new $x_0 = \sqrt{[(1.8 \times 10^{-3} / 2) \times (2 / m \times (m / 4.0))]}$	C1
	or ($E_K \propto x_0^2$ so) new $x_0 = \sqrt{[\frac{1}{2} \times (3.0 \times 10^{-2})^2]}$	
	$= 2.12 \times 10^{-2} \text{ m}$	A1
3(e)	flux linked to block changes / flux is cut by block which induces an e.m.f. in block	B1
	(eddy) currents induced in block cause heating	B1
	thermal / heat energy comes from (kinetic / potential) energy of oscillations / block	B1

34.

2(a)	e.g. period = 3 / 2.5	C1
	frequency = 0.83 Hz	A1
2(b)	light (damping)	B1
2(c)	at 2.7 s, $A_0 = 1.5$ (cm)	B1
	energy = $\frac{1}{2} m \times 4\pi^2 f^2 A_0^2$	B1
	= $\frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (1.5 \times 10^{-2})^2$	C1
	= 5.51×10^{-4} (J)	
	at 7.5 s, $A_0 = 0.75$ (cm)	B1
	energy = $\frac{1}{4} \times 5.51 \times 10^{-4}$ or energy = $\frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (0.75 \times 10^{-2})^2$	C1
	energy = 1.38×10^{-4} (J) change = $(5.51 \times 10^{-4} - 1.38 \times 10^{-4}) = 4.13$ J	A1

35.

3(a)(i)	e.g. period = 6 / 2.5	C1
	frequency = 0.42 Hz	A1
3(a)(ii)	energy = $\frac{1}{2} m \times 4\pi^2 f^2 y_0^2$	C1
	= $\frac{1}{2} \times 0.25 \times 4\pi^2 \times 0.42^2 \times (1.5 \times 10^{-2})^2$	C1
	= 2.0×10^{-4} J	A1
3(b)(i)	(induced) e.m.f. proportional to rate of change of magnetic flux (linkage) or cutting of magnetic flux	M1 A1
3(b)(ii)	coil cuts flux/field (of moving magnet) <u>inducing</u> e.m.f. in coil	B1
	(induced) current in resistor causes heating (effect)	M1
	thermal energy/heat derived from energy of oscillations (of magnet)	A1

36.

2(a)	(angular frequency =) $2\pi \times$ frequency or 2π /period	B1
2(b)(i)	1. displacement = 2.0 cm	A1
	2. amplitude = 1.5 cm	A1
2(b)(ii)	reference to displacement of oscillations or displacement from equilibrium position or displacement from 2.0 cm	B1
	straight line indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1
2(b)(iii)	$\omega^2 = (-)1/\text{gradient}$ or $\omega^2 = (-)\Delta a/\Delta s$ or $a = (-)\omega^2 x$ and correct value of x	C1
	= e.g. $(1.8/0.03)$ or $(0.9/0.015)$ or $(1.2/0.02)$ etc. or $0.9 = \omega^2 \times 0.015$	C1
	= 60	
	$f = \sqrt{60}/2\pi$ = 1.2 Hz	A1

37.

3(a)(i)	angle (subtended) where arc (length) is equal to radius	M1
	(angle subtended) at the centre of a circle	A1
3(a)(ii)	angular frequency = $2\pi \times$ frequency or 2π / period	B1
3(b)(i)	c/ML^3 is a constant so acceleration is proportional to displacement	B1
	minus sign shows that acceleration and displacement are in opposite <u>directions</u>	B1
3(b)(ii)	$c/ML^3 = (2\pi f)^2$	C1
	$c = 4\pi^2 \times 3.2^2 \times 0.24 \times 0.65^3$	C1
	$= 27 \text{ kg m}^3 \text{ s}^{-2}$	A1

38.

4(a)	frequency at which body will vibrate when there is no (resultant external) resistive force acting on it OR frequency at which body will vibrate when there is no driving force / external force acting on it	B1
4(b)(i)	resonance	B1
4(b)(ii)	peak is not sharp / peak not infinite height	M1
	so damped	A1
4(c)	e.g. (quartz crystal) to produce ultrasound (quartz crystal) in watch to keep timing NMR / MRI microwave ovens tuning circuits	B1

39.

2(a)(i)	straight line through origin indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1
2(a)(ii)	$a = -\omega^2 y$ and $\omega = 2\pi f$	C1
	$4.5 = (2\pi \times f)^2 \times 8.0 \times 10^{-3}$ (or other valid read-off)	
	$f = 3.8 \text{ Hz}$	A1
2(b)(i)	maximum displacement upwards/above rest/above the equilibrium position	B1
2(b)(ii)	(just leaves plate when) acceleration = 9.81 ms^{-2}	C1
	$9.81 = (2\pi \times 3.8)^2 \times y_0$ or $9.81 = 563 \times y_0$	C1
	amplitude = 17 mm	A1

40.

4(a)	acceleration proportional to displacement	B1
	acceleration <u>directed</u> towards fixed point or displacement and acceleration in opposite <u>directions</u>	B1
4(b)(i)	1. amplitude decreases gradually so light damping or oscillations continue so light damping	B1
	2. loss of energy	B1
	due to friction in wheels or due to friction between wheels and surface (during slipping) or due to air resistance (on trolley)	B1
4(b)(ii)1.	$\omega^2 = 2k/m$	C1
	$= (2 \times 230) / 0.950$	C1
	$\omega = 22 \text{ rad s}^{-1}$	A1
4(b)(ii)2.	$T = 2\pi / \omega$	C1
	$T = (2\pi / 22) = 0.286 \text{ s}$	A1
	time $= 1.5 T$ $= 0.43 \text{ s}$	

41.

3(a)	$\omega^2 = 2g/L$	C1
	$T = 2\pi / \omega$	C1
	$\omega^2 = (2 \times 9.81) / 0.19$ $\omega = 10.2 \text{ (rad s}^{-1}\text{)}$ $T = 2\pi / 10.2$ $= 0.62 \text{ s}$	A1
3(b)(i)	e.g. viscosity of liquid/friction within the liquid/viscous drag/friction between walls of tube and liquid	B1
3(b)(ii)	(maximum) $\text{KE} = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy $= \frac{1}{2}m\omega^2 x_0^2$	C1
	change $= \frac{1}{2} \times 18 \times 10^{-3} \times 103 \times [(2.0 \times 10^{-2})^2 - (0.95 \times 10^{-2})^2]$	C1
	$= 2.9 \times 10^{-4} \text{ J}$	A1

42.

4(a)	(defining equation of s.h.m. is) $a = -kx$ where k is a constant or $a \propto -x$	B1
	g and L are constant (so $a \propto -x$ and hence s.h.m.)	B1
4(b)	$T = 0.50 \text{ s}$ and $T = 2\pi / \omega$	C1
	$\omega^2 = 2g/L$	C1
	$L = (2 \times 9.81 \times 0.50^2) / 4\pi^2$ $= 0.12 \text{ m}$	A1

4(c)(i)	Any one from: - viscosity of liquid - friction within the liquid - viscous drag - friction/resistance between walls of tube and liquid	B1
4(c)(ii)	(maximum) $KE = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy = $\frac{1}{2}m\omega^2 x_0^2$	C1
	ratio = $(1.3/2.0)^2$ = 0.42	A1

43.

3(a)(i)	mention of upthrust and weight	B1
3(a)(ii)	upthrust is greater than the weight	B1
	(resultant force is) upwards	B1
3(b)	A, ρ, g and M are constant	B1
	either acceleration $\propto$ – displacement or acceleration $\propto$ displacement <u>and</u> (– sign indicates) a and x in opposite directions	B1
3(c)(i)	either $\omega = 2\pi / T$ or $\omega = 2\pi f$ <u>and</u> $f = 1 / T$	C1
	$\omega = 2\pi / 1.3$ = 4.8 rad s ⁻¹	A1
3(c)(ii)	$\omega^2 = A\rho g / m$	C1
	$4.83^2 = (4.5 \times 10^{-4} \times \rho \times 9.81) / 0.17$	C1
	$\rho = 900 \text{ kg m}^{-3}$	A1

44.

3(a)(i)	amplitude = 0.020 m	A1
3(a)(ii)	$T = 0.60 \text{ s}$	C1
	$f = 1 / T$ = 1.7 Hz	A1
3(a)(iii)	$a = (-)\omega^2 x$ and [$\omega = 2\pi f$ or $\omega = 2\pi / T$]	C1
	$a = (4\pi^2 / 0.60^2) \times 2.0 \times 10^{-2}$ = 2.2 m s ⁻²	A1
3(b)	$1.67 = (1 / 2\pi) \times [(24 \times 10^{-4} \times \rho \times 9.81) / 0.23]^{1/2}$	C1
	$\rho = 1.1 \times 10^3 \text{ kg m}^{-3}$	A1
3(c)	wave starting with a peak at (0,6)	B1
	wave with same period (or slightly greater)	B1
	peak height decreasing successively	B1

45.

3(a)(i)	0.10 s or 0.30 s or 0.50 s or 0.70 s or 0.90 s	A1
3(a)(ii)	0 or 0.40 s or 0.80 s	A1
3(b)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / 0.40$	A1
	$= 16 \text{ rad s}^{-1}$	
3(b)(ii)	$v_0 = \omega x_0$	C1
	$= 15.7 \times 2.5 \times 10^{-2}$	A1
	$= 0.39 \text{ m s}^{-1}$	
	or	
	tangent drawn at steepest part and working to show attempted calculation of gradient	(C1)
	leading to $v_0 = 0.39 \text{ m s}^{-1}$ (allow $\pm 0.15 \text{ m s}^{-1}$)	(A1)
3(b)(iii)	$a_0 = \omega^2 x_0$	C1
	$a_0 = (15.7^2 \times 2.5 \times 10^{-2})$	A1
	$= 6.2 \text{ m s}^{-2}$	
	or	
	$a_0 = \omega v_0$	(C1)
	$a_0 = 15.7 \times 0.39$	(A1)
	$= 6.2 \text{ m s}^{-2}$	
3(c)	period is shorter/lower	B1
	Any one from: - greater spring constant/stiffness (restoring) force is greater (for any given extension) - acceleration is greater (for any given extension) - greater energy/maximum speed (for a given amplitude)	B1

46.

4(a)(i)	distance from a (reference) point in a given direction	B1
4(a)(ii)	line is not straight or gradient is not constant	B1
4(b)(i)	0.85–0.90 cm	A1
4(b)(ii)	$a = -(2\pi f)^2 x$	C1
	e.g. $1.2 = 4\pi^2 \times f^2 \times (0.90 \times 10^{-2})$	C1
	$f = 1.8 \text{ Hz}$	A1
4(c)	complete circle/ellipse enclosing the origin	B1
	closed shape passing through $(0, \pm v_0)$ and $(\pm x_0, 0)$	B1

47.

4(a)(i)	loss of energy	B1
4(a)(ii)	amplitude (of oscillations) decreases (with time)	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$T = 0.80 \text{ s}$, so $\omega = 2\pi / 0.80$	A1
	$\omega = 7.9 \text{ rad s}^{-1}$	
4(b)(ii)	$\omega^2 = 2k / M$	C1
	$7.9^2 = 2k / 1.2$	A1
	$k = 37 \text{ N m}^{-1}$	
4(c)(i)	(one) smooth curve, not touching the f -axis, with two concave sides meeting at a peak in between them	B1
	(one) peak at 1.0ω	B1
4(c)(ii)	- lower peak/(whole) line is lower - flatter <u>peak/peak</u> is less sharp - peak at (slightly) lower angular frequency/peak moves to left any two points, one mark each	B2

48.

3(a)(i)	0.050 m	A1
3(a)(ii)	$\omega = v_o / x_o$	C1
	$T = 2\pi / \omega$	C1
	$0.42 = (2\pi \times 0.050) / T$	
	$T = \mathbf{0.75 \text{ s}}$	A1
3(a)(iii)	one point labelled P where ellipse crosses displacement axis marked	A1
3(b)(i)	(induced) e.m.f. proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
3(b)(ii)	(there is) current in the circuit	B1
	<i>either</i>	
	current causes thermal energy (dissipated) in resistor	B1
	thermal energy comes from energy of magnet	B1
	<i>or</i>	
	current causes magnetic field around coil	(B1)
	two fields cause an opposing force on magnet	(B1)

49.

3(a)(i)	amplitude = 4.9 cm	A1
3(a)(ii)	frequency = $2700 / 60$ = 45 Hz	A1
3(a)(iii)	$v_0 = x_0 \omega$ and $\omega = 2\pi f$	C1
	$v_0 = 4.9 \times 10^{-2} \times 2\pi \times 45$ = 14 m s^{-1}	A1
3(a)(iv)	$v = \omega (x_0^2 - x^2)^{1/2}$ = $2\pi \times 45 \times [(4.9 \times 10^{-2})^2 - (2.6 \times 10^{-2})^2]^{1/2}$ = 12 m s^{-1}	C1
		A1
3(b)	$F = ma$ and $a_0 = v_0 \omega$ or $a_0 = x_0 \omega^2$	C1
	$F = 0.64 \times 13.9 \times 2\pi \times 45$ or $0.64 \times 4.9 \times (2\pi \times 45)^2$	C1
	= 2500 N	A1

50.

4(a)	$(\omega = 2\pi / T \text{ and } T = 2.2 \text{ s so})$ $\omega = 2\pi / 2.2 = 2.9 \text{ rad s}^{-1}$	A1
4(b)(i)	$\omega^2 = g / R$	C1
	$R = 9.81 / 2.86^2$ = 1.2 m	A1
4(b)(ii)	$v_0 = \omega x_0$	C1
	= $2.9 \times 3.0 \times 10^{-2}$ = 0.087 m s^{-1}	A1
4(c)	smooth wave starting at 3.0 cm when $t = 0$	B1
	positions of peaks and troughs show same period (or slightly longer)	B1
	each peak and trough at lower amplitude than the previous one	B1

51.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration in opposite <u>direction</u> to displacement or acceleration (directed) towards equilibrium position	B1
3(b)	$v = \omega (x_0^2 - x^2)^{1/2}$ and $\omega = 2\pi f$ or $v_0 = x_0 \omega$ and $\omega = 2\pi f$	C1
	substitution of any correct point from graph, e.g. for $x = 0$: $0.25 = 2\pi f \times 8.8 \times 10^{-2}$ $f = 0.45 \text{ Hz}$	C1
		A1
3(c)	$1 / 0.45 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 1.2 \text{ m}$	A1
3(d)	ellipse about the origin with same intercepts on x-axis	B1
	ellipse about the origin crossing v-axis inside original loop	B1

52.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration is in opposite <u>direction</u> to displacement or acceleration is (directed) towards a fixed point	B1
3(b)(i)	zero	B1
3(b)(ii)	E_T is maximum potential energy = mgh $E_T = 94 \times 10^{-3} \times 9.81 \times 0.90 \times 10^{-2}$ $= 8.3 \times 10^{-3} \text{ J}$	C1 A1
	$E_{\text{MAX}} = \frac{1}{2} m v_0^2$ and $v_0 = \omega x_0$ or $E_{\text{MAX}} = \frac{1}{2} m (\omega x_0)^2$ $8.3 \times 10^{-3} = \frac{1}{2} \times 94 \times 10^{-3} \times \omega^2 \times (12.7 \times 10^{-2})^2$...leading to $\omega = 3.3 \text{ rad s}^{-1}$	C1 A1
3(c)	$T = 2\pi / \omega$	C1
	$2\pi / 3.3 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 0.90 \text{ m}$	A1

53.

4(a)	acceleration and displacement are in opposite directions	B1
4(b)(i)	$F = kx$ $= 8.0 \times (0.060 - 0.048)$ or $8.0 \times (0.060 + 0.048)$ or 8.0×0.012 or 8.0×0.108	M1
	$\Sigma F = (8.0 \times 0.012) - (8.0 \times 0.108) = 0.77 \text{ N}$ or $\Sigma F = 0.864 - 0.096 = 0.77 \text{ N}$	A1
4(b)(ii)	$a = \frac{F}{m}$ $= \frac{0.77}{0.25}$ $= 3.1 \text{ ms}^{-2}$	A1
4(b)(iii)	$a = -\omega^2 x$ $\omega = \sqrt{\frac{3.1}{0.048}}$ $\omega = 8.04$	C1
	$T = 2\pi / \omega$	C1
	$T = 2\pi / 8.04$ $= 0.78 \text{ s}$	A1
4(b)(iv)	(resultant) force halved and distance halved	B1
	same T	B1

54.

3(a)	acceleration (directly) proportional to displacement	B1
	acceleration is in opposite <u>direction</u> to displacement	B1
3(b)	$\omega^2 = 2k / m$ and $\omega = 2\pi f$	C1
	$(2\pi f)^2 = (2 \times 130) / 0.84$	C1
	$f = 2.8 \text{ Hz}$	A1
3(c)(i)	resonance	B1
3(c)(ii)	oscillator supplies energy (continuously)	B1
	energy of trolley constant so energy must be dissipated or without loss of energy the amplitude would continuously increase	B1

55.

3(a)	acceleration in opposite <u>direction</u> to displacement shown by – sign	B1
	g / L is constant	M1
	(so) acceleration is (directly) proportional to displacement	A1
3(b)	$\omega^2 = g / L$	C1
	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$(2\pi / T)^2 = 9.81 / 0.18$	A1
	$T = 0.85 \text{ s}$	
3(c)	energy $\propto x_0^2$	C1
	(after 3 cycles,) amplitude = $(0.94)^3 x_0$ = $0.83x_0$	C1
	ratio final energy / initial energy = 0.83^2 = 0.69	A1

56.

4(a)(i)	5.0 cm	A1
4(a)(ii)	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$\omega = 2\pi / 4.0$ = 1.6 rad s^{-1}	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	= 1.57×5.0	A1
	= 7.9 cm s^{-1}	
4(b)	- initial pull was to the right - distance from X to trolley (at equilibrium) is 20 cm - period is 4.0 s - initial motion undamped - motion becomes damped at/from 12 s - damping is light - maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc. Any three points, 1 mark each	B3

4(c)	sketch: closed loop encircling (20, 0)	B1
	minimum L shown as 15 cm <u>and</u> maximum L shown as 25 cm	B1
	minimum v shown as -7.9 cm s^{-1} <u>and</u> maximum v shown as $+7.9 \text{ cm s}^{-1}$	B1

57.

4(a)	straight line through the origin	B1
	negative gradient	B1
4(b)	$a = (-)\omega^2 x$ <u>and</u> $T = 2\pi / \omega$	C1
	e.g. $\omega = \sqrt{(0.80 / 0.12)}$ (any correct pair of values of a and x) ($= 2.58 \text{ rad s}^{-1}$)	C1
	$T = 2\pi / 2.58$ $= 2.4 \text{ s}$	A1
4(c)(i)	Point labelled P at one end of the line	B1
4(c)(ii)	Point labelled Q at displacement with magnitude more than half but less than maximum	B1

58.

3(a)	upthrust, weight	B1
3(b)	upthrust greater than weight so (resultant force is) upwards	B1
3(c)(i)	A , g and ρ all constant so $F \propto x$	B1
	minus sign means F and x are in opposite directions	B1
3(c)(ii)	$(a = \frac{F}{m} \text{ so } a = (-)\frac{A\rho g x}{m})$	M1
	so $\omega^2 = \frac{A\rho g}{m}$ hence $\omega = \sqrt{\frac{A\rho g}{m}}$	A1
3(d)(i)	damping due to viscous forces	B1
3(d)(ii)	$(E =) \frac{1}{2} m \omega^2 x_0^2$	C1
	$\omega^2 = (-)$ gradient	C1
	$(E =) \frac{1}{2} m \omega^2 (x_1^2 - x_2^2)$ $= \frac{1}{2} \times 0.57 \times (2.3 / 0.020)(0.020^2 - 0.016^2)$ $= 4.7 \times 10^{-3} \text{ J}$	A1

59.

4(a)	straight line through origin shows that a is proportional to x	B1
	negative gradient shows that a is in opposite direction to x	B1
4(b)(i)	$a_0 = \omega^2 x_0$ or $a = -\omega^2 x$ or $\omega^2 = -\text{gradient}$	C1
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8 \text{ rad s}^{-1}$	A1
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	C1
	$= 9.7 \text{ m s}^{-2}$	A1
4(c)	(increasing L causes) ω to decrease or energy $(= \frac{1}{2} m \omega^2 x_0^2) = \frac{1}{2} m k x_0^2 / L$ (and L increases)	M1
	so amplitude increases	A1

60.

4(a)	oscillations (of object) at maximum amplitude	B1
	when driving frequency equals natural frequency (of object)	B1
4(b)(i)	$T = 2\pi / \omega$	C1
	$= 2\pi / 5.0\pi$ $= 0.40 \text{ s}$	A1
4(b)(ii)	displacement scale labelled $-1.0, -0.5, (0), 0.5, 1.0$ on the 2 cm tick marks	B1
	t scale labelled $0.2, 0.4, 0.6, 0.8, 1.0, 1.2$ on the 2 cm tick marks	B1
4(b)(iii)	$\phi = 2\pi \Delta t / T$ $= 2\pi \times 0.10 / 0.40$ or $2\pi \times 0.30 / 0.40$	C1
	$= 1.6 \text{ rad}$ or 4.7 rad	A1

61.

3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12 \text{ m}$	A1
3(b)(ii)	$v = \omega \sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be $(x_0, 0)$ or value of x_0 from (i))	C1
	e.g. $0.20 = \omega \sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7 \text{ rad s}^{-1}$	A1

3(b)(iii)	$E = \frac{1}{2}M\omega^2x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$	A1
	$M = 2.5 \text{ kg}$	
	or	
	$(E_K)_{\max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$	(A1)
	$M = 2.5 \text{ kg}$	
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at $\pm 0.060 \text{ m}$ passing through $v = 0$	B1
	maximum velocity shown as $\pm 0.10 \text{ m s}^{-1}$ passing through $x = 0$	B1

62.

4(a)(i)	$x_0 = 8.0 \text{ cm}$	A1
4(a)(ii)	$\omega = 2\pi / T$	C1
	$= 2\pi / 4.0 = 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$E = \frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$	C1
	$= 0.29 \text{ J}$	A1
4(b)	dome-shaped curve, starting and ending at $E_K = 0$	B1
	maximum E_K shown as 0.29 J	B1
	position of peak shown at $h = 10.0 \text{ cm}$	B1
	line intercepts h -axis at $h = 2.0 \text{ cm}$ and at $h = 18.0 \text{ cm}$	B1

63.

3(a)	P: total energy Q: potential energy R: kinetic energy	B2
3(b)	$E = \frac{1}{2}m\omega^2x_0^2$ or $E = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$	C1
	$6.4 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times 0.015^2$ ($\omega^2 = 438$) ($\omega = 20.9$)	C1
	$T = 2\pi / \omega$	C1
	$= 2\pi / 20.9$	A1
	$= 0.30 \text{ s}$	
3(c)(i)	resistive forces	B1
3(c)(ii)	0.92^6	C1
	decrease in energy $= 6.4 - (6.4 \times 0.92^6)$ $= 2.5 \text{ mJ}$	A1
3(c)(iii)	light damping because the amplitude of oscillations gradually reduces or light damping because the system still oscillates	B1

64.

4(a)(i)	$\omega = 2\pi f$	C1
	$f = 9.7 / 2\pi$	A1
	$= 1.5 \text{ Hz}$	
4(a)(ii)	amplitude $= \sqrt{11.6} = 3.4 \text{ cm}$	A1
4(a)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 9.7^2 \times 3.4 \times 10^{-2}$	A1
	$= 3.2 \text{ m s}^{-2}$	
4(b)	sketch: straight line through the origin with negative gradient	B1
	line with negative gradient passing through $(+3.4, -a_0)$ and $(-3.4, +a_0)$	B1
	line with ends at $x = \pm 3.4 \text{ cm}$ and $a = \pm a_0$	B1
4(c)	sum of potential energy and kinetic energy is constant	B1
	at maximum displacement, kinetic energy is zero or at maximum displacement, potential energy is maximum	B1
	at zero displacement, kinetic energy is maximum or at zero displacement, potential energy is minimum	B1

65.

4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) directed towards a fixed point	B1
4(b)(i)	$\omega = 2\pi / T$	C1
	$\omega = 2\pi / 3.0$	A1
	$= 2.1 \text{ rad s}^{-1}$	
4(b)(ii)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$= \frac{1}{2} \times 0.81 \times 2.1^2 \times 0.036^2$	A1
	$= 2.3 \times 10^{-3} \text{ J}$	
4(c)	sketch: line starting at $(0, 0.036)$ and not reaching $x = \pm 0.036 \text{ m}$ at any other time	B1
	smooth curve, with no sudden changes in gradient, showing continuously decreasing magnitude of x from maximum displacement at $t = 0$ to final displacement of zero where the gradient is also zero	B1
	displacement reaches final value of zero between $t = 0.75 \text{ s}$ and $t = 3.0 \text{ s}$ at the latest	B1

66.

4(a)(i)	amplitude = $\frac{1}{2} \times 7.2 \times 10^{-15}$ = $3.6 \times 10^{-15} \text{ m}$	A1
4(a)(ii)	$\omega = 2\pi / (0.20 \times 10^{-6})$ = $3.1 \times 10^7 \text{ rad s}^{-1}$	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$v_0 = 3.1 \times 10^7 \times 3.6 \times 10^{-15} = 1.1 \times 10^{-7} \text{ m s}^{-1}$	A1
4(b)(i)	$I_0 = nA v_0 e$ = $8.5 \times 10^{28} \times 4.3 \times 10^{-4} \times 1.1 \times 10^{-7} \times 1.60 \times 10^{-19}$	C1
	= 0.64 A	A1
4(b)(ii)	sketch: two cycles of sinusoidal curve of amplitude I_0 and period $0.20 \mu\text{s}$	B1
	correct phase, with $I = +I_0$ at $t = 0$	B1
4(b)(iii)	equation of form $I = I_0 \cos \omega t$	M1
	value of I_0 used matches answer to (b)(i) and value of ω used matches answer to (a)(ii) [if (a)(ii) and (b)(i) correct then $I = 0.64 \cos (3.1 \times 10^7 t)$]	A1
4(b)(iv)	$I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ = $0.64 / \sqrt{2}$ = 0.45 A	A1